A SPECTRAL TOOLKIT FOR CGSS: STRUCTURED PROOFS

Abstract. We give structured proofs of the spectral toolkit for Cameron, Goethals, Seidel and Shult. Every proof follows the argument of the ordinary version, with explicit hypotheses, eigenvalue multiplicities and finite-dimensional coordinate maps. We retain its proposed Lean names and definitions; these proof trees guided the formalization.

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1. Conventions and checked facts

We follow the conventions of the blueprint (`blueprint/conventions.tex`) and the proof format of Lamport [L95]. The statements use numbered assumptions and a conclusion; the level-one steps follow the ordinary proofs in `spectraltools.tex`. A proof leaf names its inputs. An assumption at level k is cited as $\langle k\rangle$, and its numbered part i as $\langle k\rangle{:}i$. The tags `\lean` and `\uses` are metadata and print nothing. The ordinary source-status lines distinguish existing project declarations from proposed interfaces; names alone do not assert a Lean proof. The ordinary version already contains two corrections, the hypothesis $A \succeq 0$ for principal minors in Theorem 3.3 and the symmetry guard for graph certificates in Theorem 9.8, so they are not repairs here. Where the ordinary version attributes a result to Mathlib, the corresponding leaf applies that declaration. No step departs from the argument of the ordinary version, so no step is tagged **[new]** or **[repair]**. The proof of interlacing uses the ordinary proof's dimension witnesses directly; it does not need a maximum over a space of subspaces.

All sets indexing a matrix are finite, and we will choose decidable equality and decidable adjacency classically. Entries use functional notation $A(u, v)$ and $\mathbf{v}(u)$. Vectors are columns, I is the identity, J is the all-ones matrix and $\mathbf{j}$ is the all-ones vector, on the indicated index set. In $\mathbb{R}^U$, the inner product is $\langle \mathbf{x}, \mathbf{y}\rangle = \sum_u \mathbf{x}(u)\mathbf{y}(u)$. Matrix multiplication, trace, transpose and polynomial evaluation have their usual coordinate definitions. Rank means dimension of the image. Empty sums are zero and empty products are one, including the determinant of a zero-order matrix. Connectedness includes nonemptiness. Graphs are simple; we repeat finiteness

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in every result. All spectral multisets include repetitions. Dimension differences in formulas for multiplicities are integer differences proved nonnegative, not unexplained truncated subtraction.

The following register specifies the exact library facts used in leaves. Each path is relative to `Mathlib/`, at commit `082e2d37` (4 September 2026), the version pinned in `lake-manifest.json`. Elementary coordinate expansions, induction, finite-set bijections and field arithmetic in the proofs are deductions written out below.

For a finite real symmetric matrix, Hermitian symmetry gives an orthonormal eigenbasis, a decreasing eigenvalue list, the product formula for its characteristic polynomial, its root multiset, and its set spectrum.

- Mathlib: `Analysis/Matrix/Spectrum.lean`. `Matrix.IsHermitian.eigenvalues`$_0$, `Matrix.IsHermitian.eigenvalues`$_0$`_antitone`, `Matrix.IsHermitian.eigenvalues`, `Matrix.IsHermitian.eigenvectorBasis`, `Matrix.IsHermitian.mulVec_eigenvectorBasis`, `Matrix.IsHermitian.charpoly_eq`, `Matrix.IsHermitian.roots_charpoly_eq_eigenvalues`, `Matrix.IsHermitian.roots_charpoly_eq_eigenvalues`$_0$, `Matrix.IsHermitian.spectrum_real_eq_range_eigenvalues`, `Matrix.IsHermitian.trace_eq_sum_eigenvalues`, `Matrix.IsHermitian.det_eq_prod_eigenvalues`, `Matrix.IsHermitian.rank_eq_card_non_zero_eigs`.
- Mathlib: `Data/Fintype/EquivFin.lean`. `Fintype.equivOfCardEq`.

For finite-dimensional real inner product spaces, orthonormal bases give coordinates, and symmetric endomorphisms have orthonormal eigenbases. The function and Euclidean coordinate models have inverse linear maps; both matrix actions are the same coordinate sum. The eigenspace predicate is the eigenvector equation, allowing the zero vector.

- Mathlib: `Analysis/InnerProductSpace/PiL2.lean`. `EuclideanSpace`, `Matrix.toEuclideanLin`, `OrthonormalBasis.sum_repr`, `OrthonormalBasis.reindex`, `exists_orthonormalBasis`.
- Mathlib: `Analysis/Normed/Lp/WithLp.lean`. `WithLp.toLp`, `WithLp.linearEquiv`, `WithLp.ofLp_toLp`, `WithLp.toLp_ofLp`.
- Mathlib: `Analysis/Normed/Lp/Matrix.lean`. `Matrix.toLpLin_apply`.
- Mathlib: `LinearAlgebra/Matrix/ToLin.lean`. `Matrix.toLin'`, `Matrix.toLin'_apply`.
- Mathlib: `Analysis/Matrix/Hermitian.lean`. `Matrix.isSymmetric_toEuclideanLin_iff`.
- Mathlib: `Analysis/InnerProductSpace/Spectrum.lean`. `LinearMap.IsSymmetric.apply_eigenvectorBasis`, `LinearMap.IsSymmetric.card_filter_eigenvalues_eq`.
- Mathlib: `LinearAlgebra/Eigenspace/Basic.lean`. `Module.End.eigenspace`, `Module.End.mem_eigenspace_iff`.

Over a field, finite basis cardinalities are dimensions; linear equivalences preserve dimension. Subspace inclusion bounds dimension. Rank-nullity and the dimension formula for a sum and intersection apply in finite dimension. A finite-dimensional endomorphism is injective iff it is surjective.

- Mathlib: `LinearAlgebra/Dimension/StrongRankCondition.lean`. `Module.finrank_eq_card_basis`.
- Mathlib: `LinearAlgebra/Dimension/Finrank.lean`. `LinearEquiv.finrank_eq`.
- Mathlib: `LinearAlgebra/Dimension/Constructions.lean`. `Submodule.finrank_le`, `Submodule.finrank_mono`.
- Mathlib: `LinearAlgebra/FiniteDimensional/Lemmas.lean`. `Submodule.finrank_sup_add_finrank_inf_eq`, `LinearMap.finrank_range_add_finrank_ker`.

- Mathlib: `LinearAlgebra/FiniteDimensional/Basic.lean`. `LinearMap.injective_iff_surjective`.

Over a commutative ring, determinants multiply, reindexing both axes by a bijection preserves determinant, and an upper triangular block determinant is the product of its diagonal-block determinants. A square matrix is a unit iff its determinant is a unit; over a field this means a nonzero determinant.

- Mathlib: `LinearAlgebra/Matrix/Determinant/Basic.lean`. `Matrix.det_mul`, `Matrix.det_reindex_self`, $\texttt{Matrix.det_fromBlocks_zero}_{21}$.
- Mathlib: `LinearAlgebra/Matrix/NonsingularInverse.lean`. `Matrix.isUnit_iff_isUnit_det`.

Polynomial roots over a domain count algebraic multiplicity, and a product of linear factors has exactly their roots. For a nonzero product, root multiplicities add; the nonzero qualification is necessary. Two finite multisets are equal iff every count agrees.

- Mathlib: `Algebra/Polynomial/Roots.lean`. `Polynomial.count_roots`, `Polynomial.roots_mul`, `Polynomial.roots_multiset_prod_X_sub_C`.
- Mathlib: `Algebra/Polynomial/RingDivision.lean`. `Polynomial.rootMultiplicity_mul`.
- Mathlib: `Data/Multiset/Count.lean`. `Multiset.ext`.

For finite real square matrices, positive semidefiniteness means symmetry and a nonnegative quadratic form. The zero-form criterion requires positive semidefiniteness. Principal submatrices retain it.

- Mathlib: `LinearAlgebra/Matrix/PosDef.lean`. `Matrix.posSemidef_iff_dotProduct_mulVec`, `Matrix.PosSemidef.submatrix`.
- Mathlib: `Analysis/Matrix/PosDef.lean`. `Matrix.IsHermitian.posSemidef_iff_eigenvalues_nonneg`, `Matrix.IsHermitian.posDef_iff_eigenvalues_pos`, `Matrix.PosSemidef.det_nonneg`.
- Mathlib: `Analysis/InnerProductSpace/Positive.lean`. `Matrix.posSemidef_iff_eq_sum_vecMulVec`.
- Mathlib: `Analysis/Matrix/Order.lean`. `Matrix.PosSemidef.dotProduct_mulVec_zero_iff`.

For finite real rectangular matrices, transpose preserves rank and both Gram products have the rank of the original matrix.

- Mathlib: `LinearAlgebra/Matrix/Rank.lean`. `Matrix.rank_transpose`, `Matrix.rank_transpose_mul_self`, `Matrix.rank_self_mul_transpose`.

A finite sum of nonnegative real numbers is zero iff every term is zero, and is positive iff one term is positive. In particular this applies to squares. For a nonnegative real number, the square of its nonnegative square root is itself; its square root is positive iff the number is positive. The two finite-sum declarations are generated by the indicated additive attributes in the source.

- Mathlib: `Algebra/Order/BigOperators/Group/Finset.lean`. `Finset.sum_eq_zero_iff_of_nonneg`, `Finset.sum_pos_iff_of_nonneg`.
- Mathlib: `Analysis/Real/Sqrt.lean`. `Real.sq_sqrt`, `Real.sqrt_pos`.

Characteristic polynomials of finite square matrices over a commutative ring are monic, commute with coefficient ring maps and are invariant under conjugation by a unit; a diagonal matrix has the product of its linear factors. For rectangular products of sizes p, q, the identity is $t^q \chi_{MN} = t^p \chi_{NM}$.

- Mathlib: `LinearAlgebra/Matrix/Charpoly/Coeff.lean`. `Matrix.charpoly_monic`.
- Mathlib: `LinearAlgebra/Matrix/Charpoly/Basic.lean`. `Matrix.charpoly_map`, `Matrix.charpoly_mul_comm'`, `Matrix.charpoly_units_conj`, `Matrix.charpoly_diagonal`.

For finite simple graphs with classical decidable adjacency, the real adjacency matrix is symmetric, its powers count walks, and graph isomorphisms reindex it. The complement adjacency matrix is $J - I - A$. The Mathlib adjacency-matrix structure bundles zero-one entries, symmetry and zero diagonal, and its graph has adjacency matrix the given matrix. The line graph has actual edges as vertices, adjacent when distinct and intersecting. Component and reachability terminology has its usual simple-graph meaning.

- Mathlib: `Combinatorics/SimpleGraph/AdjMatrix.lean`. `SimpleGraph.adjMatrix`, `SimpleGraph.adjMatrix_pow_apply_eq_card_walk`, `SimpleGraph.Iso.reindex_adjMatrix`, `SimpleGraph.adjMatrix_mulVec_const_apply_of_regular`, `SimpleGraph.trace_adjMatrix`, `SimpleGraph.adjMatrix_mul_self_apply_self`, `SimpleGraph.compl_adjMatrix_eq_adjMatrix_compl`, `SimpleGraph.one_add_adjMatrix_add_compl_adjMatrix_eq_of_one`, `Matrix.IsAdjMatrix`, `Matrix.IsAdjMatrix.toGraph`, `Matrix.IsAdjMatrix.adjMatrix_toGraph_eq`.
- Mathlib: `Combinatorics/SimpleGraph/LapMatrix.lean`. `SimpleGraph.isHermitian_adjMatrix`.
- Mathlib: `Combinatorics/SimpleGraph/LineGraph.lean`. `SimpleGraph.lineGraph`.

An idempotent endomorphism is a projection onto its range. Over $\mathbb{R}$, the finite-dimensional image and kernel are free, so its trace equals its range dimension. Matrix trace and rank agree with the corresponding coordinate linear map.

- Mathlib: `LinearAlgebra/Projection.lean`. `LinearMap.isProj_range_iff_isIdempotentElem`.
- Mathlib: `LinearAlgebra/Trace.lean`. `LinearMap.IsProj.trace`, `Matrix.trace_toLin'_eq`.
- Mathlib: `LinearAlgebra/Matrix/Rank.lean`. `Matrix.rank_eq_finrank_range_toLin`.

On a nontrivial finite-dimensional real inner product space, a symmetric linear map has eigenvalues equal to the supremum and infimum of its Rayleigh quotient over nonzero vectors.

- Mathlib: `Analysis/InnerProductSpace/Rayleigh.lean`. `LinearMap.IsSymmetric.hasEigenvalue_iSup_of_finiteDimensional`, `LinearMap.IsSymmetric.hasEigenvalue_iInf_of_finiteDimensional`.

For a finite simple graph the real Laplacian is positive semidefinite; its kernel functions are constant on reachable pairs, and its kernel dimension is the number of components, including isolated vertices.

- Mathlib: `Combinatorics/SimpleGraph/LapMatrix.lean`. `SimpleGraph.posSemidef_lapMatrix`, `SimpleGraph.lapMatrix_mulVec_eq_zero_iff_forall_reachable`, `SimpleGraph.card_connectedComponent_eq_finrank_ker_toLin'_lapMatrix`.

The incidence matrix on all unordered pairs, with zero columns off the edge set, has product equal to the degree matrix plus adjacency. Restricting away the zero columns is a separate finite-sum argument.

- Mathlib: `Combinatorics/SimpleGraph/IncMatrix.lean`. `SimpleGraph.incMatrix_mul_transpose`.

The strongly regular matrix equation holds over any semiring; its parameter equation is in natural numbers with truncated subtraction and assumes positive order. Complement parameters also use natural subtraction, so conversion to integer formulas requires bounds.

- Mathlib: `Combinatorics/SimpleGraph/StronglyRegular.lean`. `SimpleGraph.IsSRGWith.matrix_eq`, `SimpleGraph.IsSRGWith.param_eq`, `SimpleGraph.IsSRGWith.compl`.

For any number of distinct nodes over a domain, a coefficient vector with all moments of degrees below that number zero is zero. The index size may be zero.

- Mathlib: `LinearAlgebra/Vandermonde.lean`. `Matrix.eq_zero_of_forall_pow_sum_mul_pow_eq_zero`.

Over a commutative ring, the determinant of $I + uv^T$ is $1 + v^T u$; in the declaration a singleton auxiliary type indexes the replicated column and row.

- Mathlib: `LinearAlgebra/Matrix/SchurComplement.lean`. `Matrix.det_one_add_replicateCol_mul_replicateRow`.

A graph is a tree iff it has a nonempty vertex set and a unique simple path between each pair of vertices. Here a tree means connected and acyclic; an edge-preserving subgraph need not be induced.

- Mathlib: `Combinatorics/SimpleGraph/Acyclic.lean`. `SimpleGraph.isTree_iff_existsUnique_path`.

A rational has a reduced numerator and positive denominator. Coprimality is preserved on taking a positive power, and a common divisor of coprime natural numbers is one. Taking the absolute value of an integer numerator gives these natural-number statements. The monic rational-root theorem also proves the final integrality implication directly.

- Mathlib: `Data/Rat/Defs.lean`. `Rat.mk'_num_den`, `Rat.pos`.
- Mathlib: `Data/Nat/GCD/Basic.lean`. `Nat.coprime_pow_left_iff`, `Nat.eq_one_of_dvd_coprimes`.
- Mathlib: `RingTheory/Polynomial/RationalRoot.lean`. `isInteger_of_is_root_of_monic`.

For a finite type of cardinality ℓ, the unordered pairs with repetitions have cardinality $\binom{\ell+1}{2}$. These index the independent entries of a symmetric matrix.

- Mathlib: `Data/Sym/Card.lean`. `Sym2.card`.

2. Definitions and the spectral theorem

Definition 2.1.

Assume:
1. U is finite and $A : U \times U \to \mathbb{R}$.

For a real square matrix A, put

$$\chi_A(t) = \det(tI - A), \qquad \mathrm{Spec}(A) = \chi_A.\mathrm{roots}, \qquad \mathrm{mult}_A(\mu) = \mathrm{count}_{\mathrm{Spec}(A)}(\mu).$$

Here roots form a multiset over $\mathbb{R}$, with algebraic multiplicity. We will call $\ker(A - \mu I)$ the *eigenspace* $V_A(\mu)$; an eigenvector is a nonzero member. The definition of Spec makes sense without symmetry, but its cardinality can then be smaller than n.

Theorem 2.2 (Coordinates, roots, and multiplicity)**.**

Assume:
1. U is finite, $|U| = n$, and $A : U \times U \to \mathbb{R}$ is symmetric.

Prove: There is an orthonormal basis $(\mathbf{v}_i)_{i \in U}$ with $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$, where λ_i is the value at i of `Matrix.IsHermitian.eigenvalues` for A, and

$$\chi_A(t) = \prod_{i \in U} (t - \lambda_i), \qquad \mathrm{Spec}(A) = [\![\lambda_i : i \in U]\!].$$
$$\mathrm{mult}_A(\mu) = \dim V_A(\mu) \quad (\mu \in \mathbb{R}).$$

Thus $|\mathrm{Spec}(A)| = n$; membership in $\mathrm{Spec}(A)$, existence of a nonzero μ-eigenvector, and membership in Mathlib's set-valued $\mathrm{spectrum}_{\mathbb{R}}(A)$ are equivalent. Multiplicity also equals `A.charpoly.rootMultiplicitymu`. More generally, if $(\mathbf{u}_j)_{j \in I}$ is any orthonormal basis of $\mathbb{R}^U$ with $A\mathbf{u}_j = \mu_j \mathbf{u}_j$, then $\mathrm{Spec}(A) = [\![\mu_j : j \in I]\!]$. There is also a sorted orthonormal eigenbasis $(\mathbf{w}_i)_{i \in \mathrm{Fin}(n)}$ with $A\mathbf{w}_i = \nu_i \mathbf{w}_i$, where $\nu_i = \texttt{hA.eigenvalues}_0(i)$ and $\nu_i \geq \nu_j$ for $i \leq j$. Its eigenvalues list the spectrum in decreasing order: $\mathrm{Spec}(A) = [\![\nu_i : i \in \mathrm{Fin}(n)]\!]$.

Proof sketch: We will diagonalize A, read its roots with repetitions, and identify the eigenspace coordinates.

⟨1⟩1. Pick an orthonormal basis $(\mathbf{v}_i)$ with $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$; its column matrix Q satisfies $Q^T Q = QQ^T = I$.

⟨2⟩1. A is Hermitian over $\mathbb{R}$.
Proof: Conjugation on $\mathbb{R}$ is the identity; use the definition of conjugate transpose. □

⟨2⟩2. There is the asserted orthonormal eigenbasis.
Proof: Apply `Matrix.IsHermitian.eigenvectorBasis` and `Matrix.IsHermitian.mulVec_eigenvectorBasis` in Facts 1 to ⟨2⟩1. □

⟨2⟩3. $Q^T Q = I$ and $QQ^T = I$.

⟨3⟩1. $Q^T Q = I$.
Proof: Its (i, j) entry is $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij}$ by ⟨2⟩2. □

⟨3⟩2. Q is invertible.
Proof: Its columns form a basis by ⟨2⟩2; the basis coordinate map is its inverse. □

⟨3⟩3. $QQ^T = I$.

PROOF: The left inverse in ⟨3⟩1 equals the inverse of ⟨3⟩2. □

⟨3⟩4. Q.E.D.

PROOF: Use ⟨3⟩1, ⟨3⟩3. □

⟨2⟩4. Q.E.D.

PROOF: Use ⟨2⟩2, ⟨2⟩3. □

⟨1⟩2. $A = Q\,\mathrm{diag}(\lambda_i)Q^T$, $\chi_A = \prod_i (t - \lambda_i)$ and $\mathrm{Spec}(A) = [\![\lambda_i : i \in U]\!]$.

⟨2⟩1. $AQ = Q\,\mathrm{diag}(\lambda_i)$.

PROOF: Its ith column is the eigenvector equation in ⟨1⟩1. □

⟨2⟩2. $A = Q\,\mathrm{diag}(\lambda_i)Q^T$.

PROOF: Multiply ⟨2⟩1 on the right by Q^T and use ⟨1⟩1. □

⟨2⟩3. $\chi_A = \prod_i (t - \lambda_i)$.

PROOF: Apply `Matrix.IsHermitian.charpoly_eq` in Facts 1; it is the determinant of the conjugation in ⟨2⟩2. □

⟨2⟩4. $\mathrm{Spec}(A) = [\![\lambda_i : i \in U]\!]$.

PROOF: Apply `Matrix.IsHermitian.roots_charpoly_eq_eigenvalues` in Facts 1 and Definition 2.1. □

⟨2⟩5. Q.E.D.

PROOF: Use ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩3.

ASSUME: I is finite and $(\mathbf{u}_j)_{j \in I}$ is an orthonormal basis of $\mathbb{R}^U$ with $A\mathbf{u}_j = \mu_j \mathbf{u}_j$.
PROVE: $\mathrm{Spec}(A) = [\![\mu_j : j \in I]\!]$.

LET: $P : U \times I \to \mathbb{R}$ has columns $\mathbf{u}_j$, and $D \triangleq \mathrm{diag}(\mu_j)$ on I.

⟨2⟩1. $P^T P = I$ and $|I| = n$.

PROOF: The entries of $P^T P$ are $\langle \mathbf{u}_j, \mathbf{u}_k \rangle = \delta_{jk}$ by ⟨1⟩; a basis of $\mathbb{R}^U$ has n members by `Module.finrank_eq_card_basis` in Facts 3. □

⟨2⟩2. $A = PDP^T$.

PROOF: Both sides act on each $\mathbf{u}_k$ by μ_k, by ⟨1⟩ and ⟨2⟩1; a linear map is determined by its values on a basis. □

⟨2⟩3. $\chi_A = \prod_j (t - \mu_j)$.

PROOF: Apply `Matrix.charpoly_mul_comm'` in Facts 9 to P and DP^T: with ⟨2⟩2 and ⟨2⟩1, $t^n \chi_A = t^n \chi_{DP^T P} = t^n \chi_D$. Cancel t^n in the domain $\mathbb{R}[t]$ and apply `Matrix.charpoly_diagonal` in Facts 9. □

⟨2⟩4. $\mathrm{Spec}(A) = [\![\mu_j : j \in I]\!]$.

PROOF: Apply `Polynomial.roots_multiset_prod_X_sub_C` in Facts 5 to ⟨2⟩3 and Definition 2.1. □

⟨2⟩5. Q.E.D.

PROOF: Use ⟨2⟩4. □

⟨1⟩4. For each $\mu \in \mathbb{R}$, the multiset count equals the Euclidean eigenspace dimension.

⟨2⟩1. The count of μ is the number of sorted indices with eigenvalue μ.

PROOF: Apply `Matrix.IsHermitian.roots_charpoly_eq_eigenvalues₀` in Facts 1 and the definition of a mapped-multiset count. □

⟨2⟩2. This number equals the Euclidean eigenspace dimension.

PROOF: Apply `LinearMap.IsSymmetric.card_filter_eigenvalues_eq` in Facts 2 to A.toEuclideanLin, using `Matrix.isSymmetric_toEuclideanLin_iff` and $\dim \mathbb{R}^U = n$. Its eigenvalues are the sorted list by definition. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □
⟨1⟩5. The two Lean eigenspace coordinate conventions have equal dimensions.

⟨2⟩1. The map $\mathbf{x} \mapsto \texttt{WithLp.toLp}\, 2\, \mathbf{x}$ has inverse taking the underlying function.
PROOF: These are the inverse coordinate maps defining `EuclideanSpace` in Facts 2. □
⟨2⟩2. This equivalence intertwines `A.toLin'` and `A.toEuclideanLin`.
PROOF: Both underlying functions have value $\sum_v A(u,v)\mathbf{x}(v)$ at u, by Facts 2. □
⟨2⟩3. The equivalence restricts to inverse linear maps of their μ-eigenspaces.
PROOF: Use ⟨2⟩2 in the defining eigenvector equations in Facts 2. □
⟨2⟩4. Q.E.D.
PROOF: Linear equivalences preserve dimension (Facts 3); apply ⟨2⟩3. □
⟨1⟩6. The cardinality, three membership criteria and root-multiplicity statement hold, also when U is empty.

⟨2⟩1. $|\operatorname{Spec}(A)| = n$.
PROOF: Count the U-indexed multiset in ⟨1⟩2. □
⟨2⟩2. $\mu \in \operatorname{Spec}(A)$ iff there is a nonzero μ-eigenvector.
PROOF: Combine ⟨1⟩4 and ⟨1⟩5; positive dimension is equivalent to a nonzero vector by Facts 3. □
⟨2⟩3. $\mu \in \operatorname{Spec}(A)$ iff $\mu \in \operatorname{spectrum}_{\mathbb{R}}(A)$.
PROOF: Combine ⟨1⟩2 with `Matrix.IsHermitian.spectrum_real_eq_range_eigenvalues` in Facts 1. □
⟨2⟩4. $\operatorname{mult}_A(\mu) = \chi_A.\text{rootMultiplicity}(\mu)$.
PROOF: Apply `Polynomial.count_roots` in Facts 5 to Definition 2.1. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩7. The eigenvalues $\nu_i = \texttt{hA.eigenvalues}_0(i)$ have a sorted orthonormal eigenbasis $(\mathbf{w}_i)_{i \in \operatorname{Fin}(n)}$, and $\operatorname{Spec}(A) = [\![\nu_i : i \in \operatorname{Fin}(n)]\!]$.

LET: $e \triangleq \texttt{Fintype.equivOfCardEq(Fintype.card_finn)} : \operatorname{Fin}(n) \simeq U$, from Facts 1, and $\mathbf{w}_i \triangleq \texttt{hA.eigenvectorBasis}(e(i))$.
⟨2⟩1. $(\mathbf{w}_i)$ is an orthonormal basis.
PROOF: It is `hA.eigenvectorBasis.reindex` e^{-1}; apply `OrthonormalBasis.reindex` in Facts 2. □
⟨2⟩2. $A\mathbf{w}_i = \nu_i \mathbf{w}_i$.
PROOF: Apply `Matrix.IsHermitian.mulVec_eigenvectorBasis` at $e(i)$ and unfold `Matrix.IsHermitian.eigenvalues`, which is defined with this e; $e^{-1}(e(i)) = i$. □
⟨2⟩3. $\nu_i \geq \nu_j$ whenever $i \leq j$.
PROOF: Apply $\texttt{Matrix.IsHermitian.eigenvalues}_0\texttt{_antitone}$ in Facts 1. □
⟨2⟩4. $\operatorname{Spec}(A) = [\![\nu_i : i \in \operatorname{Fin}(n)]\!]$.
PROOF: Apply $\texttt{Matrix.IsHermitian.roots_charpoly_eq_eigenvalues}_0$ in Facts 1 and Definition 2.1. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩8. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6, ⟨1⟩7. □

Definition 2.3.

ASSUME:

1. U is finite and nonempty, and $A : U \times U \to \mathbb{R}$ is symmetric.

For a real symmetric matrix A on $n > 0$ indices, define

$$\lambda_{\max}(A) = \texttt{hA.eigenvalues}_0(0), \qquad \lambda_{\min}(A) = \texttt{hA.eigenvalues}_0(n-1).$$

The indices carry their proofs of being less than n. These are exactly `LineGraphs/Spectral/Basic.lean`'s definitions. They require `NonemptyU` and `hA:A.IsHermitian`; there is no extreme for the empty matrix.

Lemma 2.4 (Extremal entries and multiset bounds)**.**

ASSUME:

1. U is finite and nonempty, $|U| = n$, and $A : U \times U \to \mathbb{R}$ is symmetric.

PROVE: $\lambda_{\max}(A), \lambda_{\min}(A) \in \mathrm{Spec}(A)$, and

$$\lambda_{\min}(A) \le \mu \le \lambda_{\max}(A) \qquad (\mu \in \mathrm{Spec}(A)).$$

Thus they are the maximum and minimum of the multiset and equal the first and last entries of the decreasing list.

PROOF SKETCH: We will read the two endpoints of the decreasing list $\nu_0 \ge \cdots \ge \nu_{n-1}$ of Theorem 2.2 and compare every entry with them.

⟨1⟩1. $\mathrm{Spec}(A) = [\![\nu_i : i \in \mathrm{Fin}(n)]\!]$.
PROOF: Theorem 2.2. □

⟨1⟩2. $\lambda_{\max}(A) = \nu_0$ and $\lambda_{\min}(A) = \nu_{n-1}$ belong to $\mathrm{Spec}(A)$.
PROOF: The equalities are Definition 2.3; use indices 0 and $n-1$ in ⟨1⟩1, which are valid because $n > 0$. □

⟨1⟩3. Every $\mu \in \mathrm{Spec}(A)$ satisfies $\mu \le \nu_0$.
PROOF: By ⟨1⟩1, $\mu = \nu_i$ for some i; apply `Matrix.IsHermitian.eigenvalues`$_0$`_antitone` in Facts 1 to $0 \le i$. □

⟨1⟩4. Every $\mu \in \mathrm{Spec}(A)$ satisfies $\nu_{n-1} \le \mu$.
PROOF: By ⟨1⟩1, $\mu = \nu_i$ with $i \le n-1$; apply the same declaration. □

⟨1⟩5. Q.E.D.
PROOF: ⟨1⟩2, ⟨1⟩3 and ⟨1⟩4 are the membership and the two bounds; membership with a bound is the defining property of a maximum or minimum. □

Definition 2.5.

ASSUME:

1. X and Y are finite simple graphs; their vertex types may differ.

The spectrum and multiplicity of a graph X are those of its adjacency matrix $A(X)$. Two graphs, possibly on different vertex types, are *cospectral* when their real adjacency characteristic polynomials are equal. The least and largest graph eigenvalues require a nonempty vertex set. The *Seidel matrix* is

$$S(X) = J - I - 2A(X).$$

It is symmetric, has diagonal zero, and off-diagonal entry -1 on an edge and 1 on a nonedge.

Proposition 2.6 (Equivalent spectra)**.**

ASSUME:
1. U, W are finite; $A : U \times U \to \mathbb{R}$ and $B : W \times W \to \mathbb{R}$ are symmetric.
2. For the graph assertions, X and Y are finite simple graphs.

PROVE: For real symmetric matrices A, B, even on different finite index sets, $\chi_A = \chi_B$ iff $\mathrm{Spec}(A) = \mathrm{Spec}(B)$ iff $\mathrm{mult}_A(\mu) = \mathrm{mult}_B(\mu)$ for all real μ. Thus cospectral graphs have the same number of vertices. Graph isomorphisms preserve the spectrum.

PROOF SKETCH: We will reconstruct the monic characteristic polynomial from its roots, then relabel adjacency matrices.
⟨1⟩1. The three equalities are equivalent.

⟨2⟩1. $\chi_A = \chi_B$ implies $\mathrm{Spec}(A) = \mathrm{Spec}(B)$.
PROOF: Apply the roots function in Definition 2.1. □
⟨2⟩2. $\mathrm{Spec}(A) = \mathrm{Spec}(B)$ implies $\chi_A = \chi_B$.
PROOF: Substitute the common multiset in the product formula of Theorem 2.2. □
⟨2⟩3. Multiset equality is equivalent to equality of every count.
PROOF: Apply `Multiset.ext` in Facts 5. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □
⟨1⟩2. Cospectral graphs have equal order; isomorphic graphs are cospectral.

⟨2⟩1. Equal spectra have equal orders.
PROOF: Take cardinalities and use Theorem 2.2. □
⟨2⟩2. An isomorphism reindexes the adjacency matrix.
PROOF: Apply `SimpleGraph.Iso.reindex_adjMatrix` in Facts 10. □
⟨2⟩3. The reindexed characteristic polynomial is unchanged.
PROOF: Apply `Matrix.det_reindex_self` in Facts 4 to $tI - A$ from ⟨2⟩2. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩3. □
⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2. □

3. Polynomial calculus, traces, and positivity

Theorem 3.1 (Polynomial calculus and spectral idempotents)**.**

ASSUME:
1. U is finite, $|U| = n$, and $A : U \times U \to \mathbb{R}$ is symmetric.
2. $(\theta_r)_{r \in R}$ lists the distinct eigenvalues in decreasing order; R is empty if $n = 0$.

PROVE: There are unique symmetric matrices E_r that are the identity on $V_A(\theta_r)$ and zero on every other eigenspace. They satisfy

$$\sum_r E_r = I, \quad E_r E_s = \begin{cases} E_r & r = s, \\ 0 & r \neq s, \end{cases} \qquad A = \sum_r \theta_r E_r.$$

For every real polynomial f,

$$f(A) = \sum_r f(\theta_r) E_r, \quad \operatorname{Spec}(f(A)) = \operatorname{map}(f, \operatorname{Spec}(A)), \quad \operatorname{mult}_{f(A)}(b) = \sum_{r : f(\theta_r) = b} \operatorname{mult}_A(\theta_r).$$

In particular $f(A) = 0$ iff $f(\theta_r) = 0$ for every r, and

$$E_r = \prod_{s \neq r} \frac{A - \theta_s I}{\theta_r - \theta_s}, \quad \operatorname{tr} E_r = \operatorname{rk} E_r = \operatorname{mult}_A(\theta_r).$$

PROOF SKETCH: We will select the coordinates of each eigenspace and evaluate polynomials on the eigenbasis.

⟨1⟩1. PICK an orthonormal eigenbasis $(\mathbf{v}_i)$ of A, with eigenvalues λ_i; let Q have these columns.
PROOF: Theorem 2.2. □

⟨1⟩2. Define $E_r = Q D_r Q^T$, where $D_r(i,i) = 1$ if $\lambda_i = \theta_r$ and 0 otherwise. These are the unique asserted symmetric matrices.

⟨2⟩1. $E_r \mathbf{v}_i = \mathbf{v}_i$ if $\lambda_i = \theta_r$, and 0 otherwise.
PROOF: Orthonormality of the basis of Theorem 2.2 makes $Q^T \mathbf{v}_i$ the ith coordinate unit vector; apply $Q D_r$ to it. □

⟨2⟩2. $E_r^T = E_r$.
PROOF: Transpose $Q D_r Q^T$, using $D_r^T = D_r$. □

⟨2⟩3. Any matrix with the asserted eigenspace actions equals E_r.
PROOF: The actions in ⟨2⟩1 determine every column in a basis. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □

⟨1⟩3. $\sum_r E_r = I$, $E_r E_s = \delta_{rs} E_r$ and $A = \sum_r \theta_r E_r$.

⟨2⟩1. Exactly one E_r fixes each $\mathbf{v}_i$.
PROOF: The list of distinct values contains λ_i exactly once; use ⟨1⟩2. □

⟨2⟩2. $E_r E_s \mathbf{v}_i = \delta_{rs} E_r \mathbf{v}_i$.
PROOF: Apply the two indicator actions in ⟨1⟩2. □

⟨2⟩3. $\sum_r \theta_r E_r \mathbf{v}_i = \lambda_i \mathbf{v}_i = A \mathbf{v}_i$.
PROOF: Use the unique surviving term in ⟨2⟩1 and Theorem 2.2. □

⟨2⟩4. Q.E.D.
PROOF: Equality on a basis gives the three matrix identities from ⟨2⟩1, ⟨2⟩2 and ⟨2⟩3. □

⟨1⟩4. $f(A)\mathbf{v}_i = f(\lambda_i)\mathbf{v}_i$ and $f(A) = \sum_r f(\theta_r) E_r$.

⟨2⟩1. $A^0 \mathbf{v}_i = \lambda_i^0 \mathbf{v}_i$.
PROOF: Both sides equal $\mathbf{v}_i$, by the definition of zeroth power. □

⟨2⟩2. If $A^j \mathbf{v}_i = \lambda_i^j \mathbf{v}_i$, then $A^{j+1} \mathbf{v}_i = \lambda_i^{j+1} \mathbf{v}_i$.
PROOF: Multiply the assumed equation by A and use Theorem 2.2. □

⟨2⟩3. $A^j \mathbf{v}_i = \lambda_i^j \mathbf{v}_i$ for every $j \in \mathbb{N}$.
PROOF: Induction with ⟨2⟩1 and ⟨2⟩2. □

⟨2⟩4. $f(A)\mathbf{v}_i = f(\lambda_i)\mathbf{v}_i$.
PROOF: Take the finite linear combination of ⟨2⟩3 with the coefficients of f, as specified by polynomial evaluation. □

⟨2⟩5. $f(A) = \sum_r f(\theta_r) E_r$.
PROOF: The two matrices have the same action on each $\mathbf{v}_i$ by ⟨2⟩4 and ⟨1⟩2. □

⟨2⟩6. Q.E.D.

PROOF: Use ⟨2⟩4, ⟨2⟩5. □

⟨1⟩5. $\mathrm{Spec}(f(A))$ is the mapped multiset; the displayed multiplicities add over equal images, and $f(A) = 0$ iff all $f(\theta_r) = 0$.

⟨2⟩1. $f(A) = Q\,\mathrm{diag}(f(\lambda_i))Q^T$.
PROOF: Read the basis action in ⟨1⟩4. □
⟨2⟩2. $\mathrm{Spec}(f(A)) = [\![f(\lambda_i) : i \in U]\!]$.
PROOF: $f(A) = \sum_r f(\theta_r)E_r$ is symmetric by ⟨1⟩4 and ⟨1⟩2, and $(\mathbf{v}_i)$ is an orthonormal eigenbasis of it with eigenvalues $f(\lambda_i)$ by ⟨1⟩4; apply the arbitrary-basis assertion of Theorem 2.2. □
⟨2⟩3. $\mathrm{mult}_{f(A)}(b) = \sum_{r:f(\theta_r)=b} \mathrm{mult}_A(\theta_r)$.
PROOF: Partition the indices counted in ⟨2⟩2 by their original value λ_i. □
⟨2⟩4. $f(A) = 0$ iff $f(\theta_r) = 0$ for all r.
PROOF: The diagonal matrix in ⟨2⟩1 is zero exactly when each diagonal entry is zero. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩6. The product formula for E_r and $\mathrm{tr}\,E_r = \mathrm{rk}\,E_r = \mathrm{mult}_A(\theta_r)$ hold.

⟨2⟩1. For $s \neq r$, $\theta_r - \theta_s \neq 0$.
PROOF: This is distinctness in ⟨0⟩:2. □
⟨2⟩2. $\prod_{s\neq r}(\theta_r - \theta_s)/(\theta_r - \theta_s) = 1$.
PROOF: Each factor is 1 by ⟨2⟩1. □
⟨2⟩3. For $t \neq r$, $\prod_{s\neq r}(\theta_t - \theta_s)/(\theta_r - \theta_s) = 0$.
PROOF: The factor $s = t$ is zero. □
⟨2⟩4. The displayed product equals E_r.
PROOF: Its basis action agrees with ⟨1⟩2 by ⟨1⟩4, ⟨2⟩2 and ⟨2⟩3. □
⟨2⟩5. $\mathrm{rk}\,E_r = \#\{i : \lambda_i = \theta_r\}$.
PROOF: Its image has precisely that coordinate basis by ⟨1⟩2. □
⟨2⟩6. $\mathrm{tr}\,E_r = \sum_{i:\lambda_i=\theta_r} \sum_u \mathbf{v}_i(u)^2 = \#\{i : \lambda_i = \theta_r\}$.
PROOF: Expand QD_rQ^T on the diagonal and use unit norms from Theorem 2.2. □
⟨2⟩7. Q.E.D.
PROOF: Use ⟨2⟩4, ⟨2⟩5, ⟨2⟩6. □

⟨1⟩7. Q.E.D.
PROOF: Use ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6. □

Theorem 3.2 (Trace, determinant, rank, and moments)**.**

ASSUME:

1. $A : U \times U \to \mathbb{R}$ is symmetric, U is finite with $|U| = n$, and $(\lambda_i)_{i=1}^n$ lists its eigenvalues with multiplicity.
2. $j \in \mathbb{N}$, $f \in \mathbb{R}[t]$, and (θ_r) lists the distinct eigenvalues of A; in the last assertion $E : W \times W \to \mathbb{R}$ is idempotent and W is finite.

PROVE:

$$\mathrm{tr}\,A^j = \sum_{i=1}^{n} \lambda_i^j, \qquad \det A = \prod_{i=1}^{n} \lambda_i, \qquad \mathrm{rk}\,A = n - \mathrm{mult}_A(0).$$

For any real polynomial f, $\operatorname{tr} f(A) = \sum_i f(\lambda_i)$ and $\operatorname{rk} f(A) = \sum_{r:f(\theta_r)\neq 0} \operatorname{mult}_A(\theta_r)$. If E is any real idempotent square matrix, even if not symmetric, $\operatorname{tr} E = \operatorname{rk} E$ and this number is a nonnegative integer.

PROOF SKETCH: We will compute in diagonal coordinates; for an arbitrary idempotent we will use its image and kernel.
⟨1⟩1. The trace, determinant, rank and polynomial formulas hold.

⟨2⟩1. $\operatorname{tr} A = \sum_i \lambda_i$.
PROOF: Apply `Matrix.IsHermitian.trace_eq_sum_eigenvalues` in Facts 1 to A. □
⟨2⟩2. $\det A = \prod_i \lambda_i$.
PROOF: Apply `Matrix.IsHermitian.det_eq_prod_eigenvalues` in Facts 1. □
⟨2⟩3. $\operatorname{rk} A = n - \operatorname{mult}_A(0)$.
PROOF: Apply `Matrix.IsHermitian.rank_eq_card_non_zero_eigs` in Facts 1; Theorem 2.2 identifies the complementary zero count. □
⟨2⟩4. $f(A)$ is real symmetric and its multiset is $[\![f(\lambda_i)]\!]$.
PROOF: The symmetric spectral idempotents and multiset formula in Theorem 3.1 give both assertions. □
⟨2⟩5. $\operatorname{tr} f(A) = \sum_i f(\lambda_i)$.
PROOF: Apply `Matrix.IsHermitian.trace_eq_sum_eigenvalues` to $f(A)$ using ⟨2⟩4; sum the identified multiset. □
⟨2⟩6. $\operatorname{rk} f(A) = \sum_{r:f(\theta_r)\neq 0} \operatorname{mult}_A(\theta_r)$.
PROOF: Apply `Matrix.IsHermitian.rank_eq_card_non_zero_eigs` to $f(A)$ and group the nonzero entries of ⟨2⟩4 by original eigenvalue. □
⟨2⟩7. $\operatorname{tr} A^j = \sum_i \lambda_i^j$.
PROOF: Apply the trace declaration directly to the symmetric matrix A^j; Theorem 3.1 with t^j identifies its multiset, including $j = 0$. □
⟨2⟩8. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩5, ⟨2⟩6, ⟨2⟩7. □
⟨1⟩2. $\operatorname{tr} E = \operatorname{rk} E \in \mathbb{N}$.

⟨2⟩1. E.toLin′ projects onto its range.
PROOF: Use `LinearMap.isProj_range_iff_isIdempotentElem` in Facts 11 with the hypothesis $E^2 = E$. □
⟨2⟩2. The trace equals the dimension of that range.
PROOF: Apply `LinearMap.IsProj.trace` in Facts 11 to ⟨2⟩1; its range and kernel are finite and free over $\mathbb{R}$. □
⟨2⟩3. This is the matrix trace and matrix rank.
PROOF: Apply `Matrix.trace_toLin'_eq` and `Matrix.rank_eq_finrank_range_toLin` in Facts 11, using the standard coordinate bases for the rank equality; dimension is a natural number. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩3. □
⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2. □

Theorem 3.3 (Positivity and shifts)**.**

ASSUME:

1. U is finite and $A : U \times U \to \mathbb{R}$ is symmetric.
2. $b \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^U$. The principal-submatrix assertions assume $A \succeq 0$; the last assertion also assumes $B : U \times U \to \mathbb{R}$ is positive semidefinite.

PROVE: It is positive semidefinite iff every member of $\mathrm{Spec}(A)$ is nonnegative, and positive definite iff every member is positive. For any $b \in \mathbb{R}$,

$$A - bI \succeq 0 \iff (\forall \mu \in \mathrm{Spec}(A))\ b \le \mu, \qquad bI - A \succeq 0 \iff (\forall \mu \in \mathrm{Spec}(A))\ \mu \le b.$$

If $A \succeq 0$, then $\mathbf{x}^T A\mathbf{x} = 0$ iff $A\mathbf{x} = 0$. If $A \succeq 0$, every principal submatrix is positive semidefinite and has nonnegative determinant. For positive semidefinite real matrices A, B of the same order, $\mathrm{tr}(AB) \ge 0$.

PROOF SKETCH: We will expand the quadratic form in eigenvector coordinates and inspect its nonnegative summands.

⟨1⟩1. The two positivity equivalences hold.

⟨2⟩1. $A \succeq 0$ iff every eigenvalue is nonnegative.
PROOF: Apply `Matrix.IsHermitian.posSemidef_iff_eigenvalues_nonneg` in Facts 6; Theorem 2.2 identifies the multiset. □

⟨2⟩2. A is positive definite iff every eigenvalue is positive.
PROOF: Apply `Matrix.IsHermitian.posDef_iff_eigenvalues_pos` in Facts 6 and the same multiset identification. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □

⟨1⟩2. $A - bI \succeq 0$ iff $b \le \mu$ for every $\mu \in \mathrm{Spec}(A)$, and $bI - A \succeq 0$ iff $\mu \le b$ for every $\mu \in \mathrm{Spec}(A)$.
PROOF: Apply `Matrix.IsHermitian.posSemidef_iff_eigenvalues_nonneg` in Facts 6 separately to the symmetric matrices $A - bI$ and $bI - A$; Theorem 3.1 with $t - b$ and $b - t$ gives their multisets $[\![\mu - b]\!]$ and $[\![b - \mu]\!]$. □

⟨1⟩3. For $A \succeq 0$, $\mathbf{x}^T A\mathbf{x} = 0$ iff $A\mathbf{x} = 0$.
PROOF: Apply `Matrix.PosSemidef.dotProduct_mulVec_zero_iff` in Facts 6; real conjugation is trivial. □

⟨1⟩4. For $A \succeq 0$, every principal submatrix is positive semidefinite with nonnegative determinant.

⟨2⟩1. The principal submatrix is positive semidefinite.
PROOF: Apply `Matrix.PosSemidef.submatrix` in Facts 6 to the inclusion of the principal index set. □

⟨2⟩2. Its determinant is nonnegative.
PROOF: Apply `Matrix.PosSemidef.det_nonneg` in Facts 6 to ⟨2⟩1. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □

⟨1⟩5. For $A, B \succeq 0$, $\mathrm{tr}(AB) \ge 0$.

⟨2⟩1. PICK an orthonormal eigenbasis $(\mathbf{v}_i)$ of A with eigenvalues $\lambda_i \ge 0$.
PROOF: Theorem 2.2 supplies the basis; the spectral criterion in ⟨1⟩1 makes its eigenvalues nonnegative. □

⟨2⟩2. $\mathrm{tr}(AB) = \sum_i \lambda_i \mathbf{v}_i^T B \mathbf{v}_i$.
PROOF: Expand $A = \sum_i \lambda_i \mathbf{v}_i \mathbf{v}_i^T$ and the defining trace sum. □

⟨2⟩3. Every summand is nonnegative.
PROOF: Use ⟨1⟩1 for A and the defining form inequality for B. □

⟨2⟩4. Q.E.D.

PROOF: Use ⟨2⟩2, ⟨2⟩3. □
⟨1⟩6. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5. □

Corollary 3.4 (The least-eigenvalue bridge)**.**

ASSUME:
1. X is a finite simple graph on U, and $A = A(X)$.
2. The assertions involving largest or least eigenvalues assume $U \neq \varnothing$.

PROVE:

$$(2I + A) \succeq 0 \iff (\forall \mu \in \operatorname{Spec}(X)) \ -2 \leq \mu, \qquad -2 \in \operatorname{Spec}(X) \iff \det(2I + A) = 0.$$

If X has a vertex, the first right-hand side is $\lambda_{\min}(X) \geq -2$. Its least eigenvalue equals -2 iff both the positive semidefiniteness and determinant-zero conditions hold. Also $2I - A \succeq 0$ iff $\lambda_{\max}(X) \leq 2$ for a nonempty graph.

PROOF SKETCH: We will shift every eigenvalue by two and then take the minimum or maximum of the finite list.
⟨1⟩1. The two assertions that do not require a vertex hold.

⟨2⟩1. $2I + A \succeq 0$ iff every eigenvalue of A is at least -2.
PROOF: Use adjacency symmetry (Facts 10) and Theorem 3.3 with $b = -2$. □
⟨2⟩2. $\det(2I + A) = \prod_i (2 + \lambda_i)$.
PROOF: Apply Theorems 3.1 and 3.2 to $2 + t$. □
⟨2⟩3. The product is zero iff -2 occurs in $\operatorname{Spec}(A)$.
PROOF: A finite real product vanishes iff one factor vanishes; use ⟨2⟩2. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩3. □
⟨1⟩2. The remaining equivalences hold when $U \neq \varnothing$.

⟨2⟩1. All entries are at least -2 iff the minimum is at least -2.
PROOF: Use the least-element property in Lemma 2.4. □
⟨2⟩2. The minimum is -2 iff all entries are at least -2 and -2 occurs.
PROOF: A finite nonempty list contains its minimum; apply Lemma 2.4. □
⟨2⟩3. $2I - A \succeq 0$ iff the maximum is at most 2.
PROOF: Use the upper shift of Theorem 3.3 and Lemma 2.4. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □
⟨1⟩3. When $U = \varnothing$, both sides of the determinant-zero equivalence are false.
PROOF: The empty determinant is 1 and the empty spectrum has no member, by Definition 2.1. □
⟨1⟩4. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3. □

Theorem 3.5 (Gram matrices with rank control)**.**

ASSUME:
1. U is finite and $G : U \times U \to \mathbb{R}$ is symmetric, with $\operatorname{rk} G = \rho$.

2. B is any real matrix on finite row and column sets. Vector families below are finite and lie in a finite-dimensional real inner product space.
3. For the graph assertions, X is a finite simple graph with n vertices; the lower-shift occurrence assumes a representation in dimension $\ell < n$, while the upper-shift rank bound allows any ℓ.

PROVE: A real symmetric matrix G is positive semidefinite iff it is a Gram matrix. If $G \succeq 0$ and $\operatorname{rk} G = \rho$, it has a Gram representation in $\mathbb{R}^\rho$ whose vectors span that space. For any real matrix B,

$$\ker(B^T B) = \ker B, \qquad \operatorname{rk}(B^T B) = \operatorname{rk} B = \operatorname{rk}(B B^T).$$

Thus the rank of the Gram matrix of a finite vector family equals the dimension of its span. In particular $2I + A(X) \succeq 0$ iff X has a representation by vectors of squared norm 2 with inner product 1 on edges and 0 on distinct nonedges. If such vectors lie in dimension $\ell < n$, then -2 occurs, with multiplicity at least $n - \ell$. For a Gram representation of $2I - A(X)$ in dimension ℓ, every eigenvalue is at most 2 and $\operatorname{mult}_X(2) \geq n - \ell$. If $n > \ell$, the largest eigenvalue is 2.

PROOF SKETCH: We will take square roots of positive eigendirections, then identify the kernel and image of the coordinate matrix.

⟨1⟩1. Positive semidefiniteness is equivalent to some Gram representation.
PROOF: Apply `Matrix.posSemidef_iff_eq_sum_vecMulVec` in Facts 6; the vectors in its sum are the rows of a factor matrix, whose columns have the asserted Gram matrix. □

⟨1⟩2. If $G \succeq 0$, it equals $B^T B$ for a ρ-row matrix whose columns span $\mathbb{R}^\rho$.

⟨2⟩1. PICK an orthonormal eigenbasis $(\mathbf{v}_i)$ for G, with eigenvalues λ_i.
PROOF: Theorem 2.2. □

⟨2⟩2. Exactly ρ eigenvalues of G are positive.
PROOF: Use Theorems 3.3 and 3.2. □

⟨2⟩3. For those indices set $B(i,u) = \sqrt{\lambda_i}\mathbf{v}_i(u)$; then $B^T B = G$.
PROOF: Expand the product; $(\sqrt{\lambda_i})^2 = \lambda_i$ by Facts 8, and the omitted eigenvalues are zero. □

⟨2⟩4. The rows of B are independent.
PROOF: Divide a zero linear combination by the nonzero square-root coefficients and use independence of $(\mathbf{v}_i)$. □

⟨2⟩5. The columns span $\mathbb{R}^\rho$.
PROOF: Row rank equals column rank by Facts 7; use ⟨2⟩4. □

⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩3, ⟨2⟩5. □

⟨1⟩3. Every $B^T B$ is positive semidefinite and $\ker(B^T B) = \ker B$.

⟨2⟩1. $\mathbf{x}^T B^T B \mathbf{x} = \|B\mathbf{x}\|^2$.
PROOF: Expand matrix multiplication and the Euclidean inner product. □

⟨2⟩2. $B^T B \succeq 0$.
PROOF: It is symmetric and the form in ⟨2⟩1 is nonnegative; apply Facts 6. □

⟨2⟩3. $B^T B\mathbf{x} = 0$ implies $B\mathbf{x} = 0$.
PROOF: In ⟨2⟩1 the left side vanishes, so a sum of coordinate squares is zero (Facts 8). □

⟨2⟩4. $B\mathbf{x} = 0$ implies $B^T B\mathbf{x} = 0$.
PROOF: Apply B^T to the assumed zero vector. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩4. $\operatorname{rk}(B^T B) = \operatorname{rk} B = \operatorname{rk}(B B^T)$; a Gram rank is the dimension of the vector span.

⟨2⟩1. $\operatorname{rk}(B^T B) = \operatorname{rk} B$.
PROOF: Apply `Matrix.rank_transpose_mul_self` in Facts 7 to the real rectangular matrix B. □
⟨2⟩2. $\operatorname{rk}(BB^T) = \operatorname{rk} B$.
PROOF: Apply `Matrix.rank_self_mul_transpose` in Facts 7 to B. □
⟨2⟩3. The coordinate matrix of a family has image equal to its span.
PROOF: The matrix image consists exactly of linear combinations of its columns. □
⟨2⟩4. Its Gram rank equals the dimension of that span.
PROOF: Apply `Matrix.rank_transpose_mul_self` to the coordinate matrix of the family in an orthonormal basis (Facts 2); its image is the vector span. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩4. □
⟨1⟩5. The graph representation criterion, the multiplicity bounds for both $2I + A$ and $2I - A$, and the stated occurrences hold.

⟨2⟩1. $2I + A(X)$ has diagonal 2, edge entries 1 and other off-diagonal entries 0.
PROOF: Use the definition of adjacency matrix in Facts 10. □
⟨2⟩2. Its positivity is equivalent to the specified Gram representation.
PROOF: Apply the equivalence in `Matrix.posSemidef_iff_eq_sum_vecMulVec` (Facts 6) directly to $2I + A$; its entries in ⟨2⟩1 are exactly the required inner products. □
⟨2⟩3. $\operatorname{rk}(2I + A) \le \ell$.
PROOF: Choose orthonormal coordinates for the representing family. Its coordinate matrix has at most ℓ rows; `Matrix.rank_transpose_mul_self` in Facts 7 bounds the Gram rank by ℓ. □
⟨2⟩4. $\operatorname{mult}_X(-2) = n - \operatorname{rk}(2I + A) \ge n - \ell$.
PROOF: Apply Theorems 3.1 and 3.2 to the shift, then ⟨2⟩3. □
⟨2⟩5. For a representation of $2I - A$ in dimension ℓ, $\operatorname{mult}_X(2) \ge n - \ell$.
PROOF: The coordinate Gram matrix has rank at most ℓ by `Matrix.rank_transpose_mul_self` (Facts 7); Theorems 3.1 and 3.2 identify its nullity with $\operatorname{mult}_X(2)$. □
⟨2⟩6. Every eigenvalue is at most 2, with maximum 2 if $n > \ell$.
PROOF: The Gram form is positive semidefinite, so Theorem 3.3 gives the bound. The positive multiplicity in ⟨2⟩5 and Lemma 2.4 give equality. □
⟨2⟩7. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩4, ⟨2⟩5, ⟨2⟩6. □
⟨1⟩6. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5. □

Lemma 3.6 (Trace and rank inequality)**.**

ASSUME:
1. G is a real positive semidefinite matrix on a finite index set, of rank ρ.

PROVE: $(\operatorname{tr} G)^2 \le \rho \operatorname{tr}(G^2)$, with equality iff all nonzero eigenvalues are equal; equality includes $G = 0$.

PROOF SKETCH: We will write ρ times the sum of squares of the nonzero eigenvalues, minus the square of their sum, as a sum of squared differences.

LET: $\eta_1, \dots, \eta_\rho$ are the nonzero members of $\mathrm{Spec}(G)$, with repetition; there are ρ of them by Theorem 3.2.

⟨1⟩1. $\rho \sum_i \eta_i^2 - (\sum_i \eta_i)^2 = \sum_{i<j} (\eta_i - \eta_j)^2$.
PROOF: Expand the square: each η_i^2 occurs $\rho - 1$ times on the right and each pair contributes $-2\eta_i\eta_j$. □

⟨1⟩2. The right side is nonnegative and vanishes iff all η_i are equal.
PROOF: Apply the nonnegative zero-sum criterion in Facts 8 to ⟨1⟩1. □

⟨1⟩3. $\sum_i \eta_i^2 = \mathrm{tr}(G^2)$ and $\sum_i \eta_i = \mathrm{tr}\, G$; when $\rho = 0$, $G = 0$.
PROOF: Theorem 3.2, since zero eigenvalues add nothing; for $\rho = 0$ every eigenvalue is zero, and Theorem 2.2 gives $G = Q\, 0\, Q^T = 0$. □

⟨1⟩4. Q.E.D.
PROOF: Substitute ⟨1⟩3 in ⟨1⟩1 and apply ⟨1⟩2; for $\rho = 0$ both sides are zero. □

Theorem 3.7 (Trace bounds and tight frames)**.**

ASSUME:
1. $G : U \times U \to \mathbb{R}$ is positive semidefinite on a finite set, with rank ρ.
2. For the relative bound, n unit vectors lie in $\mathbb{R}^\ell$, their distinct-pair inner products have absolute value α, $0 \le \alpha \le 1$, and $\ell\alpha^2 < 1$.
3. For the frame assertion, B is real $\ell \times n$, $\ell > 0$, $b > 0$, and $BB^T = bI$.

PROVE: $(\mathrm{tr}\, G)^2 \le \rho\, \mathrm{tr}(G^2)$; equality holds iff all its nonzero eigenvalues are equal, with equality also allowed when $G = 0$. If n unit vectors in $\mathbb{R}^\ell$ have pairwise inner products of absolute value α, where $0 \le \alpha \le 1$ and $\ell\alpha^2 < 1$, then

$$n \le \frac{\ell(1-\alpha^2)}{1-\ell\alpha^2}.$$

If B is a real $\ell \times n$ matrix, $\ell > 0$, and $BB^T = bI$ with $b > 0$, then $G = B^T B$ has spectrum $b^\ell, 0^{n-\ell}$. In particular, with $C = G - 2I$,

$$(C + 2I)(C - (b-2)I) = 0.$$

PROOF SKETCH: We will expand a sum of squared differences, apply it to a Gram matrix, and compute the frame polynomial.

⟨1⟩1. $(\mathrm{tr}\, G)^2 \le \rho\, \mathrm{tr}(G^2)$ with the stated equality criterion.
PROOF: Apply Lemma 3.6 to G. □

⟨1⟩2. $n \le \ell(1-\alpha^2)/(1-\ell\alpha^2)$.

⟨2⟩1. The unit-vector Gram matrix has $\mathrm{tr}\, G = n$ and $\mathrm{tr}(G^2) = n + \alpha^2 n(n-1)$.
PROOF: Count its n diagonal squares 1 and $n(n-1)$ off-diagonal squares α^2 in the defining trace sum. □

⟨2⟩2. $\rho \le \ell$.
PROOF: Apply Theorem 3.5 to the given family. □

⟨2⟩3.
CASE: $n > 0$.

⟨3⟩1. $n(1-\ell\alpha^2) \le \ell(1-\alpha^2)$.
PROOF: Apply Lemma 3.6 to the Gram matrix of the given unit vectors. Substitute ⟨2⟩1 and ⟨2⟩2; divide by $n > 0$ and collect terms. □

⟨3⟩2. Q.E.D.

PROOF: Divide ⟨3⟩1 by the positive number $1-\ell\alpha^2$. □

⟨2⟩4.

CASE: $n=0$.

PROOF: The numerator $\ell(1-\alpha^2)$ is nonnegative and the denominator is positive. □

⟨2⟩5. Q.E.D.

PROOF: The cases $n>0$ and $n=0$ exhaust $n\in\mathbb{N}$. □

⟨1⟩3. The frame spectrum and polynomial identity hold.

⟨2⟩1. $\operatorname{rk} B=\ell$ and $\operatorname{rk}(B^TB)=\ell\le n$.

PROOF: Apply Theorem 3.5 to $BB^T=bI$ with $b\neq 0$. □

⟨2⟩2. For $G=B^TB$, $G^2=B^T(BB^T)B=bG$.

PROOF: Associativity and the frame hypothesis. □

⟨2⟩3. Every eigenvalue of G is 0 or b.

PROOF: Apply the zero-polynomial criterion of Theorem 3.1 to ⟨2⟩2. □

⟨2⟩4. There are ℓ copies of b and $n-\ell$ zeros.

PROOF: Count the nonzero eigenvalues using ⟨2⟩1, ⟨2⟩3 and Theorem 3.2. □

⟨2⟩5. $(C+2I)(C-(b-2)I)=0$ for $C=G-2I$.

PROOF: Substitute $G=C+2I$ in $G(G-bI)=0$ from ⟨2⟩2. □

⟨2⟩6. Q.E.D.

PROOF: Use ⟨2⟩4, ⟨2⟩5. □

⟨1⟩4. Q.E.D.

PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3. □

4. Rayleigh quotients and symmetric Perron–Frobenius

Theorem 4.1 (Rayleigh extrema and equality)**.**

ASSUME:

1. U is finite and nonempty, $|U|=n$, $A:U\times U\to\mathbb{R}$ is symmetric, and $\mathbf{x}\in\mathbb{R}^U\setminus\{0\}$.
2. For the coordinate formulas, $(\mathbf{v}_i)_{i\in U}$ is any orthonormal eigenbasis of A, $A\mathbf{v}_i=\lambda_i\mathbf{v}_i$, and $\mathbf{x}=\sum_i b_i\mathbf{v}_i$.
3. For the orthogonality assertion, $\mathbf{y},\mathbf{z}\in\mathbb{R}^U$ satisfy $A\mathbf{y}=s\mathbf{y}$, $A\mathbf{z}=t\mathbf{z}$, where $s,t\in\mathbb{R}$ and $s\neq t$.

PROVE: For $\mathbf{x}\neq 0$, put $R_A(\mathbf{x})=\mathbf{x}^TA\mathbf{x}/\|\mathbf{x}\|^2$. Then

$$\lambda_{\min}(A)\le R_A(\mathbf{x})\le\lambda_{\max}(A).$$

In any orthonormal eigenbasis, writing $\mathbf{x}=\sum_i b_i\mathbf{v}_i$ gives

$$R_A(\mathbf{x})=\frac{\sum_i\lambda_ib_i^2}{\sum_i b_i^2},\qquad \sum_i b_i^2=\|\mathbf{x}\|^2>0,$$

$$\lambda_{\max}\|\mathbf{x}\|^2-\mathbf{x}^TA\mathbf{x}=\sum_i(\lambda_{\max}-\lambda_i)b_i^2,$$

$$\mathbf{x}^TA\mathbf{x}-\lambda_{\min}\|\mathbf{x}\|^2=\sum_i(\lambda_i-\lambda_{\min})b_i^2.$$

Both bounds are attained, including on the unit sphere. Equality on the right holds iff $A\mathbf{x}=\lambda_{\max}(A)\mathbf{x}$, and equality on the left holds iff $A\mathbf{x}=\lambda_{\min}(A)\mathbf{x}$. Thus the extremes

are the supremum and infimum of the Rayleigh quotient on nonzero vectors. Eigenvectors of a symmetric matrix for distinct eigenvalues are orthogonal.

PROOF SKETCH: We will express the two gaps as sums of nonnegative terms and identify where equality can occur. Such a basis exists by Theorem 2.2, and the bounds for $\mathbf{x}$ do not depend on the choice.

⟨1⟩1. $R_A(\mathbf{x}) = \sum_i \lambda_i b_i^2 / \sum_i b_i^2$, with positive denominator $\sum_i b_i^2 = \|\mathbf{x}\|^2$.

⟨2⟩1. $\mathbf{x}^T A\mathbf{x} = \sum_i \lambda_i b_i^2$.
PROOF: Expand $A\mathbf{x} = \sum_i b_i \lambda_i \mathbf{v}_i$ by ⟨0⟩:2, and use orthonormality. □
⟨2⟩2. $\|\mathbf{x}\|^2 = \sum_i b_i^2 > 0$.
PROOF: Orthonormality gives the sum; it is positive since $\mathbf{x} \neq 0$ by ⟨0⟩:1. □
⟨2⟩3. Q.E.D.
PROOF: Divide ⟨2⟩1 by ⟨2⟩2. □

⟨1⟩2. Every λ_i belongs to $\mathrm{Spec}(A)$, so $\lambda_{\min} \leq \lambda_i \leq \lambda_{\max}$.
PROOF: $\mathbf{v}_i$ is a nonzero λ_i-eigenvector; apply the membership criterion of Theorem 2.2 and the bounds of Lemma 2.4. □

⟨1⟩3. The upper gap is $\sum_i (\lambda_{\max} - \lambda_i) b_i^2 \geq 0$; thus $R_A(\mathbf{x}) \leq \lambda_{\max}$, with equality iff $A\mathbf{x} = \lambda_{\max}\mathbf{x}$.

⟨2⟩1. $\lambda_{\max} \|\mathbf{x}\|^2 - \mathbf{x}^T A\mathbf{x} = \sum_i (\lambda_{\max} - \lambda_i) b_i^2$.
PROOF: Subtract the two expressions in ⟨1⟩1. □
⟨2⟩2. Every summand is nonnegative.
PROOF: Use ⟨1⟩2. □
⟨2⟩3. The sum is zero iff $b_i = 0$ whenever $\lambda_i < \lambda_{\max}$.
PROOF: Apply Facts 8 to ⟨2⟩2 and use that a positive factor times a square vanishes iff the coordinate does. □
⟨2⟩4. That coefficient condition is the equation $A\mathbf{x} = \lambda_{\max}\mathbf{x}$.
PROOF: Compare the coefficients of $A\mathbf{x} = \sum_i \lambda_i b_i \mathbf{v}_i$ and $\lambda_{\max}\mathbf{x}$ in the basis of ⟨0⟩:2. □
⟨2⟩5. Q.E.D.
PROOF: Divide the nonnegative gap by the positive denominator in ⟨1⟩1; the zero criterion is ⟨2⟩3, ⟨2⟩4. □

⟨1⟩4. The lower gap is $\sum_i (\lambda_i - \lambda_{\min}) b_i^2 \geq 0$; thus $\lambda_{\min} \leq R_A(\mathbf{x})$, with equality iff $A\mathbf{x} = \lambda_{\min}\mathbf{x}$.

⟨2⟩1. $\mathbf{x}^T A\mathbf{x} - \lambda_{\min} \|\mathbf{x}\|^2 = \sum_i (\lambda_i - \lambda_{\min}) b_i^2$.
PROOF: Subtract using ⟨1⟩1. □
⟨2⟩2. Every summand is nonnegative.
PROOF: Use ⟨1⟩2. □
⟨2⟩3. The sum is zero iff $b_i = 0$ whenever $\lambda_i > \lambda_{\min}$.
PROOF: Use the zero-sum criterion in Facts 8 and ⟨2⟩2. □
⟨2⟩4. That condition is $A\mathbf{x} = \lambda_{\min}\mathbf{x}$.
PROOF: Compare coefficients in the basis of ⟨0⟩:2. □
⟨2⟩5. Q.E.D.
PROOF: Divide the gap by ⟨1⟩1; use ⟨2⟩3 and ⟨2⟩4 for equality. □

⟨1⟩5. Both bounds are attained on unit vectors and are the supremum and infimum on nonzero vectors.

⟨2⟩1. Unit eigenvectors attain each extreme.
PROOF: The sorted basis vectors $\mathbf{w}_0$ and $\mathbf{w}_{n-1}$ of Theorem 2.2 have eigenvalues $\nu_0 = \lambda_{\max}$ and $\nu_{n-1} = \lambda_{\min}$ by Definition 2.3; their quotients are these eigenvalues. □

⟨2⟩2. For every nonzero $\mathbf{w}$, $\lambda_{\min} \le R_A(\mathbf{w}) \le \lambda_{\max}$.
PROOF: Expand $\mathbf{w} = \sum_i c_i \mathbf{v}_i$ in the basis of ⟨0⟩:2. Orthonormality and the eigenvector equations make its quotient $\sum_i \lambda_i c_i^2 / \sum_i c_i^2$, with positive denominator since $\mathbf{w} \neq 0$, and ⟨1⟩2 bounds this weighted average. □
⟨2⟩3. The nonzero-vector supremum is an eigenvalue of A.toEuclideanLin.
PROOF: Apply `LinearMap.IsSymmetric.hasEigenvalue_iSup_of_finiteDimensional` in Facts 12, using matrix symmetry from Facts 2; $n > 0$ makes the space nontrivial. □
⟨2⟩4. The nonzero-vector infimum is also an eigenvalue.
PROOF: Apply `LinearMap.IsSymmetric.hasEigenvalue_iInf_of_finiteDimensional` in Facts 12 to the same symmetric map. □
⟨2⟩5. These two eigenvalues are $\lambda_{\max}$ and $\lambda_{\min}$.
PROOF: The Euclidean inner product in Facts 2 is the coordinate sum, so the quotient in ⟨2⟩3 and ⟨2⟩4 is R_A. The bounds in ⟨2⟩2 are attained by ⟨2⟩1, so the supremum is $\lambda_{\max}$ and the infimum is $\lambda_{\min}$. □
⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩5. □
⟨1⟩6. Eigenvectors at distinct eigenvalues are orthogonal.

⟨2⟩1. $s\langle \mathbf{y}, \mathbf{z}\rangle = t\langle \mathbf{y}, \mathbf{z}\rangle$.

⟨3⟩1. $s\langle \mathbf{y}, \mathbf{z}\rangle = \langle A\mathbf{y}, \mathbf{z}\rangle$.
PROOF: Substitute $A\mathbf{y} = s\mathbf{y}$ from ⟨0⟩:3 in the first inner-product argument. □
⟨3⟩2. $\langle A\mathbf{y}, \mathbf{z}\rangle = \langle \mathbf{y}, A\mathbf{z}\rangle$.
PROOF: Expand both coordinate sums and use $A(u, v) = A(v, u)$. □
⟨3⟩3. $\langle \mathbf{y}, A\mathbf{z}\rangle = t\langle \mathbf{y}, \mathbf{z}\rangle$.
PROOF: Substitute $A\mathbf{z} = t\mathbf{z}$ in the second argument. □
⟨3⟩4. Q.E.D.
PROOF: Transitivity of the three equalities. □
⟨2⟩2. $\langle \mathbf{y}, \mathbf{z}\rangle = 0$.
PROOF: Cancel $s - t \neq 0$ in ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩2. □
⟨1⟩7. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5, ⟨1⟩6. □

Lemma 4.2 (Rayleigh subspace witnesses)**.**

ASSUME:
1. U is finite, $|U| = n > 0$, and $A : U \times U \to \mathbb{R}$ is symmetric.
2. $(\mathbf{w}_j)_{j \in \mathrm{Fin}(n)}$ and $\nu_0 \ge \cdots \ge \nu_{n-1}$ are the sorted eigenbasis and eigenvalues of Theorem 2.2, and $0 \le i < n$.
3. The head is $\mathrm{span}\{\mathbf{w}_0, \ldots, \mathbf{w}_i\}$ and the tail is $\mathrm{span}\{\mathbf{w}_i, \ldots, \mathbf{w}_{n-1}\}$.

PROVE: For $0 \le i < n$, the head span of directions $0, \ldots, i$ has dimension $i + 1$ and quotient at least ν_i on nonzero vectors. The tail span of directions $i, \ldots, n - 1$ has dimension $n - i$ and quotient at most ν_i. Every subspace of dimension $i + 1$ contains a nonzero vector with quotient at most ν_i; every subspace of dimension $n - i$ contains one with quotient at least ν_i.

PROOF SKETCH: We will bound the quotient on the head and on the tail by the weighted-average formula of Theorem 4.1, and we will find both witnesses in intersections, by the dimension formula for a sum of subspaces.

⟨1⟩1.

ASSUME: P, Q are subspaces of $\mathbb{R}^U$ with $\dim P + \dim Q > n$.
PROVE: There is $\mathbf{x} \neq 0$ in $P \cap Q$.

⟨2⟩1. $\dim(P+Q) + \dim(P \cap Q) = \dim P + \dim Q$.
PROOF: Apply `Submodule.finrank_sup_add_finrank_inf_eq` in Facts 3. □

⟨2⟩2. $\dim(P \cap Q) > 0$.
PROOF: Use $\dim(P+Q) \leq n$ (Facts 3) and ⟨1⟩ in ⟨2⟩1. □

⟨2⟩3. There is $\mathbf{x} \neq 0$ in $P \cap Q$.
PROOF: A positive-dimensional subspace has a nonempty basis, by Facts 3. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩3. □

⟨1⟩2. The tail has dimension $n - i$, and $R_A(\mathbf{x}) \leq \nu_i$ for every nonzero $\mathbf{x}$ in it.
PROOF: The $\mathbf{w}_j$ are independent, by Theorem 2.2. In the coordinate formula of Theorem 4.1 for the basis $(\mathbf{w}_j)$, only indices $j \geq i$ have nonzero coefficients, and $\nu_j \leq \nu_i$ for them. □

⟨1⟩3. The head has dimension $i+1$, and $R_A(\mathbf{x}) \geq \nu_i$ for every nonzero $\mathbf{x}$ in it.
PROOF: In the same coordinate formula only indices $j \leq i$ have nonzero coefficients, and $\nu_j \geq \nu_i$ for them; the $\mathbf{w}_j$ are independent. □

⟨1⟩4. Every subspace of dimension $i+1$ contains a nonzero $\mathbf{x}$ with $R_A(\mathbf{x}) \leq \nu_i$.
PROOF: Apply ⟨1⟩1 to that subspace and the tail, since $(i+1) + (n-i) = n+1$; use ⟨1⟩2. □

⟨1⟩5. Every subspace of dimension $n - i$ contains a nonzero $\mathbf{x}$ with $R_A(\mathbf{x}) \geq \nu_i$.
PROOF: Apply ⟨1⟩1 to that subspace and the head, since $(n-i) + (i+1) = n+1$; use ⟨1⟩3. □

⟨1⟩6. Q.E.D.
PROOF: Use ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5. □

Theorem 4.3 (Cauchy interlacing)**.**

ASSUME:

1. U, W are finite with $|U| = n$, $|W| = m$ and $1 \leq m \leq n$.
2. $A : U \times U \to \mathbb{R}$ is symmetric, $\phi : W \hookrightarrow U$ is injective, and $B(u,v) = A(\phi(u), \phi(v))$.
3. $\lambda_i = \nu_i(A)$ and $\beta_i = \nu_i(B)$ are the decreasing eigenvalue lists of the sorted eigenbases of A and B in Theorem 2.2; by that theorem they list the two spectra with multiplicities.

PROVE:

$$\lambda_i \geq \beta_i \geq \lambda_{i+n-m} \qquad (0 \leq i < m).$$

In particular the largest eigenvalue cannot increase, and the least eigenvalue cannot decrease, on taking a nonempty principal submatrix.

PROOF SKETCH: We will use the two subspace comparisons in the ordinary proof. Their nonzero intersection witnesses give the inequalities directly, without choosing extrema over a Grassmannian.

⟨1⟩1. B is symmetric.
PROOF: For every u, v, $B(u,v) = A(\phi(u), \phi(v)) = A(\phi(v), \phi(u)) = B(v,u)$ by ⟨0⟩:2. □

⟨1⟩2. Extension by zero $\iota : \mathbb{R}^W \to \mathbb{R}^U$ preserves dimension, norm and the quadratic form.

⟨2⟩1. ι is linear and injective.
PROOF: Restricting back to $\phi(W)$ is a left inverse, by the definition of extension by zero. □

⟨2⟩2. $\|\iota\mathbf{x}\|^2 = \|\mathbf{x}\|^2$.
PROOF: Only the coordinates $\phi(w)$ contribute to the sum of squares. □
⟨2⟩3. $(\iota\mathbf{x})^T A(\iota\mathbf{x}) = \mathbf{x}^T B\mathbf{x}$.
PROOF: Only pairs $(\phi(u), \phi(v))$ contribute to the double sum; use ⟨0⟩:2. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1 and dimension invariance (Facts 3); the two sum identities give the remaining assertions. □
⟨1⟩3. For $0 \le i < m$, $\beta_i \le \lambda_i$.

⟨2⟩1. The extended span of the first $i+1$ eigendirections of B has dimension $i+1$ and quotient at least β_i.
PROOF: Apply the head assertion of Lemma 4.2 to B, which is symmetric by ⟨1⟩1, with its sorted basis from Theorem 2.2 and $\beta_i = \nu_i(B)$; then transfer by ⟨1⟩2. □
⟨2⟩2. Some nonzero vector there has quotient at most λ_i.
PROOF: Apply the first witness assertion of Lemma 4.2 to A and the subspace in ⟨2⟩1; its bound is $\nu_i(A) = \lambda_i$ by ⟨0⟩:3. □
⟨2⟩3. Q.E.D.
PROOF: The witness in ⟨2⟩2 has quotient between β_i and λ_i by ⟨2⟩1. □
⟨1⟩4. For $0 \le i < m$, $\lambda_{i+n-m} \le \beta_i$.

⟨2⟩1. The extended span of B-directions $i, \dots, m-1$ has dimension $m-i$ and quotient at most β_i.
PROOF: Apply the tail assertion of Lemma 4.2 to B, symmetric by ⟨1⟩1, with $\beta_i = \nu_i(B)$, and transfer by ⟨1⟩2. □
⟨2⟩2. Some nonzero vector there has quotient at least λ_{i+n-m}.
PROOF: Apply the second witness assertion of Lemma 4.2 to A at $i+n-m$, since $m-i = n-(i+n-m)$; its bound is λ_{i+n-m} by ⟨0⟩:3. □
⟨2⟩3. Q.E.D.
PROOF: The witness in ⟨2⟩2 has quotient at most β_i by ⟨2⟩1. □
⟨1⟩5. The largest cannot increase and the least cannot decrease.
PROOF: Take $i = 0$ in ⟨1⟩3 and $i = m-1$ in ⟨1⟩4; Lemma 2.4 identifies the endpoints. □
⟨1⟩6. Q.E.D.
PROOF: Use ⟨1⟩3, ⟨1⟩4, ⟨1⟩5. □

Definition 4.4.

ASSUME:
1. U is finite and $A : U \times U \to \mathbb{R}$ is symmetric and entrywise nonnegative.

For a symmetric entrywise nonnegative matrix A, its *positive support graph* has the same indices as vertices and an edge uv when $u \ne v$ and $A(u,v) > 0$. Connectedness means that the vertex set is nonempty and any two vertices are joined by a path. In particular a single vertex with no edges is connected.

Theorem 4.5 (Symmetric Perron–Frobenius)**.**

ASSUME:
1. U is finite and nonempty; $A : U \times U \to \mathbb{R}$ is symmetric and entrywise nonnegative.

2. $\rho = \lambda_{\max}(A)$. The strict-positivity and simplicity assertions also assume that the positive support graph of A is connected.

PROVE: $\rho \geq 0$, there is a nonzero entrywise nonnegative vector $\mathbf{x}$ with $A\mathbf{x} = \rho\mathbf{x}$, and $|\mu| \leq \rho$ for every eigenvalue μ. If the positive support graph is connected, this vector is strictly positive in every coordinate, $\text{mult}_A(\rho) = 1$, and every nonzero nonnegative ρ-eigenvector is strictly positive.

PROOF SKETCH: We will replace a top eigenvector by its absolute values, propagate zero coordinates along paths, and subtract a minimum ratio.

⟨1⟩1. $\rho \geq 0$ and a nonnegative unit ρ-eigenvector exists.

⟨2⟩1. $0 \leq A(u,u) \leq \rho$ for any $u \in U$.
PROOF: Test Theorem 4.1 on the coordinate unit vector; entrywise nonnegativity gives the first inequality. □

⟨2⟩2. PICK a unit ρ-eigenvector $\mathbf{y}$, and let $\mathbf{x}(u) = |\mathbf{y}(u)|$; then $\|\mathbf{x}\| = 1$.
PROOF: $\rho \in \text{Spec}(A)$ by Lemma 2.4, and Theorem 2.2 supplies a nonzero eigenvector, which we normalize; absolute value preserves each coordinate square. □

⟨2⟩3. $\rho = \mathbf{y}^T A\mathbf{y} \leq \mathbf{x}^T A\mathbf{x}$.
PROOF: For each u, v, multiply $\mathbf{y}(u)\mathbf{y}(v) \leq |\mathbf{y}(u)||\mathbf{y}(v)|$ by $A(u,v) \geq 0$ and sum. □

⟨2⟩4. $A\mathbf{x} = \rho\mathbf{x}$.
PROOF: By Theorem 4.1, $\mathbf{x}^T A\mathbf{x} \leq \rho$; with ⟨2⟩3 this is equality, and the equality criterion of Theorem 4.1 applies. □

⟨2⟩5. Q.E.D.
PROOF: ⟨2⟩1 gives $\rho \geq 0$; ⟨2⟩2 and ⟨2⟩4 give the nonnegative unit ρ-eigenvector $\mathbf{x}$. □

⟨1⟩2. Every eigenvalue μ satisfies $|\mu| \leq \rho$.

⟨2⟩1. For a nonzero μ-eigenvector $\mathbf{w}$, $|\mu| \|\mathbf{w}\|^2 = |\mathbf{w}^T A\mathbf{w}|$.
PROOF: Take absolute values of its quadratic eigenvector equation. □

⟨2⟩2. $|\mathbf{w}^T A\mathbf{w}| \leq |\mathbf{w}|^T A|\mathbf{w}| \leq \rho \|\mathbf{w}\|^2$.
PROOF: The finite-sum triangle inequality gives the first bound; Theorem 4.1 gives the second, since absolute values preserve the norm. □

⟨2⟩3. $|\mu| \leq \rho$.
PROOF: Combine ⟨2⟩1, ⟨2⟩2 and divide by $\|\mathbf{w}\|^2 > 0$. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩3. □

⟨1⟩3.
ASSUME: The positive support graph of A is connected; $\mathbf{z}$ is entrywise nonnegative, $A\mathbf{z} = b\mathbf{z}$ for some $b \in \mathbb{R}$, and $\mathbf{z}(u_0) = 0$.
PROVE: $\mathbf{z} = 0$.

⟨2⟩1. If $\mathbf{z}(u) = 0$, then $\sum_v A(u,v)\mathbf{z}(v) = 0$.
PROOF: Evaluate the eigenvector equation of ⟨1⟩ at u. □

⟨2⟩2. If $\mathbf{z}(u) = 0$, then $\mathbf{z}(v) = 0$ for every support neighbour v of u.
PROOF: Each term in ⟨2⟩1 is nonnegative; its vth term is zero by Facts 8, and $A(u,v) > 0$ by Definition 4.4. □

⟨2⟩3. $\mathbf{z}$ vanishes at every vertex of a support path starting at u_0.

⟨3⟩1. It holds for a path of length zero.
PROOF: Its only vertex is u_0, and $\mathbf{z}(u_0) = 0$ by ⟨1⟩. □

⟨3⟩2. If it holds for a path, it holds for its extension by one edge.
PROOF: Apply ⟨2⟩2 to the last vertex of the path and the new edge. □
⟨3⟩3. Q.E.D.
PROOF: Induction on path length using ⟨3⟩1 and ⟨3⟩2. □
⟨2⟩4. Every coordinate of $\mathbf{z}$ is zero.
PROOF: Connectedness in ⟨1⟩ supplies a support path from u_0 to each vertex; apply ⟨2⟩3. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩4. □
⟨1⟩4.
ASSUME: The positive support graph of A is connected.
PROVE: Every nonzero nonnegative ρ-eigenvector is positive, and $\mathrm{mult}_A(\rho) = 1$.

⟨2⟩1. Every coordinate of a nonzero nonnegative eigenvector is positive.
PROOF: A zero coordinate would make the vector zero, by ⟨1⟩3 and ⟨1⟩. □
⟨2⟩2. For a positive ρ-eigenvector $\mathbf{x}$ and any ρ-eigenvector $\mathbf{y}$, PICK u_0 with $\mathbf{y}(u_0)/\mathbf{x}(u_0) = b \triangleq \min_u \mathbf{y}(u)/\mathbf{x}(u)$.
PROOF: A positive ρ-eigenvector exists by ⟨1⟩1 and ⟨2⟩1; a finite nonempty set of real ratios has an attained minimum. □
⟨2⟩3. $\mathbf{y} - b\mathbf{x}$ is a nonnegative top eigenvector with a zero coordinate.

⟨3⟩1. $A(\mathbf{y} - b\mathbf{x}) = \rho(\mathbf{y} - b\mathbf{x})$.
PROOF: Linearity and the two top-eigenvector equations. □
⟨3⟩2. Every coordinate of $\mathbf{y} - b\mathbf{x}$ is nonnegative.
PROOF: By ⟨2⟩2, $b \le \mathbf{y}(u)/\mathbf{x}(u)$; multiply by $\mathbf{x}(u) > 0$. □
⟨3⟩3. One coordinate is zero.
PROOF: At an attaining index in ⟨2⟩2, the ratio equals b. □
⟨3⟩4. Q.E.D.
PROOF: Use ⟨3⟩1, ⟨3⟩2, ⟨3⟩3. □
⟨2⟩4. $\mathbf{y} = b\mathbf{x}$.
PROOF: Apply ⟨1⟩3, with ⟨1⟩, to the vector of ⟨2⟩3. □
⟨2⟩5. The multiplicity of ρ is one.
PROOF: The nonzero vector $\mathbf{x}$ spans the eigenspace by ⟨2⟩4; use Theorem 2.2. □
⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩5. □
⟨1⟩5. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩4. □

Lemma 4.6 (A positive supervector bounds each eigenvalue)**.**

ASSUME:
1. A is real, entrywise nonnegative on a finite nonempty set, $\mathbf{x} > 0$, and $A\mathbf{x} \le b\mathbf{x}$.
2. $A\mathbf{y} = \theta\mathbf{y}$ and $\mathbf{y} \neq 0$.

PROVE: $\theta \le b$. Symmetry of A is not required.

PROOF SKETCH: We will follow the maximum-ratio proof in `LineGraphs/Spectral/PositiveVector.lean`.
⟨1⟩1. PICK u maximizing $|\mathbf{y}(u)|/\mathbf{x}(u)$, and denote the maximum by c.
PROOF: A finite nonempty set of real numbers has an attained maximum. □

⟨1⟩2. $c > 0$ and $|\mathbf{y}(u)| = c\mathbf{x}(u) > 0$.
PROOF: A nonzero coordinate of $\mathbf{y}$ gives a positive ratio; use maximality and the positivity of $\mathbf{x}$. □
⟨1⟩3. $|\mathbf{y}(v)| \leq c\mathbf{x}(v)$ for every v.
PROOF: Multiply the maximal-ratio inequality by $\mathbf{x}(v) > 0$. □
⟨1⟩4. $\theta|\mathbf{y}(u)| \leq |(A\mathbf{y})(u)|$.
PROOF: Use $\theta \leq |\theta|$ and the eigenvector equation. □
⟨1⟩5. $|(A\mathbf{y})(u)| \leq \sum_v A(u,v)|\mathbf{y}(v)| \leq c(A\mathbf{x})(u)$.
PROOF: The finite-sum triangle inequality and nonnegative entries give the first inequality; sum ⟨1⟩3 with these coefficients for the second. □
⟨1⟩6. $\theta \leq b$.
PROOF: Combine ⟨1⟩4, ⟨1⟩5 and $c(A\mathbf{x})(u) \leq cb\mathbf{x}(u) = b|\mathbf{y}(u)|$; divide by the positive number in ⟨1⟩2. □
⟨1⟩7. Q.E.D.
PROOF: Use ⟨1⟩6. □

Corollary 4.7 (Positive-vector certificates)**.**

ASSUME:
1. A is real symmetric and entrywise nonnegative on a finite nonempty set; $\mathbf{x} > 0$ and $b \in \mathbb{R}$.

PROVE: If $A\mathbf{x} = b\mathbf{x}$, then $b = \lambda_{\max}(A)$, whether or not its support is connected. More generally, $A\mathbf{x} \leq b\mathbf{x}$ coordinatewise implies $\lambda_{\max}(A) \leq b$.

PROOF SKETCH: We will apply Lemma 4.6 to a top eigenvector for the upper bound, and use the eigenvector $\mathbf{x}$ itself for the reverse bound.
⟨1⟩1. If $A\mathbf{x} \leq b\mathbf{x}$, then $\lambda_{\max}(A) \leq b$.
PROOF: Apply Lemma 4.6 to the sorted basis vector $\mathbf{w}_0$ of Theorem 2.2, whose eigenvalue is $\nu_0 = \lambda_{\max}(A)$ by Definition 2.3. □
⟨1⟩2. If $A\mathbf{x} = b\mathbf{x}$, then $b \in \mathrm{Spec}(A)$.
PROOF: $\mathbf{x} > 0$ on a nonempty set implies $\mathbf{x} \neq 0$; apply Theorem 2.2. □
⟨1⟩3. In that case $b = \lambda_{\max}(A)$.
PROOF: Lemma 2.4 gives $b \leq \lambda_{\max}$ from ⟨1⟩2; ⟨1⟩1 gives the reverse inequality. □
⟨1⟩4. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩3. □

Theorem 4.8 (Strict monotonicity)**.**

ASSUME:
1. U, W are finite and nonempty. $A : U \times U \to \mathbb{R}$ is symmetric, nonnegative and has connected positive support.
2. $B : W \times W \to \mathbb{R}$ is symmetric and nonnegative; $\phi : W \hookrightarrow U$ is injective and $B(u,v) \leq A(\phi(u), \phi(v))$.
3. For the graph assertion, $A = A(X)$ for a connected finite simple graph X on U, and $B = A(X')$ for a nonempty subgraph X' on W, given by the injective vertex map ϕ preserving edges; X' is proper if it omits a vertex or an edge.

PROVE: $\lambda_{\max}(B) \leq \lambda_{\max}(A)$. The inequality is strict if ϕ is not surjective, or if one of these entry inequalities is strict. In particular every nonempty proper subgraph of a

connected finite graph has strictly smaller largest eigenvalue. A subgraph here may omit edges as well as vertices.

PROOF SKETCH: We will extend a top eigenvector by zero. Equality in both quadratic-form inequalities forces full vertex support and equality of every entry.

⟨1⟩1. PICK a nonnegative unit top eigenvector $\mathbf{y}$ of B; let $\mathbf{z}$ be its extension by zero along ϕ.
PROOF: Theorem 4.5 gives a nonzero nonnegative top eigenvector; divide by its positive norm. □

⟨1⟩2. $\mathbf{y}^T B\mathbf{y} = \lambda_{\max}(B) \le \mathbf{z}^T A\mathbf{z} \le \lambda_{\max}(A)$, and $\|\mathbf{z}\| = 1$.

⟨2⟩1. The chosen $\mathbf{z}$ is nonnegative and has norm one.
PROOF: In ⟨1⟩1, extension by zero along an injection preserves each coordinate square and its sum. □

⟨2⟩2. $\lambda_{\max}(B) = \mathbf{y}^T B\mathbf{y} \le \mathbf{z}^T A\mathbf{z} \le \lambda_{\max}(A)$.
PROOF: The entry inequalities may be multiplied by $\mathbf{y}(u)\mathbf{y}(v) \ge 0$ and summed; the final bound is Theorem 4.1, with extremes defined in Definition 2.3. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □

⟨1⟩3. If the two largest eigenvalues are equal, ϕ is surjective and every entry inequality is equality.

⟨2⟩1. $\mathbf{z}$ is a top eigenvector of A.
PROOF: Equality in ⟨1⟩2 gives the equality case of Theorem 4.1. □

⟨2⟩2. $\mathbf{z}(u) > 0$ for all u, so ϕ is surjective.
PROOF: Connectedness and Theorem 4.5 make ⟨2⟩1 positive; extension by zero cannot omit a coordinate. □

⟨2⟩3. $0 = \sum_{u,v}(A(\phi(u), \phi(v)) - B(u, v))\mathbf{y}(u)\mathbf{y}(v)$.
PROOF: Subtract the equal quadratic forms in ⟨1⟩2. □

⟨2⟩4. $A(\phi(u), \phi(v)) = B(u, v)$ for every pair.
PROOF: The summands in ⟨2⟩3 are nonnegative; each vanishes by Facts 8, and ⟨2⟩2 makes its coordinate product positive. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩4. □

⟨1⟩4. Either strictness hypothesis implies strict inequality; the graph assertion follows.

⟨2⟩1. An omitted vertex or a strict entry excludes equality.
PROOF: Take the contrapositive of ⟨1⟩3. □

⟨2⟩2. In the graph assertion, $A = A(X)$ and $B = A(X')$ satisfy ⟨0⟩:1 and ⟨0⟩:2.
PROOF: Adjacency matrices are symmetric with entries $0, 1$ (Facts 10); the positive support graph of $A(X)$ is X, which is connected; an edge-preserving injection gives $B(u, v) \le A(\phi(u), \phi(v))$. □

⟨2⟩3. A proper subgraph gives one of the two strictness conditions.

⟨3⟩1.
CASE: The subgraph omits a vertex.
PROOF: The omitted vertex is not in the image of the vertex injection, which is thus not surjective. □

⟨3⟩2.
CASE: The subgraph contains every vertex and omits an edge.
PROOF: At the endpoints of that edge the smaller adjacency entry is 0 and the larger one is 1, so an entry inequality is strict. □

⟨3⟩3. Q.E.D.
PROOF: By ⟨0⟩:3 a proper subgraph omits a vertex or an edge, so the two cases are exhaustive. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □
⟨1⟩5. Q.E.D.
PROOF: Use ⟨1⟩2, ⟨1⟩4. □

5. Walks, Laplacians, and regular graphs

Theorem 5.1 (Walks and moments)**.**

ASSUME:
1. X is a finite simple graph on U, $|U| = n$, $|E(X)| = m$, $A = A(X)$, and $j \in \mathbb{N}$.
2. t denotes the number of three-element vertex sets inducing a triangle.

PROVE: For $j \geq 0$, $(A^j)(u, v)$ counts walks of length j from u to v, and $\operatorname{tr} A^j$ counts closed walks with a distinguished starting vertex. In particular

$$\sum_{\mu \in \operatorname{Spec}(X)} 1 = n, \qquad \sum_{\mu \in \operatorname{Spec}(X)} \mu = 0, \qquad \sum_{\mu \in \operatorname{Spec}(X)} \mu^2 = 2m = \sum_u \deg(u), \qquad \sum_{\mu \in \operatorname{Spec}(X)} \mu^3 = 6t,$$

where t is the number of triangles and all sums count multiplicity. At each vertex u, $(A^3)(u, u) = 2\,\#\{\text{triangles containing } u\}$. The total number of walks of length j is $\mathbf{j}^T A^j \mathbf{j}$; it need not equal $\operatorname{tr} A^j$.

PROOF SKETCH: We will count walks recursively, then count closed walks of lengths zero through three.
⟨1⟩1. $(A^j)(u, v)$ counts walks of length j from u to v.
PROOF: Apply `SimpleGraph.adjMatrix_pow_apply_eq_card_walk` in Facts 10 over $\mathbb{R}$. □
⟨1⟩2. Trace counts closed walks with a starting vertex; $\mathbf{j}^T A^j \mathbf{j}$ counts all walks.

⟨2⟩1. $\operatorname{tr} A^j = \sum_u (A^j)(u, u)$.
PROOF: This is the definition of trace. □
⟨2⟩2. $\mathbf{j}^T A^j \mathbf{j} = \sum_{u,v} (A^j)(u, v)$.
PROOF: Expand multiplication by the all-ones vectors. □
⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □
⟨1⟩3. The first four closed-walk counts are $n, 0, 2m, 6t$.

⟨2⟩1. There are n length-zero closed walks.
PROOF: Choose their single vertex. □
⟨2⟩2. There are no length-one closed walks.
PROOF: Apply `SimpleGraph.trace_adjMatrix` in Facts 10. □
⟨2⟩3. There are $2m = \sum_u \deg(u)$ length-two closed walks.
PROOF: Apply `SimpleGraph.adjMatrix_mul_self_apply_self` in Facts 10, sum over vertices, and count incident vertex-edge pairs by their two endpoints. □
⟨2⟩4. There are $6t$ length-three closed walks.

⟨3⟩1. In u, v, w, u, the first three vertices are distinct.
PROOF: Any equality would make one of its three required edges a loop. □
⟨3⟩2. Each such walk gives a triangle and a cyclic ordering with a distinguished start.
PROOF: The edges uv, vw, wu in the walk are precisely the three triangle edges. □
⟨3⟩3. Each triangle gives exactly $3 \cdot 2 = 6$ such walks.
PROOF: Choose one of its three starting vertices and then one of the two neighbours as the next vertex. □
⟨3⟩4. Q.E.D.
PROOF: Use ⟨3⟩1, ⟨3⟩2, ⟨3⟩3. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩4. The four moment identities follow, with multiplicities.
PROOF: Apply Theorem 3.2 to ⟨1⟩3; the zeroth moment counts n entries by Theorem 2.2. □
⟨1⟩5. $(A^3)(u, u)$ is twice the number of triangles containing u.
PROOF: Apply `SimpleGraph.adjMatrix_pow_apply_eq_card_walk` in Facts 10 at power 3: the entry counts closed walks u, v, w, u. Looplessness makes u, v, w distinct; each triangle through u supplies precisely the two possible orders of v, w. □
⟨1⟩6. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩5, ⟨1⟩4. □

Theorem 5.2 (Laplacian and signless Laplacian kernels)**.**

ASSUME:

1. X is a finite simple graph on U, $A = A(X)$ and D is its diagonal degree matrix.
2. $L = D - A$, $K = D + A$, and $\mathbf{x} \in \mathbb{R}^U$. A component is called bipartite if its vertices admit a partition crossed by every edge.

PROVE:

$$\mathbf{x}^T L\mathbf{x} = \sum_{\{u,v\}\in E(X)} (\mathbf{x}(u) - \mathbf{x}(v))^2, \qquad \mathbf{x}^T K\mathbf{x} = \sum_{\{u,v\}\in E(X)} (\mathbf{x}(u) + \mathbf{x}(v))^2.$$

Both matrices are positive semidefinite. Their kernels consist, respectively, of functions constant across each edge and functions whose values sum to zero across each edge. Thus

$$\dim\ker L = \#\{\text{components of } X\}, \qquad \dim\ker K = \#\{\text{bipartite components of } X\}.$$

An isolated vertex counts as a bipartite component.

PROOF SKETCH: We will expand edge sums of squares and construct bases of the two kernels component by component.
⟨1⟩1. The two displayed quadratic-form identities hold and both matrices are positive semidefinite.

⟨2⟩1. $\sum_{\{u,v\}\in E}(\mathbf{x}(u)^2 + \mathbf{x}(v)^2) = \sum_u \deg(u)\mathbf{x}(u)^2$.
PROOF: Count one occurrence of $\mathbf{x}(u)^2$ for each edge incident with u. □
⟨2⟩2. $\mathbf{x}^T A\mathbf{x} = 2\sum_{\{u,v\}\in E} \mathbf{x}(u)\mathbf{x}(v)$.
PROOF: The ordered adjacency sum has the two orientations of each edge. □
⟨2⟩3. $\mathbf{x}^T L\mathbf{x} = \sum_{\{u,v\}\in E}(\mathbf{x}(u) - \mathbf{x}(v))^2$.
PROOF: Subtract ⟨2⟩2 from ⟨2⟩1 and expand the square. □
⟨2⟩4. $\mathbf{x}^T K\mathbf{x} = \sum_{\{u,v\}\in E}(\mathbf{x}(u) + \mathbf{x}(v))^2$.

PROOF: Add ⟨2⟩2 to ⟨2⟩1 and expand the square. □

⟨2⟩5. $L, K \succeq 0$.

⟨3⟩1. $L \succeq 0$.
PROOF: Apply `SimpleGraph.posSemidef_lapMatrix` in Facts 13 over $\mathbb{R}$. □

⟨3⟩2. $K \succeq 0$.
PROOF: K is symmetric and its form in ⟨2⟩4 is nonnegative; apply `Matrix.posSemidef_iff_dotProduct_mulVec` in Facts 6. □

⟨3⟩3. Q.E.D.
PROOF: Use ⟨3⟩1, ⟨3⟩2. □

⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩3, ⟨2⟩4, ⟨2⟩5. □

⟨1⟩2. The kernels are exactly the functions satisfying the stated edge relations.

⟨2⟩1. $L\mathbf{x} = 0$ iff $\mathbf{x}(u) = \mathbf{x}(v)$ on every edge.

⟨3⟩1. $L\mathbf{x} = 0$ iff $\mathbf{x}$ is constant on reachable pairs.
PROOF: Apply `SimpleGraph.lapMatrix_mulVec_eq_zero_iff_forall_reachable` in Facts 13. □

⟨3⟩2. Constancy on reachable pairs implies constancy across edges.
PROOF: An edge is a path of length one. □

⟨3⟩3. Constancy across edges implies constancy along paths.
PROOF: Induct on path length: the empty path has equal endpoints, and each successive edge preserves the value. □

⟨3⟩4. Q.E.D.
PROOF: Use ⟨3⟩1, ⟨3⟩2, ⟨3⟩3. □

⟨2⟩2. $K\mathbf{x} = 0$ iff $\mathbf{x}(u) = -\mathbf{x}(v)$ on every edge.
PROOF: Apply the zero-form criterion of Theorem 3.3 and the zero-sum-of-squares criterion in Facts 8 to the plus-square identity in ⟨1⟩1. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □

⟨1⟩3. $\dim \ker L$ equals the number of components.
PROOF: Apply `SimpleGraph.card_connectedComponent_eq_finrank_ker_toLin'_lapMatrix` in Facts 13; its kernel already uses the function-coordinate linear map. □

⟨1⟩4. On a fixed connected component the signless kernel has dimension one if bipartite and zero otherwise.

⟨2⟩1. Along a path of length j from u to v, $\mathbf{x}(v) = (-1)^j \mathbf{x}(u)$.
PROOF: Induction on j using the sign relation in ⟨1⟩2. □

⟨2⟩2. Evaluation at a fixed vertex is injective on the component kernel.
PROOF: By ⟨2⟩1, a zero value there forces all values to be zero. □

⟨2⟩3. A nonzero kernel function gives a bipartition into its positive and negative vertices.
PROOF: By ⟨2⟩1 all absolute values equal a positive number; the edge relation in ⟨1⟩2 reverses the sign. □

⟨2⟩4. A bipartition gives a nonzero kernel function equal to 1 and -1 on the two parts.
PROOF: Each edge has opposite signs, so ⟨1⟩2 applies; a component has a vertex. □

⟨2⟩5. The component dimension is one in the bipartite case and zero otherwise.

⟨3⟩1. The dimension is at most one.

PROOF: Evaluation injects it into $\mathbb{R}$ by ⟨2⟩2; apply dimension monotonicity in Facts 3. □
⟨3⟩2.
CASE: The component is bipartite.
PROOF: The nonzero vector in ⟨2⟩4 gives dimension at least one; with ⟨3⟩1 the dimension is one. □
⟨3⟩3.
CASE: The component is not bipartite.
PROOF: By ⟨2⟩3 any nonzero kernel function would give a bipartition; thus the kernel is the zero space. □
⟨3⟩4. Q.E.D.
PROOF: The two cases are exhaustive. □
⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩5. □
⟨1⟩5. The two kernel dimensions are the numbers of components and bipartite components.

⟨2⟩1. Extend one signed function from each bipartite component by zero; these functions span $\ker K$.
PROOF: Use the one-dimensional or zero component spaces in ⟨1⟩4 and restrict any kernel function to each component. □
⟨2⟩2. The extended functions are independent.
PROOF: Evaluate a zero combination at the chosen vertex of each component; supports are disjoint. □
⟨2⟩3. An isolated vertex contributes one dimension to each kernel.
PROOF: There is no edge relation at that vertex, and its singleton component has a bipartition. □
⟨2⟩4. The dimension formulas hold, including an empty vertex set.
PROOF: Use ⟨1⟩3 for L. For K, count the basis given by ⟨2⟩1 and ⟨2⟩2; an empty basis has dimension zero. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩4, ⟨2⟩3. □
⟨1⟩6. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩5. □

Theorem 5.3 (SP1: regular graphs)**.**

ASSUME:
1. X is a finite simple k-regular graph on a nonempty set U, where $k \in \mathbb{N}$.
PROVE: Its largest eigenvalue is k, every eigenvalue lies in $[-k, k]$, and

$$\mathrm{mult}_X(k) = \#\{\text{connected components of } X\}.$$

PROOF SKETCH: We will use the constant vector for occurrence of k and the two Laplacians for bounds and multiplicity.
⟨1⟩1. k occurs and every eigenvalue lies in $[-k, k]$.

⟨2⟩1. $A\mathbf{j} = k\mathbf{j}$ and $\mathbf{j} \neq 0$.
PROOF: Apply `SimpleGraph.adjMatrix_mulVec_const_apply_of_regular` in Facts 10 with constant 1; nonemptiness makes $\mathbf{j} \neq 0$. □
⟨2⟩2. $kI - A \succeq 0$ and $kI + A \succeq 0$.

PROOF: These are L, K in Theorem 5.2 for constant degree. □

⟨2⟩3. $-k \le \mu \le k$ for every eigenvalue, and the largest is k.
PROOF: Apply the shifts in Theorem 3.3 to ⟨2⟩2. By ⟨2⟩1 and Theorem 2.2, $k \in \mathrm{Spec}(X)$; Lemma 2.4 then gives $\lambda_{\max}(X) = k$. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩3. □

⟨1⟩2. $\mathrm{mult}_X(k)$ is the number of components.

⟨2⟩1. $V_A(k) = \ker(kI - A)$.
PROOF: Rearrange the defining eigenvector equation. □

⟨2⟩2. The dimension of this space is the number of components.
PROOF: Apply Theorem 5.2 to ⟨2⟩1. □

⟨2⟩3. Multiplicity is that dimension.
PROOF: Apply Theorem 2.2. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3. □

⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2. □

Theorem 5.4 (SP2: regularity detected by the spectrum)**.**

ASSUME:
1. X is a finite simple graph on $n > 0$ vertices.

PROVE:
$$X \text{ is regular} \iff \sum_{\mu \in \mathrm{Spec}(X)} \mu^2 = n\lambda_{\max}(X).$$

PROOF SKETCH: We will recognize equality in the Rayleigh bound at the all-ones vector.
⟨1⟩1. The displayed spectral identity is equivalent to regularity.

⟨2⟩1. $\|\mathbf{j}\|^2 = n$ and $\mathbf{j}^T A\mathbf{j} = \sum_u \deg(u) = \sum_{\mu \in \mathrm{Spec}(X)} \mu^2$.
PROOF: The all-ones sums and Theorem 5.1 give the equalities. □

⟨2⟩2. The identity is equivalent to $A\mathbf{j} = \lambda_{\max}\mathbf{j}$.
PROOF: Divide by $n > 0$ in ⟨2⟩1 and apply Theorem 4.1. □

⟨2⟩3. That equation forces all degrees equal to a natural number k.
PROOF: Evaluate ⟨2⟩2 at each vertex and choose one vertex to set $k = \deg(u)$. □

⟨2⟩4. If X is k-regular, the spectral identity holds.
PROOF: Use $A\mathbf{j} = k\mathbf{j}$ and $\lambda_{\max} = k$ from Theorem 5.3 in ⟨2⟩1. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩2. Q.E.D.
PROOF: Use ⟨1⟩1. □

Corollary 5.5 (Cospectral mates of regular graphs)**.**

ASSUME:
1. X is a nonempty finite k-regular simple graph and Y is a finite simple graph cospectral with X.

PROVE: Y is k-regular, has the same number of components as X, and is connected if X is connected.

PROOF SKETCH: We will transfer the order, the second moment and the largest eigenvalue to Y, and apply SP2 and SP1 to Y.

⟨1⟩1. X, Y have equal order $n > 0$, equal second moments $\sum_\mu \mu^2$ and equal largest eigenvalues.
PROOF: Proposition 2.6 gives $\mathrm{Spec}(X) = \mathrm{Spec}(Y)$; the three quantities are functions of the multiset, by Theorem 2.2 and Lemma 2.4. □

⟨1⟩2. Y is k-regular.
PROOF: Theorem 5.4 for X gives $\sum_\mu \mu^2 = n\lambda_{\max}$; by ⟨1⟩1 this identity holds for Y, and Theorem 5.4 for Y makes Y regular. Theorem 5.3 for Y and for X identifies both degrees with the common largest eigenvalue k. □

⟨1⟩3. X and Y have the same number of components.
PROOF: By Theorem 5.3 for X and for Y, using ⟨1⟩2, both numbers equal $\mathrm{mult}(k)$, which is common by Proposition 2.6. □

⟨1⟩4. If X is connected, so is Y.
PROOF: A nonempty graph is connected exactly when it has one component; use ⟨1⟩3. □

⟨1⟩5. Q.E.D.
PROOF: Use ⟨1⟩2, ⟨1⟩3, ⟨1⟩4. □

Theorem 5.6 (SP3: bipartiteness)**.**

ASSUME:
1. X is a finite connected simple k-regular graph, $k \in \mathbb{N}$.
2. The second assertion concerns any finite simple bipartite graph Y and any $\mu \in \mathbb{R}$.

PROVE: It is bipartite iff $-k \in \mathrm{Spec}(X)$; when this occurs its multiplicity is one. More generally, for every finite bipartite graph Y,

$$\mathrm{mult}_Y(\mu) = \mathrm{mult}_Y(-\mu) \qquad (\mu \in \mathbb{R}).$$

PROOF SKETCH: We will count the signless kernel, then reverse signs on one part of a bipartition.

⟨1⟩1. X is bipartite iff $-k$ occurs, and then its multiplicity is one.

⟨2⟩1. $V_A(-k) = \ker(kI + A)$.
PROOF: Rearrange the eigenvector equation. □

⟨2⟩2. That kernel has dimension one if X is bipartite and zero otherwise.
PROOF: Connectedness leaves one component in Theorem 5.2. □

⟨2⟩3. The claimed occurrence and multiplicity follow, including $k = 0$.
PROOF: Apply Theorem 2.2 to ⟨2⟩1, ⟨2⟩2; no division by k occurs. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩3. □

⟨1⟩2. For a finite bipartite graph Y, $\mathrm{mult}_Y(\mu) = \mathrm{mult}_Y(-\mu)$.

LET: $A \triangleq A(Y)$; fix a bipartition of Y, and let T be the diagonal matrix with entry 1 on the first part and -1 on the second.

⟨2⟩1. $T^2 = I$ and $TAT = -A$.

⟨3⟩1. $T^2 = I$.
PROOF: Each diagonal sign squares to 1, and the off-diagonal entries vanish. □

⟨3⟩2.

ASSUME: $u, v \in V(Y)$.
PROVE: $(TAT)(u, v) = -A(u, v)$.

⟨4⟩1.
CASE: $u = v$.
PROOF: Both entries are zero, since Y is loopless and T is diagonal. □
⟨4⟩2.
CASE: $u \neq v$ and uv is an edge.
PROOF: $(TAT)(u, v) = T(u, u)A(u, v)T(v, v)$, and the edge crosses the two parts, so the sign product is -1. □
⟨4⟩3.
CASE: $u \neq v$ and uv is not an edge.
PROOF: Both entries are zero, since $A(u, v) = 0$. □
⟨4⟩4. Q.E.D.
PROOF: The three cases are exhaustive. □
⟨3⟩3. Q.E.D.
PROOF: Use ⟨3⟩1, ⟨3⟩2. □
⟨2⟩2. $\mathbf{x} \mapsto T\mathbf{x}$ maps $V_A(\mu)$ to $V_A(-\mu)$.
PROOF: From ⟨2⟩1, $AT = -TA$; apply this to $A\mathbf{x} = \mu\mathbf{x}$. □
⟨2⟩3. The reverse map is also multiplication by T.
PROOF: For $A\mathbf{y} = -\mu\mathbf{y}$, the identity $AT = -TA$ from ⟨2⟩1 gives $A(T\mathbf{y}) = \mu T\mathbf{y}$ directly. Both composites are $T^2 = I$. □
⟨2⟩4. The multiplicities agree.
PROOF: The linear equivalence in ⟨2⟩3 preserves dimensions (Facts 3); use Theorem 2.2. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩4. □
⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2. □

Theorem 5.7 (The all-ones direction)**.**

ASSUME:
1. U is finite with $|U| = n > 0$, $A : U \times U \to \mathbb{R}$ is symmetric, and $A\mathbf{j} = k\mathbf{j}$ for $k \in \mathbb{R}$.
2. $b, s, t \in \mathbb{R}$. For the complement identity X is any finite simple graph; for the two graph spectra X is a nonempty finite simple k-regular graph on U and $A = A(X)$.

PROVE: For every $\mathbf{x}$, $J\mathbf{x} = (\sum_u \mathbf{x}(u))\mathbf{j}$; thus $J\mathbf{x} = 0$ on $\mathbf{j}^\perp$. There is an orthogonal decomposition $\mathbb{R}^U = \operatorname{span}(\mathbf{j}) \oplus \mathbf{j}^\perp$, of dimensions $1, n-1$. The orthogonal complement $\mathbf{j}^\perp$ is invariant under A, so one can choose an orthonormal eigenbasis starting with $\mathbf{j}/\sqrt{n}$. Write the remaining eigenvalues, with multiplicity, as $\mu_1, \ldots, \mu_{n-1}$. For real b, s, t, the eigenvalues of $bI + sA + tJ$ are

$$b + sk + tn, \qquad b + s\mu_i \quad (1 \leq i < n).$$

For every finite simple graph X, $A(\overline{X}) = J - I - A(X)$. Thus for a nonempty k-regular graph,

$$\operatorname{Spec}(\overline{X}) = [\![n - 1 - k, -1 - \mu_1, \ldots, -1 - \mu_{n-1}]\!],$$
$$\operatorname{Spec}(S(X)) = [\![n - 1 - 2k, -1 - 2\mu_1, \ldots, -1 - 2\mu_{n-1}]\!].$$

PROOF SKETCH: We will separate the constant direction and diagonalize on its orthogonal complement.

⟨1⟩1. The orthogonal decomposition into span($\mathbf{j}$) and $\mathbf{j}^\perp$ has dimensions $1, n-1$; the complement is invariant and admits the asserted eigenbasis.

⟨2⟩1. For $\langle \mathbf{x}, \mathbf{j}\rangle = 0$, $\langle A\mathbf{x}, \mathbf{j}\rangle = 0$.

⟨3⟩1. $\langle A\mathbf{x}, \mathbf{j}\rangle = \langle \mathbf{x}, A\mathbf{j}\rangle$.
PROOF: Symmetry of A, by exchanging its two coordinate summation indices. □

⟨3⟩2. $\langle \mathbf{x}, A\mathbf{j}\rangle = k\langle \mathbf{x}, \mathbf{j}\rangle$.
PROOF: Substitute $A\mathbf{j} = k\mathbf{j}$ and use linearity of the inner product. □

⟨3⟩3. $k\langle \mathbf{x}, \mathbf{j}\rangle = 0$.
PROOF: The assumed inner product is zero. □

⟨3⟩4. Q.E.D.
PROOF: Transitivity of the three equalities. □

⟨2⟩2. Every $\mathbf{x}$ splits as $\frac{\langle \mathbf{x},\mathbf{j}\rangle}{n}\mathbf{j} + (\mathbf{x} - \frac{\langle \mathbf{x},\mathbf{j}\rangle}{n}\mathbf{j})$ into the constant direction and its orthogonal complement.
PROOF: The second term has inner product $\langle \mathbf{x}, \mathbf{j}\rangle - \langle \mathbf{x}, \mathbf{j}\rangle = 0$ with $\mathbf{j}$, since $\|\mathbf{j}\|^2 = n$. □

⟨2⟩3. The restriction to $\mathbf{j}^\perp$ is symmetric and has an orthonormal eigenbasis.
PROOF: Use ⟨2⟩1 and the finite-dimensional linear-map spectral theorem in Facts 2. □

⟨2⟩4. Adjoining $\mathbf{j}/\sqrt{n}$ gives the asserted basis.
PROOF: It has norm one by Facts 8; ⟨2⟩2 is an orthogonal direct sum. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩4. □

⟨1⟩2. $J\mathbf{x} = (\sum_u \mathbf{x}(u))\mathbf{j}$, so J kills $\mathbf{j}^\perp$, and the displayed list is the spectrum of $bI + sA + tJ$.

⟨2⟩1. $J\mathbf{x} = (\sum_u \mathbf{x}(u))\mathbf{j}$.
PROOF: Expand the all-ones matrix product. □

⟨2⟩2. On $\mathbf{j}/\sqrt{n}$ the scalar is $b + sk + tn$.
PROOF: Apply the three summands and ⟨2⟩1. □

⟨2⟩3. On a μ_i-eigenvector in $\mathbf{j}^\perp$ the scalar is $b + s\mu_i$.
PROOF: The coordinate sum is zero, so the J term vanishes by ⟨2⟩1. □

⟨2⟩4. The list counts every eigenvalue with multiplicity.
PROOF: By ⟨2⟩2 and ⟨2⟩3, the basis in ⟨1⟩1 is an orthonormal eigenbasis of the symmetric matrix $bI + sA + tJ$ with the displayed eigenvalues; apply the arbitrary-basis assertion of Theorem 2.2. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩4. □

⟨1⟩3. The complement identity and the complement and Seidel formulas hold.

⟨2⟩1. $A(\overline{X}) = J - I - A(X)$ for every finite simple graph X.
PROOF: Apply `SimpleGraph.one_add_adjMatrix_add_compl_adjMatrix_eq_of_one` in Facts 10, which gives $I + A(X) + A(X).\mathrm{compl} = J$, and `SimpleGraph.compl_adjMatrix_eq_adjMatrix_compl`, which identifies $A(X).\mathrm{compl}$ with $A(\overline{X})$. □

⟨2⟩2. $S(X) = J - I - 2A(X)$.
PROOF: This is Definition 2.5. □

⟨2⟩3. For $A = A(X)$ as in ⟨0⟩:2, the two lists follow from ⟨1⟩2.

PROOF: By ⟨2⟩1 and ⟨2⟩2, $A(\overline{X})$ and $S(X)$ are $bI + sA + tJ$ for $(b, s, t) = (-1, -1, 1)$ and $(-1, -2, 1)$, respectively. □

⟨2⟩4. Q.E.D.

PROOF: Use ⟨2⟩1, ⟨2⟩3. □

⟨1⟩4. Q.E.D.

PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3. □

Corollary 5.8 (Removing the constant direction)**.**

ASSUME:

1. G is real positive semidefinite of order $n > 0$, $G\mathbf{j} = b\mathbf{j}$, and $b > 0$.

PROVE: $G - (b/n)J \succeq 0$ and its rank is $\operatorname{rk} G - 1$.

PROOF SKETCH: We will apply Theorem 5.7 to G; the correction by J replaces the eigenvalue b on the constant direction by zero and leaves the other eigenvalues unchanged.

⟨1⟩1. The eigenvalues of $G - (b/n)J$ are 0 on $\mathbf{j}/\sqrt{n}$ and the eigenvalues $\mu_1, \ldots, \mu_{n-1}$ of G on $\mathbf{j}^\perp$.

PROOF: G is symmetric (Facts 6); apply Theorem 5.7 to G with $k = b$ and coefficients $0, 1, -b/n$, so that $0 + b - (b/n)n = 0$. □

⟨1⟩2. $G - (b/n)J \succeq 0$.

PROOF: By Theorem 5.7 for G with coefficients $0, 1, 0$, $\operatorname{Spec}(G) = [\![b, \mu_1, \ldots, \mu_{n-1}]\!]$, so each $\mu_i \geq 0$ by Theorem 3.3; with ⟨1⟩1, Theorem 3.3 applies to $G - (b/n)J$. □

⟨1⟩3. $\operatorname{rk}(G - (b/n)J) = \operatorname{rk} G - 1$.

PROOF: By Theorem 3.2, $\operatorname{rk} G$ counts the nonzero entries of $[\![b, \mu_1, \ldots, \mu_{n-1}]\!]$, and $\operatorname{rk}(G - (b/n)J)$ counts those of ⟨1⟩1; the two lists differ only in $b > 0$. □

⟨1⟩4. Q.E.D.

PROOF: Use ⟨1⟩2, ⟨1⟩3. □

Corollary 5.9 (A scalar matrix plus the all-ones matrix)**.**

ASSUME:

1. U is finite with $|U| = n > 0$, $b, t \in \mathbb{R}$, and I, J are the identity and all-ones matrices on U.

PROVE:

$$\operatorname{Spec}(bI + tJ) = [\![(b + tn)^1, b^{n-1}]\!],$$
$$\chi_{bI+tJ}(z) = (z - b - tn)(z - b)^{n-1},$$
$$\det(bI + tJ) = (b + tn)b^{n-1}.$$

The multiset convention combines the two values when $t = 0$.

PROOF SKETCH: We will compute on the constant direction and its orthogonal complement.

⟨1⟩1. The two actions have scalars $b + tn$ and b, in dimensions one and $n - 1$.

⟨2⟩1. $(bI + tJ)\mathbf{j} = (b + tn)\mathbf{j}$.

PROOF: $J\mathbf{j} = n\mathbf{j}$ by the defining row sum. □

⟨2⟩2. For $\mathbf{x} \perp \mathbf{j}$, $(bI + tJ)\mathbf{x} = b\mathbf{x}$.

PROOF: $J\mathbf{x} = 0$ by the coordinate-sum formula asserted in Theorem 5.7, applied to the zero matrix with $k = 0$. □

⟨2⟩3. The two orthogonal spaces have dimensions $1, n - 1$.

PROOF: Use the orthogonal decomposition asserted in Theorem 5.7, applied to the zero matrix with $k = 0$. □

⟨2⟩4. Q.E.D.

PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □

⟨1⟩2. All three formulas, including coincident values and $n = 1$, hold.

⟨2⟩1. $\mathrm{Spec}(bI + tJ) = [\![(b + tn)^1, b^{n-1}]\!]$.

PROOF: The vector $\mathbf{j}/\sqrt{n}$ and an orthonormal basis of $\mathbf{j}^\perp$ form an orthonormal eigenbasis of $bI + tJ$ with eigenvalues $b + tn$ and b, by ⟨1⟩1; apply the arbitrary-basis assertion of Theorem 2.2. □

⟨2⟩2. $\chi(z) = (z - b - tn)(z - b)^{n-1}$.

PROOF: Apply the product formula in Theorem 2.2 to ⟨2⟩1. □

⟨2⟩3. $\det(bI + tJ) = (b + tn)b^{n-1}$.

⟨3⟩1.

CASE: $b \neq 0$.

PROOF: Apply `Matrix.det_one_add_replicateCol_mul_replicateRow` in Facts 17 to the constant vectors $(t/b)\mathbf{j}$ and $\mathbf{j}$: $\det(I + (t/b)J) = 1 + tn/b$. Then $\det(bI + tJ) = b^n(1 + tn/b) = (b + tn)b^{n-1}$, since $n > 0$. □

⟨3⟩2.

CASE: $b = 0$.

⟨4⟩1.

CASE: $n = 1$.

PROOF: By Theorem 3.2 and ⟨2⟩1, the determinant is the single eigenvalue tn, which is $(b + tn)b^0$. □

⟨4⟩2.

CASE: $n > 1$.

PROOF: By Theorem 3.2 and ⟨2⟩1, the determinant is a product with a factor $b = 0$; the right side $(b + tn)b^{n-1}$ is also zero. □

⟨4⟩3. Q.E.D.

PROOF: The cases $n = 1$ and $n > 1$ exhaust $n > 0$. □

⟨3⟩3. Q.E.D.

PROOF: The cases $b \neq 0$ and $b = 0$ are exhaustive. □

⟨2⟩4. Q.E.D.

PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □

⟨1⟩3. Q.E.D.

PROOF: Use ⟨1⟩2. □

6. Rectangular products and line graphs

Theorem 6.1 (Rectangular characteristic polynomials)**.**

ASSUME:

1. P, Q are finite with cardinalities p, q; $M : P \times Q \to R$ and $N : Q \times P \to R$, where R is a commutative ring.

2. For the multiplicity and eigenspace assertions $R = \mathbb{R}$ and $\mu \in \mathbb{R} \setminus \{0\}$. For the final assertion $B : P \times Q \to \mathbb{R}$ has rank ρ.

PROVE:

$$t^q \chi_{MN}(t) = t^p \chi_{NM}(t).$$

Over $\mathbb{R}$, the nonzero roots have equal algebraic multiplicities and

$$q + \mathrm{mult}_{MN}(0) = p + \mathrm{mult}_{NM}(0).$$

For $\mu \neq 0$, multiplication by N is a linear equivalence $V_{MN}(\mu) \simeq V_{NM}(\mu)$, with inverse $\mu^{-1}M$. If B is a real $p \times q$ matrix of rank ρ, both BB^T and B^TB are positive semidefinite; their positive spectra agree with multiplicity, and their zero multiplicities are $p - \rho$ and $q - \rho$.

PROOF SKETCH: We will compare two block determinants, transfer eigenspaces, and count the zero modes of Gram products.

⟨1⟩1. $t^q \chi_{MN} = t^p \chi_{NM}$.

PROOF: Apply `Matrix.charpoly_mul_comm'` in Facts 9 over the commutative coefficient ring. □

⟨1⟩2. Nonzero algebraic multiplicities agree and $q + \mathrm{mult}_{MN}(0) = p + \mathrm{mult}_{NM}(0)$.

⟨2⟩1. Both characteristic polynomials and the powers of t are nonzero.

PROOF: Characteristic polynomials are monic (Facts 9); powers of t over $\mathbb{R}$ are nonzero. □

⟨2⟩2. Root multiplicities in ⟨1⟩1 add over the two factors on each side.

PROOF: Apply the product-root multiplicity fact in Facts 5, using ⟨2⟩1. □

⟨2⟩3. t^h has multiplicity h at zero and zero at every nonzero number.

PROOF: Its roots are h copies of the unique root zero of t. □

⟨2⟩4. The two claimed equations hold.

PROOF: Substitute ⟨2⟩3 in ⟨2⟩2 and use Definition 2.1. □

⟨2⟩5. Q.E.D.

PROOF: Use ⟨2⟩4. □

⟨1⟩3. N and $\mu^{-1}M$ restrict to inverse linear maps of the two μ-eigenspaces.

⟨2⟩1. $NM(N\mathbf{x}) = \mu N\mathbf{x}$ when $MN\mathbf{x} = \mu\mathbf{x}$.

PROOF: Associativity gives $NM(N\mathbf{x}) = N(MN\mathbf{x})$; substitute the hypothesis. □

⟨2⟩2. $MN(\mu^{-1}M\mathbf{y}) = \mu(\mu^{-1}M\mathbf{y})$ when $NM\mathbf{y} = \mu\mathbf{y}$.

PROOF: Commute the scalar with M, use associativity, and substitute $NM\mathbf{y} = \mu\mathbf{y}$. □

⟨2⟩3. $\mu^{-1}MN\mathbf{x} = \mathbf{x}$ on $V_{MN}(\mu)$.

PROOF: Cancel $\mu \neq 0$ in its eigenvector equation. □

⟨2⟩4. $N(\mu^{-1}M\mathbf{y}) = \mathbf{y}$ on $V_{NM}(\mu)$.

PROOF: Cancel $\mu \neq 0$ in the NM eigenvector equation. □

⟨2⟩5. Q.E.D.

PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩4. The Gram products have common positive spectrum and zero multiplicities $p - \rho, q - \rho$.

⟨2⟩1. $BB^T, B^TB \succeq 0$ and both ranks are ρ.

PROOF: Theorem 3.5. □

⟨2⟩2. Their zero multiplicities are their orders minus ρ.

PROOF: Theorem 3.2 applied to ⟨2⟩1. □

⟨2⟩3. All remaining eigenvalues are positive, with equal counts.

PROOF: Use Theorem 3.3 for positivity. Apply `Matrix.charpoly_mul_comm'` directly to B, B^T (Facts 9); at nonzero roots the factors t^p, t^q have zero multiplicity, by the product rule in Facts 5. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3. □

⟨1⟩5. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4. □

Theorem 6.2 (Incidence identities)**.**

ASSUME:
1. X is a finite simple graph with vertex set U, $|U| = n$, and $|E(X)| = m$.
2. $M : U \times E(X) \to \mathbb{R}$ is the endpoint incidence matrix, with actual edges as columns; D is the diagonal degree matrix.
3. b is the number of bipartite connected components of X.

PROVE:
$$MM^T = D + A(X), \qquad M^T M = 2I + A(L(X)).$$
If b is the number of bipartite connected components of X, then
$$\operatorname{rk} M = n - b, \qquad \operatorname{mult}_{L(X)}(-2) = m - n + b.$$
The right side is nonnegative. Every line graph has least eigenvalue at least -2 whenever it is nonempty.

PROOF SKETCH: We will count common incidences in both products and apply the signless kernel formula.

⟨1⟩1. $MM^T = D + A(X)$.

⟨2⟩1. Mathlib's full incidence product has degree diagonal and adjacency off the diagonal.
PROOF: Apply `SimpleGraph.incMatrix_mul_transpose` in Facts 14 over $\mathbb{R}$. □

⟨2⟩2. Restricting its columns to actual edges leaves the product unchanged.
PROOF: For every nonedge column all incidence entries are zero; remove precisely these zero summands from each product entry. □

⟨2⟩3. The resulting product is $D + A(X)$.
PROOF: The diagonal in ⟨2⟩1 is D, the off-diagonal is A, and ⟨2⟩2 identifies the restricted product. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩3. □

⟨1⟩2. $M^T M = 2I + A(L(X))$.

⟨2⟩1. $(M^T M)(e, e) = 2$.
PROOF: Each simple edge has exactly two distinct endpoints. □

⟨2⟩2. For $e \neq f$, $(M^T M)(e, f)$ is 1 if they intersect and 0 if they do not.
PROOF: Distinct unordered two-element sets have intersection of size at most one. □

⟨2⟩3. These off-diagonal entries equal line-graph adjacency.
PROOF: This is `SimpleGraph.lineGraph` in Facts 10. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □

⟨1⟩3. $\operatorname{rk} M = n - b$ and $\operatorname{mult}_{L(X)}(-2) = m - n + b \geq 0$.

⟨2⟩1. $\dim\ker(MM^T) = b$.
PROOF: Use ⟨1⟩1 and Theorem 5.2. □
⟨2⟩2. $\operatorname{rk} M = \operatorname{rk}(MM^T) = n - b \le m$.
PROOF: Rank-nullity (Facts 3) and Theorem 3.5. □
⟨2⟩3. M^TM has zero multiplicity $m - (n - b)$.
PROOF: Theorem 6.1 and ⟨2⟩2. □
⟨2⟩4. That multiplicity equals $\text{mult}_{L(X)}(-2)$.
PROOF: Use ⟨1⟩2 and Theorem 3.1 with the shift by two. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩4. Every nonempty line graph has least eigenvalue at least -2.
PROOF: $M^TM \succeq 0$ by Theorem 3.5; apply Corollary 3.4 to ⟨1⟩2. □
⟨1⟩5. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4. □

Lemma 6.3 (Connectivity of a line graph)**.**

ASSUME:
1. X is a finite simple graph with at least one edge and no isolated vertices.
2. $\{u, v\}$ is an edge for the degree assertion.

PROVE: X is connected iff $L(X)$ is connected. For any edge $\{u, v\}$, its degree in the line graph is $\deg_X(u) + \deg_X(v) - 2$.

PROOF SKETCH: We will turn vertex paths into edge walks and edge walks into reachability, then count the two disjoint sets of neighbouring edges.
⟨1⟩1. Connectedness of X implies connectedness of $L(X)$.

⟨2⟩1. Between any two edges, a path between chosen endpoints gives an edge sequence starting and ending with those edges.
PROOF: Connectedness supplies the endpoint path; prepend and append the chosen edges. □
⟨2⟩2. Consecutive distinct members of this sequence are adjacent in $L(X)$.
PROOF: They share the intervening endpoint; use the line-graph definition (Facts 10). □
⟨2⟩3. Removing consecutive repetitions gives a walk between the chosen edges.
PROOF: Each retained transition is one of the adjacencies in ⟨2⟩2; if all entries coincide, the zero-length walk suffices. □
⟨2⟩4. Q.E.D.
PROOF: The line graph has a vertex because X has an edge, and ⟨2⟩3 proves every pair reachable. □
⟨1⟩2. Connectedness of $L(X)$ implies connectedness of X.

⟨2⟩1. Given vertices u, v, choose incident edges e, f.
PROOF: No vertex is isolated. □
⟨2⟩2. There is a sequence $e = e_0, \ldots, e_h = f$ whose consecutive edges intersect.
PROOF: Connectedness gives a walk in $L(X)$ and the definition gives the intersections. □
⟨2⟩3. Both endpoints of every e_i are reachable from u.
PROOF: Induct on i: the base uses the edge at u; an intersection vertex and the next edge give the inductive extension. □
⟨2⟩4. v is reachable from u.

PROOF: It is an endpoint of e_h in ⟨2⟩3. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩4. □
⟨1⟩3. $\deg_{L(X)}(\{u, v\}) = \deg_X(u) + \deg_X(v) - 2$.

⟨2⟩1. Its neighbours are the other edges at u, together with the other edges at v.
PROOF: A line-graph neighbour shares one endpoint with $\{u, v\}$. □
⟨2⟩2. Those two sets are disjoint.
PROOF: An edge containing both u, v is the original edge, which was removed. □
⟨2⟩3. The two cardinalities are $\deg(u) - 1$ and $\deg(v) - 1$.
PROOF: Remove the single original edge from each incident-edge set; both degrees are positive. □
⟨2⟩4. Q.E.D.
PROOF: Add the disjoint cardinalities in ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □
⟨1⟩4. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3. □

Theorem 6.4 (SP4: regular line graphs)**.**

ASSUME:
1. X is a finite simple graph with at least one edge and no isolated vertices.
2. $L(X)$ is connected and h-regular, with $h \in \mathbb{N}$.

PROVE: X is either regular or bipartite with constant degree on each part. In the latter case, writing the part sizes as p, q and the respective degrees as k_1, k_2, we have $k_1 + k_2 = h + 2$ and $pk_1 = qk_2 = |E(X)|$.

PROOF SKETCH: We will propagate the sum of endpoint degrees along paths and split according to whether its two alternating values agree.
⟨1⟩1. X is connected and every degree is one of two positive integers k_1, k_2 with $k_1 + k_2 = h + 2$.

⟨2⟩1. X is connected and $\deg(u) + \deg(v) = h + 2$ on every edge.
PROOF: Apply Lemma 6.3. □
⟨2⟩2. PICK an edge u_0v_0, and let $k_1 = \deg(u_0)$, $k_2 = \deg(v_0)$; they are positive and $k_1+k_2 = h+2$.
PROOF: The edge hypothesis supplies u_0v_0; its endpoints have positive degree, and ⟨2⟩1 applies to it. □
⟨2⟩3. Along a path from u_0, the degrees alternate k_1, k_2.
PROOF: Induct on its length: the edge equation in ⟨2⟩1 determines the next degree as $h + 2$ minus the current one. □
⟨2⟩4. Every vertex has degree k_1 or k_2.
PROOF: Connectedness in ⟨2⟩1 supplies a path from u_0; apply ⟨2⟩3. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩4. □
⟨1⟩2. X is regular, or bipartite with degrees k_1, k_2 on parts of sizes p, q, where $k_1 + k_2 = h + 2$ and $pk_1 = qk_2 = |E(X)|$.

⟨2⟩1.
CASE: $k_1 = k_2$.
PROOF: Every degree has that value by ⟨1⟩1, so X is regular. □
⟨2⟩2.

CASE: $k_1 \neq k_2$.

⟨3⟩1. The two degree classes form a bipartition of X into nonempty parts, of sizes p, q say.
PROOF: The endpoint equation in ⟨1⟩1 forces opposite degree classes across every edge, since $k_1 \neq k_2$; the edge u_0v_0 meets both classes. □
⟨3⟩2. $pk_1 = |E(X)|$.
PROOF: By ⟨3⟩1, each edge has exactly one endpoint in the first part; count the edges at the vertices of that part. □
⟨3⟩3. $qk_2 = |E(X)|$.
PROOF: Count the edges at the vertices of the second part in the same way. □
⟨3⟩4. Q.E.D.
PROOF: Use ⟨3⟩1, ⟨3⟩2, ⟨3⟩3, and $k_1 + k_2 = h + 2$ from ⟨1⟩1. □
⟨2⟩3. Q.E.D.
PROOF: The cases $k_1 = k_2$ and $k_1 \neq k_2$ are exhaustive. □
⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩2. □

Theorem 6.5 (Recovering a regular graph's spectrum)**.**

ASSUME:
1. X is a finite simple k-regular graph with at least one edge; $n = |V(X)|$, $m = |E(X)|$.
2. $h = \lambda_{\max}(L(X))$ and $\rho = \operatorname{rk}(2I + A(L(X)))$.
3. For the last assertion X' is a second finite simple regular graph with at least one edge, and $L(X')$ is cospectral with $L(X)$.

PROVE: the spectrum of $L(X)$ determines k, n and the full spectrum of X. In detail, with $h = \lambda_{\max}(L(X))$ and $\rho = \operatorname{rk}(2I + A(L(X)))$,

$$k = (h + 2)/2, \qquad n = 2m/k,$$

and $\operatorname{Spec}(X)$ is obtained by taking all nonzero eigenvalues of $2I + A(L(X))$, adjoining $n - \rho$ zeros, and subtracting k from every entry. Equivalently, without negative exponents,

$$(t + 2)^n \chi_{L(X)}(t) = (t + 2)^m \chi_X(t + 2 - k).$$

Thus two regular graphs with an edge and cospectral line graphs are cospectral.

PROOF SKETCH: We will recover the degree from line-graph regularity and then add precisely the missing zero modes of the incidence product. We will prove this for an arbitrary regular graph Y with an edge, so that the reconstruction can be applied to both X and X'.
⟨1⟩1.
ASSUME: Y is a finite simple k_Y-regular graph with at least one edge, $n_Y = |V(Y)|$, $m_Y = |E(Y)|$, $h_Y = \lambda_{\max}(L(Y))$ and $\rho_Y = \operatorname{rk}(2I + A(L(Y)))$.
PROVE: $m_Y = |\operatorname{Spec}(L(Y))|$, $k_Y = (h_Y + 2)/2$, $n_Y = 2m_Y/k_Y$, $\operatorname{Spec}(Y)$ is obtained from $\operatorname{Spec}(L(Y))$ by the stated reconstruction, and $(t+2)^{n_Y}\chi_{L(Y)}(t) = (t+2)^{m_Y}\chi_Y(t + 2 - k_Y)$.

⟨2⟩1. $m_Y = |\operatorname{Spec}(L(Y))|$, $k_Y = (h_Y + 2)/2 > 0$ and $n_Y = 2m_Y/k_Y$.

⟨3⟩1. $k_Y > 0$ and Y has no isolated vertex.
PROOF: An edge of ⟨1⟩ has an endpoint of positive degree, and all degrees equal k_Y. □
⟨3⟩2. $h_Y = 2k_Y - 2$.

PROOF: Lemma 6.3 makes $L(Y)$ $(2k_Y - 2)$-regular, and $L(Y)$ is nonempty; Theorem 5.3 gives its largest eigenvalue. □

⟨3⟩3. $m_Y = |\operatorname{Spec}(L(Y))|$ and $n_Y k_Y = 2m_Y$.
PROOF: Theorem 2.2 counts the vertices of $L(Y)$; Theorem 5.1 gives the handshake identity. □

⟨3⟩4. $k_Y = (h_Y + 2)/2$ and $n_Y = 2m_Y/k_Y$.
PROOF: Solve ⟨3⟩2, ⟨3⟩3, using ⟨3⟩1. □

⟨3⟩5. Q.E.D.
PROOF: Use ⟨3⟩1, ⟨3⟩3, ⟨3⟩4. □

⟨2⟩2. For the incidence matrix M of Y, $MM^T = k_Y I + A(Y)$ and $M^T M = 2I + A(L(Y))$.
PROOF: Theorem 6.2 and regularity. □

⟨2⟩3. $\operatorname{Spec}(Y)$ is obtained by taking the nonzero eigenvalues of $2I + A(L(Y))$, adjoining $n_Y - \rho_Y$ zeros, and subtracting k_Y.

⟨3⟩1. $k_Y I + A(Y)$ has the nonzero spectrum of $2I + A(L(Y))$ and $n_Y - \rho_Y$ zeros.
PROOF: Apply Theorem 6.1 to M; by ⟨2⟩2 the rank of both products is ρ_Y. □

⟨3⟩2. Subtracting k_Y from every entry yields $\operatorname{Spec}(Y)$.
PROOF: Apply Theorem 3.1 to $k_Y I + A(Y)$ and the polynomial $t - k_Y$, using ⟨3⟩1. □

⟨3⟩3. Q.E.D.
PROOF: Use ⟨3⟩2. □

⟨2⟩4. $(t+2)^{m_Y}\chi_Y(t + 2 - k_Y) = (t+2)^{n_Y}\chi_{L(Y)}(t)$.
PROOF: Apply Theorem 6.1 to M and M^T, whose products are given by ⟨2⟩2, and substitute $t + 2$ for the indeterminate. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩2. The displayed formulas and the reconstruction hold for X.
PROOF: Apply ⟨1⟩1 to $Y = X$ with ⟨0⟩:1 and ⟨0⟩:2. □

⟨1⟩3. X and X' are cospectral.
PROOF: Apply ⟨1⟩1 to $Y = X$ and to $Y = X'$. By ⟨0⟩:3 and Proposition 2.6, $L(X)$ and $L(X')$ have the same spectrum, so the quantities m, h, ρ, and thus the degree, the order and the reconstructed multiset, agree; the last is $\operatorname{Spec}(X) = \operatorname{Spec}(X')$. Apply Proposition 2.6 once more. □

⟨1⟩4. Q.E.D.
PROOF: Use ⟨1⟩2, ⟨1⟩3. □

Theorem 6.6 (Bipartite block spectra)**.**

ASSUME:
1. P, Q are finite of sizes p, q; $B : P \times Q \to \mathbb{R}$ has rank ρ.
2. $\lambda_1, \dots, \lambda_\rho$ are the positive eigenvalues of BB^T, with repetition, and $s, t \in \mathbb{R}$.

PROVE: the matrix

$$C = \begin{pmatrix} sI_p & B \\ B^T & tI_q \end{pmatrix}$$

has the two eigenvalues

$$\gamma_j^{\pm} = \frac{s + t \pm \sqrt{(s-t)^2 + 4\lambda_j}}{2} \quad (1 \le j \le \rho),$$

as well as s repeated $p-\rho$ times and t repeated $q-\rho$ times. This is a multiset assertion, so coinciding displayed values add their multiplicities. For $s=t=0$, it gives $\pm\sqrt{\lambda_j}$ and $p+q-2\rho$ zeros. Furthermore

$$(C-sI)(C-tI)=\begin{pmatrix} BB^T & 0 \\ 0 & B^TB \end{pmatrix}.$$

PROOF SKETCH: We will pair positive Gram eigendirections into orthogonal two-dimensional spaces and keep both kernel spaces.

⟨1⟩1. PICK orthonormal $\mathbf{u}_j$ with $BB^T\mathbf{u}_j=\lambda_j\mathbf{u}_j$, and put $\mathbf{v}_j=B^T\mathbf{u}_j/\sqrt{\lambda_j}$. Then $(\mathbf{v}_j)$ is orthonormal and $B\mathbf{v}_j=\sqrt{\lambda_j}\mathbf{u}_j$.

⟨2⟩1. PICK the $\mathbf{u}_j$ as the positive-eigenvalue vectors of an orthonormal eigenbasis of BB^T.
PROOF: Theorem 2.2 applied to the symmetric Gram matrix. □

⟨2⟩2. Each $\sqrt{\lambda_j}>0$ and its square is λ_j.
PROOF: Facts 8 and $\lambda_j>0$. □

⟨2⟩3. $\langle\mathbf{v}_i,\mathbf{v}_j\rangle=\lambda_j\langle\mathbf{u}_i,\mathbf{u}_j\rangle/(\sqrt{\lambda_i}\sqrt{\lambda_j})$.
PROOF: Move B^T across the inner product and substitute the eigenvector equation. □

⟨2⟩4. This inner product is 1 for $i=j$ and 0 for $i\neq j$.
PROOF: Use the orthonormality of $(\mathbf{u}_j)$ and ⟨2⟩2 in ⟨2⟩3. □

⟨2⟩5. $B\mathbf{v}_j=\sqrt{\lambda_j}\mathbf{u}_j$ and $B^T\mathbf{u}_j=\sqrt{\lambda_j}\mathbf{v}_j$.
PROOF: The first is the BB^T equation divided by $\sqrt{\lambda_j}$; the second is the definition of $\mathbf{v}_j$. □

⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩4, ⟨2⟩5. □

⟨1⟩2. The ambient space is the orthogonal sum of the ρ paired planes and kernel spaces of dimensions $p-\rho, q-\rho$.

⟨2⟩1. The remaining left eigenbasis vectors span $\ker B^T$ of dimension $p-\rho$.
PROOF: Theorem 3.5 identifies $\ker BB^T=\ker B^T$; Theorem 6.1 counts its zero modes. □

⟨2⟩2. The ρ vectors $\mathbf{v}_j$ form a basis of $\operatorname{im} B^T$.
PROOF: They are independent by ⟨1⟩1, belong to the image, and that image has dimension ρ by Facts 7. □

⟨2⟩3. The orthogonal complement of this image is $\ker B$, of dimension $q-\rho$.
PROOF: $\langle B^T\mathbf{u},\mathbf{v}\rangle=\langle\mathbf{u},B\mathbf{v}\rangle$ for every $\mathbf{u}$ makes the two membership conditions equivalent; apply rank-nullity (Facts 3). □

⟨2⟩4. The dimensions add to $2\rho+(p-\rho)+(q-\rho)=p+q$.
PROOF: Finite-dimensional arithmetic; the summands are orthogonal by ⟨1⟩1, ⟨2⟩1, ⟨2⟩3. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩3. The stated multiset gives all eigenvalues of C.

⟨2⟩1. On the jth paired plane the matrix is $\begin{pmatrix} s & \sqrt{\lambda_j} \\ \sqrt{\lambda_j} & t \end{pmatrix}$.
PROOF: Apply the four blocks of C using ⟨1⟩1. □

⟨2⟩2. Its characteristic polynomial is $(z-s)(z-t)-\lambda_j$.
PROOF: Use the two-by-two determinant formula on ⟨2⟩1. □

⟨2⟩3. The two roots are $\gamma_j^{\pm}$ and are distinct.
PROOF: Complete the square: $(2z-s-t)^2=(s-t)^2+4\lambda_j>0$; use Facts 8. □

⟨2⟩4. The two kernel spaces have scalars s and t, respectively.

⟨3⟩1. For $B^T\mathbf{u} = 0$, $C(\mathbf{u}, 0) = (s\mathbf{u}, 0)$.
PROOF: The two block rows give $s\mathbf{u} + B0 = s\mathbf{u}$ and $B^T\mathbf{u} + t0 = 0$. □
⟨3⟩2. For $B\mathbf{v} = 0$, $C(0, \mathbf{v}) = (0, t\mathbf{v})$.
PROOF: The two block rows give $s0 + B\mathbf{v} = 0$ and $B^T0 + t\mathbf{v} = t\mathbf{v}$. □
⟨3⟩3. Q.E.D.
PROOF: Use ⟨3⟩1, ⟨3⟩2. □
⟨2⟩5. All eigenvalues and their multiplicities are accounted for.
PROOF: Take the union of bases of the invariant orthogonal summands in ⟨1⟩2; by ⟨2⟩3 and ⟨2⟩4 it is an orthonormal eigenbasis of the symmetric matrix C with the stated eigenvalues. Apply the arbitrary-basis assertion of Theorem 2.2. □
⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩5. □
⟨1⟩4. $(C - sI)(C - tI) = BB^T \oplus B^TB$; for $s = t = 0$ the spectrum specializes as stated.

⟨2⟩1. Both off-diagonal blocks of $C^2 - (s + t)C + stI$ vanish.
PROOF: They are $(s + t)B - (s + t)B$ and its transpose. □
⟨2⟩2. The diagonal blocks are BB^T and B^TB.
PROOF: The scalar terms are $s^2 - (s + t)s + st = 0$ and $t^2 - (s + t)t + st = 0$. □
⟨2⟩3. For $s = t = 0$ the paired roots are $\pm\sqrt{\lambda_j}$ and the other $p + q - 2\rho$ entries are zero.
PROOF: Substitute in ⟨1⟩3, using $\sqrt{4\lambda_j} = 2\sqrt{\lambda_j}$. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □
⟨1⟩5. Q.E.D.
PROOF: Use ⟨1⟩3, ⟨1⟩4. □

Theorem 6.7 (Cospectral line graphs of bipartite graphs)**.**

ASSUME:
1. X_1 is a finite connected simple bipartite graph with an edge, and $L(X_1)$ is regular.
2. X_2 is a finite simple graph with no isolated vertices and $L(X_2)$ cospectral with $L(X_1)$.

PROVE: X_2 is connected, and precisely one of the following alternatives holds:
(1) X_2 is bipartite. Order the part sizes $p \le q$ in each graph. The part sizes and respective degrees k_1, k_2 agree, and the two biadjacency matrices B_1, B_2 have cospectral products $B_1B_1^T, B_2B_2^T$.
(2) X_1 is not regular; X_2 is regular and not bipartite; $|V(X_1)| = |V(X_2)| + 1$.

PROOF SKETCH: We will recover the order from the signless rank, recover the two degrees from their sum and product, and distinguish the bipartite and nonbipartite cases.
⟨1⟩1. Both line graphs are connected and h-regular with the same $m > 0$ vertices; X_2 is connected and is regular or bipartite semiregular.

⟨2⟩1. $L(X_1)$ is connected.
PROOF: X_1 is connected with an edge, so has no isolates; apply Lemma 6.3. □
⟨2⟩2. $L(X_2)$ is connected, regular with the same degree h, and has the same order $m > 0$.
PROOF: Apply Corollary 5.5 and Proposition 2.6. □
⟨2⟩3. X_2 is connected and has the required regular-or-semiregular alternative.
PROOF: It has $m > 0$ edges and no isolates; apply Lemma 6.3 and Theorem 6.4. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □
⟨1⟩2. $2I + A(L(X_1))$ and $2I + A(L(X_2))$ have the same rank and the same nonzero spectrum.
PROOF: By ⟨1⟩1 and Proposition 2.6 the two line graphs have the same spectrum; shift it by two with Theorem 3.1 and count nonzero entries with Theorem 3.2. □
LET: $s \triangleq h + 2$, where h is the common degree in ⟨1⟩1; R is the common rank in ⟨1⟩2, and $N \triangleq R + 1$.
⟨1⟩3.
ASSUME: Z is X_1, or $Z = X_2$ and X_2 is bipartite.
PROVE: $|V(Z)| = N$, and the signless Laplacian of Z has the common nonzero spectrum of ⟨1⟩2 with one zero adjoined.

⟨2⟩1. The signless Laplacian of Z has exactly one zero eigenvalue.
PROOF: Z is connected, by ⟨0⟩:1 or ⟨1⟩1, and bipartite; apply Theorem 5.2 and Theorem 2.2. □
⟨2⟩2. $|V(Z)| = R + 1$, and the nonzero part of the signless spectrum of Z is that of ⟨1⟩2.
PROOF: By Theorem 6.2, the signless Laplacian and $2I + A(L(Z))$ are the two Gram products of the incidence matrix of Z; Theorem 6.1 gives equal nonzero spectra, and with ⟨2⟩1 the order is the rank plus one. □
⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □
⟨1⟩4.
ASSUME: Z is X_1, or $Z = X_2$ and X_2 is bipartite.
PROVE: Z has parts of sizes $p \le q$ with constant degrees $k_1 \ge k_2 > 0$, where k_1, k_2 are the ordered roots of $z^2 - sz + ms/N$, $p = m/k_1$ and $q = m/k_2$.

⟨2⟩1. The degrees of Z are constant on the two parts; write them $k_1 \ge k_2 > 0$ on parts of sizes $p \le q$.
PROOF: $L(Z)$ is connected and h-regular by ⟨0⟩:1, ⟨1⟩1; apply Theorem 6.4, and in its regular case note that each part has that common degree. An edge gives positive degrees. □
⟨2⟩2. $p + q = N$, $k_1 + k_2 = s$ and $pk_1 = qk_2 = m$.
PROOF: Theorem 6.4 gives the last two, since Z has m edges by ⟨1⟩1; ⟨1⟩3 gives the first. □
⟨2⟩3. $k_1 k_2 = ms/N$.
PROOF: From ⟨2⟩2, $N = m/k_1 + m/k_2 = ms/(k_1 k_2)$; all denominators are positive. □
⟨2⟩4. k_1, k_2 are the ordered roots of $z^2 - sz + ms/N$, and $p = m/k_1$, $q = m/k_2$.
PROOF: Use their sum and product in ⟨2⟩2, ⟨2⟩3; ordering fixes which root is first. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩4. □
⟨1⟩5.
CASE: X_2 is bipartite.

⟨2⟩1. The signless Laplacians of X_1 and X_2 are $C_i = \begin{pmatrix} k_1 I & B_i \\ B_i^T & k_2 I \end{pmatrix}$ with common p, q, k_1, k_2 and common spectrum.
PROOF: Apply ⟨1⟩4 and ⟨1⟩3 to $Z = X_1$ and $Z = X_2$, using ⟨1⟩; order the vertices by parts. □
⟨2⟩2. $B_1 B_1^T \oplus B_1^T B_1$ and $B_2 B_2^T \oplus B_2^T B_2$ have the same spectrum.
PROOF: By Theorem 6.6, $(C_i - k_1 I)(C_i - k_2 I)$ is this direct sum; apply Theorem 3.1 with $f(z) = (z - k_1)(z - k_2)$ to ⟨2⟩1. □

⟨2⟩3. Each positive multiplicity in $B_i B_i^T$ is half the corresponding multiplicity in the direct sum.
PROOF: Theorem 6.1 makes the positive counts of $B_i B_i^T$ and $B_i^T B_i$ equal; direct-sum eigenbases add counts. □
⟨2⟩4. $B_1 B_1^T$ and $B_2 B_2^T$ have the same zero multiplicity.
PROOF: Each has order p; subtract the common sum of positive multiplicities in ⟨2⟩2, ⟨2⟩3 from p. □
⟨2⟩5. $B_1 B_1^T$ and $B_2 B_2^T$ are cospectral.
PROOF: Apply Proposition 2.6 to ⟨2⟩3, ⟨2⟩4. □
⟨2⟩6. Q.E.D.
PROOF: Alternative (1) holds by ⟨2⟩1 and ⟨2⟩5; alternative (2) fails, since it requires X_2 not to be bipartite. □
⟨1⟩6.
CASE: X_2 is not bipartite.

⟨2⟩1. X_2 is regular of degree $s/2$.

⟨3⟩1. X_2 is regular, or bipartite with constant degree on each part.
PROOF: This is ⟨1⟩1. □
⟨3⟩2.
CASE: X_2 is bipartite with constant degree on each part.
PROOF: This case assumption contradicts ⟨1⟩, so the case does not occur. □
⟨3⟩3.
CASE: X_2 is regular, of degree k.
PROOF: Lemma 6.3 gives line degree $2k - 2 = h$, and $s = h + 2$, so $k = s/2$. □
⟨3⟩4. Q.E.D.
PROOF: The two cases of ⟨3⟩1 are exhaustive. □
⟨2⟩2. $|V(X_2)| = R = N - 1$.
PROOF: X_2 is connected by ⟨1⟩1 and not bipartite, so Theorem 5.2 gives a zero signless kernel. By Theorems 6.2 and 3.5, the rank of the incidence matrix of X_2 is $|V(X_2)|$ and is the rank R of $2I + A(L(X_2))$. □
⟨2⟩3. X_1 is not regular.
PROOF: If it were, of degree k, then $2k - 2 = h$ by Lemma 6.3, so $k = s/2 > 0$, which is also the degree of X_2 by ⟨2⟩1; the handshake identities $Nk = 2m = (N - 1)k$ for X_1 and X_2, from ⟨1⟩3 and ⟨2⟩2, would contradict $k > 0$. □
⟨2⟩4. Q.E.D.
PROOF: Alternative (2) holds by ⟨2⟩1, ⟨2⟩2, ⟨2⟩3; alternative (1) fails, since it requires X_2 to be bipartite. □
⟨1⟩7. Q.E.D.
PROOF: X_2 is connected by ⟨1⟩1. The cases ⟨1⟩5 and ⟨1⟩6 are exhaustive, and in each exactly one alternative holds. □

Corollary 6.8 (The regular nonbipartite alternative)**.**

ASSUME:

1. X_1 is a finite connected simple bipartite semiregular graph with part sizes $p \le q$ and degrees $k_1 > k_2 > 0$; B is its biadjacency matrix of rank ρ.

2. X_2 is finite, simple, connected, regular and nonbipartite; $L(X_2)$ is cospectral with $L(X_1)$.
3. $s = k_1 + k_2$.

PROVE: The positive eigenvalues of BB^T include k_1k_2 once. Delete this one occurrence and call the remaining list $\lambda_2, \dots, \lambda_\rho$. Then the spectrum of X_2 is

$$\left[\left[\frac{s}{2}, \quad \pm\sqrt{\lambda_j + \frac{(k_1-k_2)^2}{4}}\ (2 \le j \le \rho), \quad \left(\frac{k_1-k_2}{2}\right)^{p-\rho}, \quad \left(-\frac{k_1-k_2}{2}\right)^{q-\rho}\right]\right].$$

PROOF SKETCH: We will identify the single endpoint pair $0, s$, delete its zero member before shifting, and retain all rectangular kernel modes.

⟨1⟩1. k_1k_2 is a positive eigenvalue of BB^T of multiplicity one.

⟨2⟩1. $B\mathbf{j}_q = k_1\mathbf{j}_p$ and $B^T\mathbf{j}_p = k_2\mathbf{j}_q$.
PROOF: These are the constant row and column sums. □

⟨2⟩2. $BB^T\mathbf{j}_p = k_1k_2\mathbf{j}_p$.
PROOF: Apply B to the second equation in ⟨2⟩1 and use the first. □

⟨2⟩3. For $C = D(X_1) + A(X_1)$, $(k_1\mathbf{j}_p, k_2\mathbf{j}_q)$ is a positive s-eigenvector.
PROOF: Block multiplication with ⟨2⟩1 gives scalars $k_1^2 + k_1k_2 = sk_1$ and $k_1k_2 + k_2^2 = sk_2$. □

⟨2⟩4. s is a simple eigenvalue of C.
PROOF: Its support is the connected graph X_1; use Corollary 4.7 and Theorem 4.5. □

⟨2⟩5. Each copy of k_1k_2 gives one 0 and one s in C.
PROOF: In Theorem 6.6, $(z-k_1)(z-k_2) - k_1k_2 = z(z-s)$. □

⟨2⟩6. There is exactly one such copy.
PROOF: It occurs by ⟨2⟩2; two copies would contradict ⟨2⟩4 through ⟨2⟩5. □

⟨2⟩7. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩6. □

⟨1⟩2. The signless spectrum of X_2 is the spectrum of C with its single zero deleted.

⟨2⟩1. C has exactly one zero eigenvalue.
PROOF: Theorem 5.2 for connected bipartite X_1. □

⟨2⟩2. The signless matrix of X_2 has no zero eigenvalue and has the same nonzero spectrum as C.
PROOF: Use nonbipartiteness, Theorem 5.2, and the incidence Gram transfer from Theorems 6.2, 6.1. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □

⟨1⟩3. Subtracting $s/2$ gives exactly the stated adjacency multiset.

⟨2⟩1. X_2 has degree $s/2$.
PROOF: Lemma 6.3 gives degree $s-2$ for $L(X_1)$. Apply Theorem 6.7; X_1 has an edge since $k_2 > 0$, and connected nonbipartite X_2 has no isolates. □

⟨2⟩2. The surviving endpoint eigenvalue s becomes $s/2$.
PROOF: Use ⟨1⟩2 and subtract the degree in ⟨2⟩1. □

⟨2⟩3. Each remaining pair becomes $\pm\sqrt{\lambda_j + (k_1-k_2)^2/4}$.
PROOF: Subtract $s/2$ from the two roots in Theorem 6.6 and factor 4 inside the square root. □

⟨2⟩4. The $p-\rho$ left-kernel modes become $(k_1-k_2)/2$.
PROOF: Subtract $s/2$ from the scalar k_1 in Theorem 6.6. □

⟨2⟩5. The $q - \rho$ right-kernel modes become $-(k_1 - k_2)/2$.
PROOF: Subtract $s/2$ from the scalar k_2 in Theorem 6.6. □
⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4, ⟨2⟩5. □
⟨1⟩4. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩3. □

Error 14 of `cgss/errata.md` [Err] records the endpoint deletion and the two kernel multiplicities in the preceding corollary.

7. Strongly regular graphs and design spectra

Theorem 7.1 (SR1: parameter and matrix identities)**.**

ASSUME:
1. X is a finite simple strongly regular graph with parameters $(n, k, a, c) \in \mathbb{N}^4$: n vertices, degree k, and a, c common neighbours for adjacent and distinct nonadjacent pairs.
2. $A = A(X)$. The parameter equation assumes $n > 0$; the final bounds assume connectedness and $0 < k < n - 1$.

PROVE:
$$A^2 = (k - c)I + (a - c)A + cJ, \qquad AJ = JA = kJ.$$
If $n > 0$, then $k(k - a - 1) = c(n - k - 1)$ as an equality of integers. If X is connected and $0 < k < n - 1$, then $0 < c \le k$ and $a \le k - 1$.

PROOF SKETCH: We will count entries, apply the identity to the constant vector, and exhibit pairs on which the neighbour counts are attained.
⟨1⟩1. $A^2 = (k - c)I + (a - c)A + cJ$ and $AJ = JA = kJ$.

⟨2⟩1. $A^2 = kI + aA + cA(\overline{X})$.
PROOF: Apply `SimpleGraph.IsSRGWith.matrix_eq` in Facts 15 over $\mathbb{R}$; natural scalar multiplication becomes multiplication by the real cast. □
⟨2⟩2. $A^2 = (k - c)I + (a - c)A + cJ$.
PROOF: Substitute the complement identity $A(\overline{X}) = J - I - A$ asserted in Theorem 5.7 into ⟨2⟩1 and collect terms. □
⟨2⟩3. $AJ = JA = kJ$.
PROOF: Apply the regular constant-vector declaration in Facts 10 to every column of J; transpose using symmetry for JA. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3. □
⟨1⟩2. If $k > 0$, then $a \le k - 1$.
PROOF: An edge uv exists since $k > 0$; its a common neighbours lie in $N(u) \setminus \{v\}$, which has $k - 1$ elements. □
⟨1⟩3. For $n > 0$, $k(k - a - 1) = c(n - k - 1)$ in $\mathbb{Z}$.

⟨2⟩1. In $\mathbb{N}$, $k(k - a - 1) = (n - k - 1)c$ with truncated subtraction.
PROOF: Apply `SimpleGraph.IsSRGWith.param_eq` in Facts 15 with $n > 0$. □
⟨2⟩2. $k \le n - 1$.
PROOF: A vertex exists since $n > 0$; its k neighbours belong to the other $n - 1$ vertices. □

⟨2⟩3.
CASE: $k = 0$.
PROOF: The left side is zero over both rings. By ⟨2⟩2, the right natural difference casts to $n - k - 1$; cast ⟨2⟩1. □
⟨2⟩4.
CASE: $k > 0$.
PROOF: By ⟨1⟩2 and ⟨2⟩2, both subtractions in ⟨2⟩1 cast without truncation; commute the right factors. □
⟨2⟩5. Q.E.D.
PROOF: The cases $k = 0$ and $k > 0$ exhaust $k \in \mathbb{N}$. □
⟨1⟩4.
ASSUME: X is connected and $0 < k < n - 1$.
PROVE: $0 < c \le k$ and $a \le k - 1$.

⟨2⟩1. $a \le k - 1$.
PROOF: Apply ⟨1⟩2, since $k > 0$ by ⟨1⟩. □
⟨2⟩2. $c \le k$.
PROOF: A distinct nonadjacent pair exists since $k < n - 1$; its common-neighbour set lies in a neighbourhood of size k. □
⟨2⟩3. There is a path $u_0u_1u_2$ with u_0, u_2 distinct and nonadjacent.
PROOF: PICK a distinct nonadjacent pair, as $k < n - 1$, and a shortest path between them, as X is connected; its length is at least two, and an edge between its first and third vertices would shorten it. □
⟨2⟩4. $c \ge 1$.
PROOF: The vertex u_1 in ⟨2⟩3 is a common neighbour of u_0 and u_2. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩4. □
⟨1⟩5. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩3, ⟨1⟩4. □

Theorem 7.2 (SR3: spectrum and integral multiplicities)**.**

ASSUME:
1. X is a finite connected simple strongly regular graph with parameters $(n, k, a, c) \in \mathbb{N}^4$ and $0 < k < n - 1$.
2. $A = A(X)$, $\Delta = (a - c)^2 + 4(k - c)$, and $\theta_1 = (a - c + \sqrt{\Delta})/2$, $\theta_2 = (a - c - \sqrt{\Delta})/2$.

PROVE: $\Delta > 0$, $k > \theta_1 > \theta_2$, and

$$\operatorname{Spec}(X) = \left[\!\left[k^1, \theta_1^{m_1}, \theta_2^{m_2}\right]\!\right], \qquad m_1 = \frac{-k - (n-1)\theta_2}{\theta_1 - \theta_2}, \quad m_2 = \frac{k + (n-1)\theta_1}{\theta_1 - \theta_2}.$$

Both displayed multiplicities are positive integers. Its spectral idempotents are

$$E_0 = \frac{1}{n}J, \quad E_1 = \frac{A - \theta_2 I - \frac{k-\theta_2}{n}J}{\theta_1 - \theta_2}, \quad E_2 = \frac{A - \theta_1 I - \frac{k-\theta_1}{n}J}{\theta_2 - \theta_1}.$$

In particular $\operatorname{rk} E_i = \operatorname{tr} E_i = m_i$ for $i = 1, 2$, $1 + m_1 + m_2 = n$, and $k + m_1\theta_1 + m_2\theta_2 = 0$.

PROOF SKETCH: We will find the two roots on $\mathbf{j}^\perp$, force both to occur, solve for their dimensions, and check each idempotent on each eigenspace.

⟨1⟩1. $\Delta > 0$ and every eigenvalue on $\mathbf{j}^\perp$ is θ_1 or θ_2, with $\theta_1 > \theta_2$.

⟨2⟩1. $\Delta \geq 0$.
PROOF: Theorem 7.1 gives $k - c \geq 0$ and the other term is a square. □
⟨2⟩2. $\Delta \neq 0$.
PROOF: If the two nonnegative terms sum to zero, then $a = c = k$, contradicting $a \leq k - 1$ in Theorem 7.1. □
⟨2⟩3. On $\mathbf{j}^\perp$, $A^2 - (a - c)A - (k - c)I = 0$.
PROOF: J kills this subspace by the coordinate-sum formula of Theorem 5.7, applied to A with $A\mathbf{j} = k\mathbf{j}$ from Theorem 7.1; substitute in the matrix identity of Theorem 7.1. □
⟨2⟩4. An eigenvalue there is one of θ_1, θ_2, which are distinct and ordered.
PROOF: Apply ⟨2⟩3 to its nonzero eigenvector, then complete the square and use $\sqrt{\Delta} > 0$ (Facts 8). □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩4. □
⟨1⟩2. Both roots occur, and $k > \theta_1 > \theta_2$ with k simple.

⟨2⟩1. k is the simple largest eigenvalue.
PROOF: Theorem 5.3 and connectedness. □
⟨2⟩2. If just one value b occurred on $\mathbf{j}^\perp$, then $A = bI + (k - b)J/n$.
PROOF: The two sides act identically on the eigenbasis from Theorem 5.7. □
⟨2⟩3. That is impossible: there is an off-diagonal entry 1 and another 0.
PROOF: $k > 0$ gives an edge, $k < n - 1$ gives a distinct nonedge, while ⟨2⟩2 makes all off-diagonal entries equal. □
⟨2⟩4. Both roots are strictly below k.
PROOF: They occur in the perpendicular subspace by ⟨1⟩1, ⟨2⟩3; ⟨2⟩1 leaves no copy of k there. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩3. The displayed m_1, m_2 are positive integers and give the stated spectrum.

LET: $m_i \triangleq \dim V_A(\theta_i)$ for $i = 1, 2$; these are the multiplicities, by Theorem 2.2.
⟨2⟩1. $m_1, m_2 > 0$ and $m_1 + m_2 = n - 1$.
PROOF: The eigenbasis of Theorem 5.7 has $\mathbf{j}/\sqrt{n}$ and $n - 1$ perpendicular vectors, whose eigenvalues are θ_1 or θ_2 by ⟨1⟩1; both values occur by ⟨1⟩2. The arbitrary-basis assertion of Theorem 2.2 makes m_i the number of these vectors with eigenvalue θ_i, since $k \neq \theta_i$. □
⟨2⟩2. $m_1\theta_1 + m_2\theta_2 = -k$.
PROOF: The trace is zero by Theorem 5.1; expand it with Theorem 3.2. □
⟨2⟩3. $m_1 = (-k - (n - 1)\theta_2)/(\theta_1 - \theta_2)$.
PROOF: Substitute $m_2 = n - 1 - m_1$ in ⟨2⟩2 and divide by $\theta_1 - \theta_2 > 0$. □
⟨2⟩4. $m_2 = (k + (n - 1)\theta_1)/(\theta_1 - \theta_2)$.
PROOF: Substitute $m_1 = n - 1 - m_2$ in ⟨2⟩2 and divide by $\theta_1 - \theta_2 > 0$. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩4. The three displayed matrices are the spectral idempotents, with the asserted traces and ranks.

⟨2⟩1. J/n acts by $1, 0, 0$ on the k, θ_1, θ_2 eigenspaces.

PROOF: Theorem 5.7 and simplicity of k in ⟨1⟩2. □

⟨2⟩2. The displayed E_1 kills the k-direction.
PROOF: Its numerator there is $k - \theta_2 - (k - \theta_2)n/n = 0$. □

⟨2⟩3. E_1 fixes the θ_1-space.
PROOF: Its scalar there is $(\theta_1 - \theta_2)/(\theta_1 - \theta_2) = 1$. □

⟨2⟩4. E_1 kills the θ_2-space.
PROOF: Its numerator there is $\theta_2 - \theta_2 = 0$. □

⟨2⟩5. E_2 kills the k-direction.
PROOF: Its numerator there is $k - \theta_1 - (k - \theta_1)n/n = 0$. □

⟨2⟩6. E_2 kills the θ_1-space.
PROOF: Its numerator there is $\theta_1 - \theta_1 = 0$. □

⟨2⟩7. E_2 fixes the θ_2-space.
PROOF: Its scalar there is $(\theta_2 - \theta_1)/(\theta_2 - \theta_1) = 1$. □

⟨2⟩8. $\operatorname{rk} E_i = \operatorname{tr} E_i = m_i$ for $i = 1, 2$.
PROOF: The actions just checked identify the unique matrices of Theorem 3.1, which gives their ranks and traces. □

⟨2⟩9. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4, ⟨2⟩5, ⟨2⟩6, ⟨2⟩7, ⟨2⟩8. □

⟨1⟩5. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4. □

Theorem 7.3 (SR2: complements)**.**

ASSUME:
1. X is a finite connected simple strongly regular graph with parameters $(n, k, a, c) \in \mathbb{N}^4$, $0 < k < n - 1$.
2. $\Delta = (a - c)^2 + 4(k - c)$, $\theta_1 = (a - c + \sqrt{\Delta})/2$, $\theta_2 = (a - c - \sqrt{\Delta})/2$, $m_1 = (-k - (n - 1)\theta_2)/(\theta_1 - \theta_2)$, and $m_2 = (k + (n - 1)\theta_1)/(\theta_1 - \theta_2)$.

PROVE: Its complement has parameters

$$(n, n - k - 1, n - 2k + c - 2, n - 2k + a)$$

and spectrum

$$\left[\!\left[(n - k - 1)^1, (-1 - \theta_1)^{m_1}, (-1 - \theta_2)^{m_2}\right]\!\right],$$

where the roots and multiplicities are those of Theorem 7.2. Coincident values in this list must be combined; the complement need not be connected.

PROOF SKETCH: We will count common nonneighbours in each of the two pair cases, then use the constant-direction decomposition.
⟨1⟩1. The complement has the displayed parameters.

⟨2⟩1. $\overline{X}$ is strongly regular with Mathlib's natural complement parameters.
PROOF: Apply `SimpleGraph.IsSRGWith.compl` in Facts 15. □

⟨2⟩2. Its degree is $n - k - 1$.
PROOF: Among the other $n - 1$ vertices, remove the k original neighbours. □

⟨2⟩3. For a nonedge uv of X, the complement common-neighbour count is $n - 2 - 2k + c$.
PROOF: $N_X(u) \cup N_X(v)$ has size $2k - c$ and avoids u, v; subtract it from the other $n - 2$ vertices. □

⟨2⟩4. For an edge uv of X, the complement common-neighbour count is $n - 2k + a$.

PROOF: The union has size $2k-a$ and contains u, v; remove its other $2k-a-2$ vertices from the available $n-2$. □

⟨2⟩5. These parameters equal the displayed integer expressions.
PROOF: The actual finite common-neighbour counts in ⟨2⟩3 and ⟨2⟩4 are nonnegative. They justify replacing Mathlib's truncated subtractions by the integer differences; ⟨2⟩2 handles the degree. □

⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩5. □

⟨1⟩2. The displayed complement multiset is correct, even when values coincide.

⟨2⟩1. The spectrum of X is $[\![k^1, \theta_1^{m_1}, \theta_2^{m_2}]\!]$.
PROOF: Theorem 7.2. □

⟨2⟩2. Its complement spectrum is $[\![(n-k-1)^1, (-1-\theta_1)^{m_1}, (-1-\theta_2)^{m_2}]\!]$.
PROOF: Apply the complement formula of Theorem 5.7 to ⟨2⟩1. □

⟨2⟩3. Coincident values contribute the sum of their multiplicities.
PROOF: The displayed object in ⟨2⟩2 is a multiset, by Definition 2.1. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3. □

⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2. □

Corollary 7.4 (SR4: least eigenvalue minus two)**.**

ASSUME:
1. X is a finite connected simple strongly regular graph with parameters $(n, k, a, c) \in \mathbb{N}^4$, $0 < k < n-1$, and $\lambda_{\min}(X) = -2$.
2. $A = A(X)$, and Δ, θ_1, θ_2 and m_1 are those of Theorem 7.2.

PROVE:

$$\theta_1 = a - c + 2 \in \mathbb{Z}, \quad k = c + 2\theta_1, \quad m_1 = \frac{2n-k-2}{\theta_1+2}, \quad \operatorname{rk}(2I + A) = 1 + m_1.$$

In particular the positive integer $\theta_1 + 2$ divides $2n - k - 2$ in $\mathbb{Z}$, and the rank is one plus that quotient.

PROOF SKETCH: We will substitute the least root into its quadratic and count the positive eigenvalues after shifting by two.

⟨1⟩1. $\theta_2 = -2$, $\theta_1 = a - c + 2 \in \mathbb{Z}$ and $k = c + 2\theta_1$.

⟨2⟩1. $\theta_2 = -2$.
PROOF: By Theorem 7.2, $\operatorname{Spec}(X) = [\![k, \theta_1^{m_1}, \theta_2^{m_2}]\!]$ with $k > \theta_1 > \theta_2$, so Lemma 2.4 gives $\lambda_{\min}(X) = \theta_2$; use ⟨0⟩:1. □

⟨2⟩2. $\theta_1 - 2 = a - c$.
PROOF: The definitions give $\theta_1 + \theta_2 = a - c$; use ⟨2⟩1. □

⟨2⟩3. $-2\theta_1 = c - k$.
PROOF: The definitions give $\theta_1\theta_2 = ((a-c)^2 - \Delta)/4 = c - k$; use ⟨2⟩1. □

⟨2⟩4. θ_1 is the indicated integer.
PROOF: Rearrange ⟨2⟩2; $a, c, 2$ are integer casts. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩2. The multiplicity, rank and divisibility formulas hold.

⟨2⟩1. $m_1 = (2n - k - 2)/(\theta_1 + 2)$.
PROOF: Substitute $\theta_2 = -2$ from ⟨1⟩1 in the multiplicity formula of Theorem 7.2. □
⟨2⟩2. $2I + A$ has $1 + m_1$ positive eigenvalues and all its other eigenvalues are zero.
PROOF: Theorem 3.1 shifts the three values; $k + 2 > 0$ and $\theta_1 + 2 > 0$ by Theorem 7.2. □
⟨2⟩3. $\mathrm{rk}(2I + A) = 1 + m_1$.
PROOF: Apply Theorem 3.2 to ⟨2⟩2. □
⟨2⟩4. The positive integer $\theta_1 + 2$ divides $2n - k - 2$.
PROOF: Clear the nonzero denominator in ⟨2⟩1; m_1 is a natural number by Theorem 7.2. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2. □

Definition 7.5.

ASSUME:
1. D has finite point and block types, with the parameters and incidence axioms below.

A 2-design D consists of finite point and block types and an incidence relation, with parameters (v, k, b, r, λ): $v > 1$ points, $b > 0$ labelled blocks, $k > 1$ points in each block, r blocks at each point, and λ blocks at each pair of distinct points. Repeated blocks are allowed. In Lean the fields are `vkbrlam`. Its zero-one point-block matrix is $N = D$.incMatrix. The graph $I(D) = D$.incidenceGraph has the disjoint union of the point and block types as vertices and exactly the incidences as edges. The flag graph is $F(D) = D$.flagGraph $= L(I(D))$.

Lemma 7.6 (A distinguished constant eigendirection)**.**

ASSUME:
1. U is finite with $|U| = n > 0$, $G : U \times U \to \mathbb{R}$ is symmetric, and $G\mathbf{j} = b\mathbf{j}$.
2. Every eigenvalue of the restriction of G to $\mathbf{j}^\perp$ equals s; the restriction may have dimension zero.

PROVE:
$$G = sI + \frac{b - s}{n} J.$$

PROOF SKETCH: We will compare the two matrices on the orthonormal eigenbasis of Theorem 5.7, which starts with $\mathbf{j}/\sqrt{n}$.
⟨1⟩1. $(sI + \frac{b-s}{n} J)\mathbf{j} = b\mathbf{j} = G\mathbf{j}$.
PROOF: $J\mathbf{j} = n\mathbf{j}$ and $s + (b - s)n/n = b$; the last equality is ⟨0⟩:1. □
⟨1⟩2. $(sI + \frac{b-s}{n} J)\mathbf{v} = s\mathbf{v} = G\mathbf{v}$ for each vector $\mathbf{v}$ in $\mathbf{j}^\perp$ of the eigenbasis of G in Theorem 5.7.
PROOF: $J\mathbf{v} = 0$ by the coordinate-sum formula of Theorem 5.7; the eigenvalue of $\mathbf{v}$ is s by ⟨0⟩:2. □
⟨1⟩3. Q.E.D.
PROOF: By ⟨1⟩1 and ⟨1⟩2, the two matrices agree on the basis of Theorem 5.7, also when $\mathbf{j}^\perp = 0$; thus they are equal. □

Theorem 7.7 (Designs from two Gram eigenvalues)**.**

ASSUME:

1. For the incidence assertion N is a $v \times b$ zero-one matrix, $v > 1$, with positive constant row sum r, constant column sum $k > 1$, and $\lambda > 0$ satisfying $r(k-1) = \lambda(v-1)$.
2. Also $\operatorname{Spec}(NN^T) = [\![(rk)^1, (r-\lambda)^{v-1}]\!]$; the last bound assumes $r > \lambda$.

PROVE: If $\operatorname{Spec}(NN^T) = [\![(rk)^1, (r-\lambda)^{v-1}]\!]$, then

$$NN^T = (r-\lambda)I + \lambda J.$$

Thus any two distinct rows have inner product λ, and N is the incidence matrix of a 2-design with these parameters, with columns treated as block labels. If $r > \lambda$, then $b \geq v$.

PROOF SKETCH: We will compare actions on the two orthogonal summands and apply the identity to the incidence Gram matrix.

⟨1⟩1. $NN^T = (r-\lambda)I + \lambda J$, so N is the stated design incidence matrix with labelled blocks.

⟨2⟩1. $NN^T \mathbf{j}_v = rk\mathbf{j}_v$.
PROOF: Apply the constant column-sum equation and then the row-sum equation. □

⟨2⟩2. $rk - (r-\lambda) = \lambda v > 0$.
PROOF: Use $r(k-1) = \lambda(v-1)$ and collect terms. □

⟨2⟩3. Every eigenvalue of the restriction of NN^T to $\mathbf{j}_v^\perp$ is $r - \lambda$.
PROOF: By Theorem 5.7 for the symmetric matrix NN^T and ⟨2⟩1, $\operatorname{Spec}(NN^T) = [\![rk, \mu_1, \ldots, \mu_{v-1}]\!]$, where the μ_i are the eigenvalues on $\mathbf{j}_v^\perp$. Compare with ⟨0⟩:2: since $rk \neq r - \lambda$ by ⟨2⟩2, every μ_i is $r - \lambda$. □

⟨2⟩4. $NN^T = (r-\lambda)I + \lambda J$.
PROOF: Apply Lemma 7.6 to NN^T with $b = rk$, $s = r - \lambda$; ⟨2⟩3 gives the hypotheses and ⟨2⟩2 gives the coefficient λ. □

⟨2⟩5. Every pair of distinct rows has inner product λ.
PROOF: Take an off-diagonal entry of ⟨2⟩4; entries count common columns because N is zero-one. □

⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩4, ⟨2⟩5. □

⟨1⟩2. If $r > \lambda$, then $b \geq v$.

⟨2⟩1. NN^T has rank v.
PROOF: Both displayed eigenvalues are positive, so Theorem 3.2 counts v nonzero entries. □

⟨2⟩2. $v = \operatorname{rk} N \leq b$.
PROOF: Theorem 3.5 and the dimension of the column domain. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩2. □

⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2. □

Theorem 7.8 (Spectra of design incidence and flag graphs)**.**

ASSUME:

1. D is a design as in Definition 7.5, and $N = D.\mathrm{incMatrix}$ is its zero-one incidence matrix with $v > 1$ point rows and b labelled block columns; each column has $k > 1$ ones, each row r ones, and each distinct row pair has inner product $\lambda > 0$.
2. $r > \lambda$, $m = vr = bk$, $I(D)$ is the finite simple bipartite incidence graph, and $F(D) = L(I(D))$.

PROVE: Define

$$\gamma^{\pm} = \frac{r + k \pm \sqrt{(r-k)^2 + 4(r-\lambda)}}{2}.$$

Then $b \geq v$, $m - v - b + 1 \geq 0$, and

$$\mathrm{Spec}(I(D)) = \left[\!\left[\sqrt{rk}, -\sqrt{rk}, (\sqrt{r-\lambda})^{v-1}, (-\sqrt{r-\lambda})^{v-1}, 0^{b-v}\right]\!\right],$$

$$\mathrm{Spec}(F(D)) = \left[\!\left[r + k - 2, (\gamma^{+} - 2)^{v-1}, (\gamma^{-} - 2)^{v-1}, (k-2)^{b-v}, (-2)^{m-v-b+1}\right]\!\right].$$

Repeated displayed values are combined. Both graphs are connected, and the flag graph is $(r + k - 2)$-regular.

PROOF SKETCH: We will compute the design Gram spectrum, then the adjacency and signless block spectra, and finally the flag spectrum.

⟨1⟩1. $r(k-1) = \lambda(v-1)$, and $NN^T = (r-\lambda)I + \lambda J$ has spectrum $\left[\!\left[(rk)^1, (r-\lambda)^{v-1}\right]\!\right]$ and rank $v \leq b$.

⟨2⟩1. $vr = bk$.
PROOF: Count all incident point-block pairs by rows and by columns. □

⟨2⟩2. $r(k-1) = \lambda(v-1)$.
PROOF: Fix a point; count pairs of a block through it and another point of that block, first by blocks and then by other points. □

⟨2⟩3. $NN^T = (r-\lambda)I + \lambda J$.
PROOF: The diagonal counts r incident blocks; distinct-row entries count λ common blocks. □

⟨2⟩4. The two Gram eigenvalues are rk once and $r - \lambda$ with multiplicity $v - 1$.
PROOF: Corollary 5.9 applied to ⟨2⟩3, with $r - \lambda + \lambda v = rk$ from ⟨2⟩2. □

⟨2⟩5. $\mathrm{rk}\, N = v \leq b$.
PROOF: The eigenvalues in ⟨2⟩4 are positive; use Theorems 3.2 and 3.5. □

⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4, ⟨2⟩5. □

⟨1⟩2. The displayed spectrum of $I(D)$ holds.

⟨2⟩1. Its adjacency matrix is $\begin{pmatrix} 0 & N \\ N^T & 0 \end{pmatrix}$.
PROOF: Incidence joins the two disjoint vertex parts exactly where N has entry one. □

⟨2⟩2. The paired values and $b - v$ zeros are precisely the stated list.
PROOF: Apply Theorem 6.6 with $s = t = 0$ to ⟨1⟩1, ⟨2⟩1. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩2. □

⟨1⟩3. Both graphs are connected, and $F(D)$ is $(r + k - 2)$-regular.

⟨2⟩1. Any two distinct point vertices are joined by a path of length two.
PROOF: Their $\lambda > 0$ common blocks supply an intermediate vertex. □

⟨2⟩2. Every block is joined to a point.
PROOF: Its size is $k > 0$. □

⟨2⟩3. $I(D)$ is connected and has an edge.

PROOF: The paths in ⟨2⟩1, ⟨2⟩2 join all vertices; $v > 0, r > 0$ supplies an incidence. □

⟨2⟩4. The flag graph is connected and has degree $r + k - 2$.

PROOF: Apply both assertions of Lemma 6.3 to ⟨2⟩3; endpoints of an incidence have degrees r, k. □

⟨2⟩5. Q.E.D.

PROOF: Use ⟨2⟩3, ⟨2⟩4. □

⟨1⟩4. The displayed spectrum of $F(D)$ holds and $m - v - b + 1 \geq 0$.

⟨2⟩1. The signless matrix of $I(D)$ has scalar diagonal blocks rI_v, kI_b.

PROOF: These are the point and block degrees. □

⟨2⟩2. The Gram value rk gives the pair $0, r + k$; each $r - \lambda$ gives $\gamma^{\pm}$; the remaining $b - v$ values are k.

PROOF: Apply Theorem 6.6 to ⟨2⟩1 and ⟨1⟩1. □

⟨2⟩3. Both $\gamma^{\pm}$ are positive.

PROOF: They are real with positive sum $r + k$ and positive product $rk - r + \lambda = \lambda v$ by ⟨1⟩1. □

⟨2⟩4. $2I + A(F(D))$ has the preceding nonzero list and $m - v - b + 1$ zeros.

PROOF: Theorems 6.2 and 6.1, using the single zero in ⟨2⟩2, ⟨2⟩3. □

⟨2⟩5. The asserted list follows, and its final exponent is nonnegative.

PROOF: Subtract two from ⟨2⟩4 using Theorem 3.1; a kernel dimension is nonnegative. □

⟨2⟩6. Q.E.D.

PROOF: Use ⟨2⟩5. □

⟨1⟩5. Q.E.D.

PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4. □

Corollary 7.9 (The trivial design and complete bipartite graphs)**.**

ASSUME:

1. $p, q \in \mathbb{N}$ satisfy $p, q \geq 1$; $K_{p,q}$ is the finite simple complete bipartite graph.
2. The design consequence concerns a zero-one incidence matrix of a 2-design with $v > 1$ points, $b \geq 1$ labelled blocks, constant block size $k > 1$, and replication $r = \lambda > 0$.

PROVE:

$$\operatorname{Spec}(L(K_{p,q})) = \left[\!\!\left[p + q - 2, (p-2)^{q-1}, (q-2)^{p-1}, (-2)^{(p-1)(q-1)} \right]\!\!\right].$$

Values that coincide are combined. This covers the design case $r = \lambda > 0$, in which every point belongs to every block.

PROOF SKETCH: We will use the one-dimensional image of the all-ones biadjacency matrix.

⟨1⟩1. The biadjacency matrix has rank one and positive Gram spectrum $[\![pq]\!]$.

⟨2⟩1. The image of $B = J_{p \times q}$ is span($\mathbf{j}_p$) of dimension one.

PROOF: Every column equals the nonzero vector $\mathbf{j}_p$, and a column exists since $q \geq 1$. □

⟨2⟩2. $BB^T = qJ_p$ has the single positive eigenvalue pq.

PROOF: Apply Corollary 5.9 with $b = 0, t = q$. □

⟨2⟩3. Q.E.D.

PROOF: Use ⟨2⟩1, ⟨2⟩2. □

⟨1⟩2. The displayed line-graph multiset holds.

⟨2⟩1. The signless spectrum is $[\![0, p+q, q^{p-1}, p^{q-1}]\!]$.
PROOF: Apply Theorem 6.6 with $s = q, t = p$ and ⟨1⟩1. □
⟨2⟩2. $K_{p,q}$ is connected.
PROOF: Opposite-part vertices have an edge; same-part vertices have a two-edge path through the nonempty other part. □
⟨2⟩3. The edge incidence rank is $p+q-1$; its other Gram product has $(p-1)(q-1)$ zeros.
PROOF: Use Theorem 6.2 and the identity $pq - (p+q-1) = (p-1)(q-1)$. □
⟨2⟩4. The required list follows, including $p = 1$ or $q = 1$.
PROOF: Delete the one zero in ⟨2⟩1, add the zeros in ⟨2⟩3 by Theorem 6.1, and subtract two using Theorem 3.1. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩4. □
⟨1⟩3. The design consequence holds.

⟨2⟩1. $r(k-1) = r(v-1)$, so $k = v$.
PROOF: Fix a point and count pairs of a block through it and another point in that block: the counts are $r(k-1)$ and $\lambda(v-1)$. Substitute $\lambda = r$ and cancel $r > 0$. □
⟨2⟩2. Each block contains every point.
PROOF: A zero-one column on v rows with sum v has every entry one, by ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩2. □
⟨1⟩4. Q.E.D.
PROOF: Use ⟨1⟩2, ⟨1⟩3. □

Theorem 7.10 (The polynomial on the constant direction)**.**

ASSUME:
1. X is a finite connected simple k-regular graph on $n > 0$ vertices, with adjacency matrix A.
2. $f \in \mathbb{R}[t]$.

PROVE: If f vanishes at every eigenvalue different from k, then

$$f(A) = \frac{f(k)}{n} J.$$

Conversely this matrix identity implies that f vanishes at every eigenvalue on $\mathbf{j}^{\perp}$.

PROOF SKETCH: We will evaluate on the constant eigendirection and its orthogonal complement.
⟨1⟩1.
ASSUME: f vanishes at every eigenvalue different from k.
PROVE: $f(A) = f(k)J/n$.

⟨2⟩1. The k-eigenspace is exactly span($\mathbf{j}$).
PROOF: Theorem 5.3 makes it one-dimensional and the constant vector belongs to it. □
⟨2⟩2. Both matrices act by $f(k)$ on $\mathbf{j}$.
PROOF: Theorem 3.1 for $f(A)$ and $J\mathbf{j} = n\mathbf{j}$ for the other matrix. □
⟨2⟩3. Both matrices kill every eigenvector in $\mathbf{j}^{\perp}$.
PROOF: Its eigenvalue differs from k by ⟨2⟩1; use the vanishing hypothesis and $J\mathbf{x} = 0$. □
⟨2⟩4. Q.E.D.
PROOF: The two actions agree on the eigenbasis in Theorem 5.7. □

⟨1⟩2.
ASSUME: $f(A) = f(k)J/n$.
PROVE: f vanishes at every eigenvalue on $\mathbf{j}^{\perp}$.

⟨2⟩1. On a nonzero such eigenvector, $f(\mu)\mathbf{v} = 0$.
PROOF: Apply the assumed matrix identity and Theorem 3.1; $J\mathbf{v} = 0$. □
⟨2⟩2. $f(\mu) = 0$.
PROOF: A nonzero vector cannot be killed by a nonzero scalar; use ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩2. □
⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2. □

8. Smith graphs

Definition 8.1.

ASSUME:
1. The parameters satisfy $p, q, r > 0$, $\ell \geq 2$ for A, $\ell \geq 4$ for D, and $h \in \{6, 7, 8\}$ for E.

Write $T(p, q, r)$ for the tree consisting of one vertex of degree three and three internally disjoint arms of positive edge-lengths p, q, r. Define $\widetilde{\mathrm{A}}_\ell$ to be the cycle on $\ell + 1$ vertices, $\ell \geq 2$; $\widetilde{\mathrm{D}}_4 = K_{1,4}$; and, for $\ell \geq 5$, define $\widetilde{\mathrm{D}}_\ell$ by a path of length $\ell - 4$ with two new leaves at each endpoint. Define

$$\widetilde{\mathrm{E}}_6 = T(2, 2, 2), \quad \widetilde{\mathrm{E}}_7 = T(1, 3, 3), \quad \widetilde{\mathrm{E}}_8 = T(1, 2, 5).$$

These are the Smith graphs of [CGSS], Figure 1; the finite-type and affine-type names are kept distinct. A labelled realization uses a central vertex and arm vertices (a, j), $1 \leq j \leq h_a$, joined consecutively and to the centre at $j = 1$. For $\widetilde{\mathrm{D}}_\ell$ label the path consecutively and label each of its four added leaves separately; cycles use cyclic indices. Write `SimpleGraph.smithE`(h) for $h \in \{6, 7, 8\}$.

Theorem 8.2 (SM2: positive integer eigenvectors)**.**

ASSUME:
1. X is one of the finite simple Smith graphs just defined: $\widetilde{\mathrm{A}}_\ell$ with $\ell \geq 2$, $\widetilde{\mathrm{D}}_\ell$ with $\ell \geq 4$, or $T(2, 2, 2), T(1, 3, 3), T(1, 2, 5)$.
2. $A = A(X)$ and $N = |V(X)|$. The last assertion concerns any finite family in a real inner product space with Gram matrix $2I - A$.

PROVE: Every Smith graph has largest eigenvalue 2, of multiplicity one. For a cycle, a certificate vector is $\mathbf{j}$. For $\widetilde{\mathrm{D}}_\ell$, it has value 2 on the central path (or on the centre when $\ell = 4$) and value 1 on the four leaves. For $T(p, q, r)$ in the three exceptional cases, take central value $L = 3, 4, 6$, respectively; at distance j on an arm of length h take

$$\mathbf{x}(j) = \frac{L(h + 1 - j)}{h + 1} \qquad (1 \leq j \leq h).$$

For a Smith graph on N vertices, $2I - A$ is positive semidefinite of rank $N - 1$. Deleting any vertex from a Gram representation with this Gram matrix leaves a basis of its span.

PROOF SKETCH: We will check the neighbour sum in every vertex type, then use positivity and strict monotonicity.

⟨1⟩1. The proposed coordinates are positive integers and satisfy $A\mathbf{x} = 2\mathbf{x}$ on cycles and affine D graphs.

⟨2⟩1. At a cycle vertex the neighbour sum is $1 + 1 = 2$.
PROOF: Its two neighbours each have coordinate one. □

⟨2⟩2. At a D leaf the neighbour sum is $2 = 2 \cdot 1$.
PROOF: Its unique neighbour belongs to the central path or is the centre of D_4. □

⟨2⟩3. At an internal path vertex the sum is $2 + 2 = 4$.
PROOF: The two path neighbours have coordinate two. □

⟨2⟩4. At a nontrivial path endpoint the sum is $2 + 1 + 1 = 4$.
PROOF: Its neighbours are one path vertex and the two new leaves. □

⟨2⟩5. At the centre of D_4 the sum is $1 + 1 + 1 + 1 = 4$.
PROOF: It has precisely its four leaves as neighbours. □

⟨2⟩6. Q.E.D.
PROOF: The coordinates 1 and 2 are positive integers. Every vertex of a cycle is covered by ⟨2⟩1; every vertex of an affine D graph is a leaf, an internal path vertex, an endpoint of a nontrivial path, or the centre of $\widetilde{\mathrm{D}}_4$, covered by ⟨2⟩2, ⟨2⟩3, ⟨2⟩4, ⟨2⟩5. □

⟨1⟩2. The proposed exceptional coordinates are positive integers and satisfy $A\mathbf{x} = 2\mathbf{x}$.

⟨2⟩1. For $T(2, 2, 2)$, the centre has value 3 and each arm has values $2, 1$.
PROOF: Substitute $L = 3, h = 2$ in the stated formula. □

⟨2⟩2. For $T(1, 3, 3)$, the centre has value 4; the short arm has value 2 and each long arm has values $3, 2, 1$.
PROOF: Substitute $L = 4$ and $h = 1, 3$. □

⟨2⟩3. For $T(1, 2, 5)$, the centre has value 6 and the three arms have values 3; $4, 2$; and $5, 4, 3, 2, 1$.
PROOF: Substitute $L = 6$ and $h = 1, 2, 5$. □

⟨2⟩4. At an internal arm vertex, the two neighbour coordinates sum to twice its coordinate.
PROOF: The values $L(h + 1 - j)/(h + 1)$ form an arithmetic progression, including central value at $j = 0$. □

⟨2⟩5. At an arm leaf the single neighbour coordinate is twice the leaf coordinate.
PROOF: At $j = h$, the values at $h - 1, h$ are $2L/(h + 1), L/(h + 1)$. □

⟨2⟩6. The E_6 central neighbour sum is $2 + 2 + 2 = 6 = 2 \cdot 3$.
PROOF: Use ⟨2⟩1. □

⟨2⟩7. The E_7 central neighbour sum is $2 + 3 + 3 = 8 = 2 \cdot 4$.
PROOF: Use ⟨2⟩2. □

⟨2⟩8. The E_8 central neighbour sum is $3 + 4 + 5 = 12 = 2 \cdot 6$.
PROOF: Use ⟨2⟩3. □

⟨2⟩9. Q.E.D.
PROOF: The values in ⟨2⟩1, ⟨2⟩2, ⟨2⟩3 are positive integers. Every vertex is the centre, covered by ⟨2⟩6, ⟨2⟩7, ⟨2⟩8, an arm vertex other than a leaf, covered by ⟨2⟩4, or an arm leaf, covered by ⟨2⟩5. □

⟨1⟩3. $\lambda_{\max}(X) = 2$, its multiplicity is one, and $2I - A \succeq 0$ has rank $N - 1$.

⟨2⟩1. The largest eigenvalue is two.
PROOF: Apply Corollary 4.7 to ⟨1⟩1 or ⟨1⟩2, according to the model. □

⟨2⟩2. X is connected.

⟨3⟩1.
CASE: X is a cycle.
PROOF: Following cyclic indices reaches every vertex from any chosen vertex. □
⟨3⟩2.
CASE: X is an affine D graph.
PROOF: Each leaf is attached to the central path or centre, and that path is connected. □
⟨3⟩3.
CASE: X is one of the three exceptional trees.
PROOF: Every arm vertex has a path along its arm to the common centre. □
⟨3⟩4. Q.E.D.
PROOF: By ⟨0⟩:1 the three cases are exhaustive. □
⟨2⟩3. The eigenvalue two is simple.
PROOF: Theorem 4.5, using ⟨2⟩2. □
⟨2⟩4. The shifted matrix is positive semidefinite with exactly one zero eigenvalue.
PROOF: Theorems 3.3, 3.1, 3.2 and ⟨2⟩1, ⟨2⟩3. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩4. Deleting any representing vector leaves a basis of the original span.

⟨2⟩1. Deleting one vertex gives a nonempty proper subgraph of X.
PROOF: Every defined Smith graph has at least three vertices. □
⟨2⟩2. That graph has largest eigenvalue less than two.
PROOF: Theorem 4.8 and ⟨1⟩3. □
⟨2⟩3. The remaining Gram matrix is positive definite of order $N-1$.
PROOF: Its eigenvalues are $2-\mu>0$ by ⟨2⟩2 and Theorem 3.3. □
⟨2⟩4. The $N-1$ remaining vectors are independent and span the original space.
PROOF: Theorem 3.5 gives their span dimension $N-1$ from ⟨2⟩3; the original span also has dimension $N-1$ by ⟨1⟩3. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩4. □
⟨1⟩5. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4. □

Lemma 8.3 (Trees of maximum degree two)**.**

ASSUME:
1. X is a finite tree and each vertex has degree at most two.
PROVE: A finite tree whose vertices have degree at most two is a nonempty path.

PROOF SKETCH: We will take a longest simple path and show that no vertex lies outside it.
⟨1⟩1. PICK a simple path P of maximum length.
PROOF: Path supports are lists of distinct vertices of a finite set, so their lengths have a finite maximum; a tree is nonempty, so a singleton path exists. □
⟨1⟩2. A vertex outside P would have a path to P meeting P only at its last vertex.
PROOF: Use connectedness, and stop at the first vertex of P on a path from the outside vertex. □
⟨1⟩3. Attachment at an endpoint of P contradicts maximal length.
PROOF: Concatenate the nonempty outside segment with P; by ⟨1⟩2 no vertex repeats. □

⟨1⟩4. Attachment at an internal vertex of P contradicts degree at most two.
PROOF: The attachment edge and the two path edges at that vertex have distinct other endpoints. □
⟨1⟩5. X has no edge outside P.
PROOF: Both endpoints lie on P by ⟨1⟩2, ⟨1⟩3, ⟨1⟩4; an edge between nonconsecutive vertices of P would close a cycle. □
⟨1⟩6. Q.E.D.
PROOF: By ⟨1⟩3 and ⟨1⟩4, an attachment vertex is an endpoint or an internal vertex, and both are impossible; thus every vertex lies on P, and by ⟨1⟩5 X is the path P. □

Theorem 8.4 (SM3: Smith's theorem)**.**

ASSUME:
1. X is a finite connected simple graph; in particular its vertex set is nonempty.
2. The Smith graphs and $T(p, q, r)$ have the explicit meanings above; the paths in the conclusion are nonempty.

PROVE: A connected finite graph has largest eigenvalue 2 iff it is isomorphic to a Smith graph. It has largest eigenvalue less than 2 iff it is a path, a tree $T(1, 1, r)$ for $r \geq 1$, or one of $T(1, 2, 2), T(1, 2, 3), T(1, 2, 4)$.

PROOF SKETCH: We will first classify graphs that contain no Smith subgraph and then use strict monotonicity to distinguish equality from a strict bound.
⟨1⟩1.
ASSUME: X contains a Smith graph Z as a subgraph, through an edge-preserving vertex injection.
PROVE: $\lambda_{\max}(X) = 2$ if the injection is onto vertices and edges, and $\lambda_{\max}(X) > 2$ otherwise.

⟨2⟩1.
CASE: The injection is onto vertices and edges.
PROOF: It is then an isomorphism $Z \cong X$. By Proposition 2.6, X and Z have the same spectrum, and thus the same largest eigenvalue by Lemma 2.4; Theorem 8.2 gives $\lambda_{\max}(Z) = 2$. □
⟨2⟩2.
CASE: The injection omits a vertex or an edge.
PROOF: Z is a nonempty proper subgraph of the connected graph X; Theorem 4.8 and Theorem 8.2 give $2 = \lambda_{\max}(Z) < \lambda_{\max}(X)$. □
⟨2⟩3. Q.E.D.
PROOF: The two cases are exhaustive. □
⟨1⟩2.
ASSUME: X contains no Smith subgraph.
PROVE: X is a tree of maximum degree at most three, with at most one degree-three vertex.

⟨2⟩1. X has no cycle.
PROOF: A simple cycle has length at least three and is a Smith graph $\widetilde{\mathrm{A}}_\ell$. □
⟨2⟩2. There is a unique simple path between every two vertices.
PROOF: Connectedness and acyclicity in ⟨2⟩1; apply Facts 18. □
⟨2⟩3. No vertex has four distinct neighbours.
PROOF: That vertex and four incident edges would give the Smith subgraph $K_{1,4}$. □

⟨2⟩4. Two degree-three vertices would give a Smith subgraph.

⟨3⟩1. Pick two degree-three vertices at minimum distance; internal vertices of their path have degree two.
Proof: An internal degree-three vertex would form a shorter pair; internal path vertices already have two neighbours. □
⟨3⟩2. The two off-path neighbours at each endpoint are outside the path and distinct from all three other choices.
Proof: A repeated vertex or return to the path creates two distinct simple paths, contradicting ⟨2⟩2. □
⟨3⟩3. The connecting path and four extra edges form $\widetilde{\mathrm{D}}_{d+4}$, where $d \geq 1$ is the path length.
Proof: The explicit definition in 8.1 has that central path and four leaves; ⟨3⟩2 makes the vertex map injective. □
⟨3⟩4. Q.E.D.
Proof: The subgraph in ⟨3⟩3 contradicts the assumption of ⟨1⟩2. □
⟨2⟩5. Q.E.D.
Proof: Use ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩3.
Assume: X contains no Smith subgraph.
Prove: X is a path or $T(p, q, r)$ with $1 \leq p \leq q \leq r$.

⟨2⟩1. A tree with exactly one degree-three vertex consists of three nonempty path arms.

⟨3⟩1. Deleting the central vertex gives exactly three components, one at each neighbour.
Proof: Unique paths in ⟨1⟩2 put distinct neighbours in distinct components, and every remaining vertex reaches the centre through one of them. □
⟨3⟩2. Each component has maximum degree at most two, and its attachment vertex has degree at most one within the component.
Proof: No other vertex has degree three by ⟨1⟩2; the attachment loses its edge to the centre. □
⟨3⟩3. Each component is a path attached at an endpoint.
Proof: Each component is connected and acyclic as a subtree. Apply Lemma 8.3, using ⟨3⟩2; its attachment cannot be internal because its component degree is at most one. □
⟨3⟩4. Q.E.D.
Proof: The three components give positive arm lengths; reorder them to obtain $p \leq q \leq r$. □

⟨2⟩2.
Case: X has no vertex of degree three.
Proof: By ⟨1⟩2, X is a tree of maximum degree at most two; apply Lemma 8.3. □

⟨2⟩3.
Case: X has a vertex of degree three.
Proof: It is the only one, by ⟨1⟩2; apply ⟨2⟩1. □

⟨2⟩4. Q.E.D.
Proof: The two cases are exhaustive, and by ⟨1⟩2 no vertex has degree greater than three. □

⟨1⟩4.
Assume: $X = T(p, q, r)$, $1 \leq p \leq q \leq r$, and X contains no Smith subgraph.
Prove: The only possibilities are $(1, 1, r)$ for $r \geq 1$ and $(1, 2, r)$ for $2 \leq r \leq 4$.

⟨2⟩1. $p = 1$.

PROOF: If $p \geq 2$, truncate each arm to length two to obtain $T(2,2,2)$. □

⟨2⟩2. $q = 1$ or $q = 2$.
PROOF: If $q \geq 3$, ⟨2⟩1 and $r \geq q$ allow truncation to $T(1,3,3)$. □

⟨2⟩3. If $q = 2$, then $2 \leq r \leq 4$.
PROOF: Ordering gives $r \geq 2$; $r \geq 5$ permits truncation to $T(1,2,5)$. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □

⟨1⟩5. Every graph on the finite-type list is a proper subgraph of a Smith graph, and has largest eigenvalue less than two.

⟨2⟩1. A path on $N \geq 1$ vertices embeds properly in a cycle on $N + 2$ vertices.
PROOF: Use N consecutive cycle vertices with their $N - 1$ path edges; at least two vertices are omitted. □

⟨2⟩2. $T(1,1,r)$ embeds properly in $\widetilde{\mathrm{D}}_{r+4}$.
PROOF: Adjoin two new leaves at the far endpoint of its length-r arm; that arm becomes the central path. □

⟨2⟩3. $T(1,2,2)$ embeds properly in $T(1,2,5)$.
PROOF: Keep the first two vertices of the length-five arm and omit the last three. □

⟨2⟩4. $T(1,2,3)$ embeds properly in $T(1,2,5)$.
PROOF: Keep the first three vertices of the length-five arm and omit the last two. □

⟨2⟩5. $T(1,2,4)$ embeds properly in $T(1,2,5)$.
PROOF: Keep the first four vertices of the length-five arm and omit the last one. □

⟨2⟩6. Each listed graph has largest eigenvalue less than two.
PROOF: Apply Theorems 4.8, 8.2 to the five explicit embeddings. □

⟨2⟩7. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4, ⟨2⟩5, ⟨2⟩6. □

⟨1⟩6.
ASSUME: $\lambda_{\max}(X) = 2$.
PROVE: X is isomorphic to a Smith graph.

⟨2⟩1.
CASE: X contains a Smith subgraph.
PROOF: By ⟨1⟩1 and ⟨1⟩, the injection is onto vertices and edges, so it is an isomorphism. □

⟨2⟩2.
CASE: X contains no Smith subgraph.
PROOF: By ⟨1⟩3 and ⟨1⟩4, X is on the finite-type list, so $\lambda_{\max}(X) < 2$ by ⟨1⟩5; this contradicts ⟨1⟩. □

⟨2⟩3. Q.E.D.
PROOF: The two cases are exhaustive. □

⟨1⟩7.
ASSUME: X is isomorphic to a Smith graph.
PROVE: $\lambda_{\max}(X) = 2$.
PROOF: Use Theorem 8.2, Proposition 2.6 and Lemma 2.4. □

⟨1⟩8.
ASSUME: $\lambda_{\max}(X) < 2$.
PROVE: X is on the finite-type list.

⟨2⟩1. X contains no Smith subgraph.
PROOF: Both cases of ⟨1⟩1 give $\lambda_{\max}(X) \geq 2$, contrary to ⟨1⟩. □

⟨2⟩2. X lies on the stated list.
PROOF: Apply ⟨1⟩3 and ⟨1⟩4 with ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩2. □
⟨1⟩9.
ASSUME: X is on the finite-type list.
PROVE: $\lambda_{\max}(X) < 2$.
PROOF: This is ⟨1⟩5. □
⟨1⟩10. Q.E.D.
PROOF: Use ⟨1⟩6, ⟨1⟩7, ⟨1⟩8 and ⟨1⟩9. □

9. Exact certificates and rational transfer

Lemma 9.1 (Moments on an annihilating support)**.**

ASSUME:
1. A is real symmetric of finite order; $(\theta_i)_{i<s}$ is injective, $s \geq 0$, and $\prod_{i<s}(A - \theta_i I) = 0$.

PROVE: every eigenvalue belongs to the nodes and

$$\operatorname{tr} A^j = \sum_{i<s} \operatorname{mult}_A(\theta_i)\theta_i^j \qquad (j \geq 0).$$

PROOF SKETCH: We will evaluate the annihilator on eigenvectors, and then group the trace-power sum of Theorem 3.2 by nodes.
⟨1⟩1. Every eigenvalue μ satisfies $\prod_{i<s}(\mu - \theta_i) = 0$.
PROOF: Apply Theorem 3.1 to the given annihilator. □
⟨1⟩2. Each eigenvalue equals a node.
PROOF: A product of real numbers vanishes only if a factor vanishes; use ⟨1⟩1. □
⟨1⟩3. Each eigenvalue occurrence has exactly one node index.
PROOF: Existence is ⟨1⟩2; injectivity of the node function gives uniqueness. □
⟨1⟩4. $\operatorname{tr} A^j = \sum_{i<s} \operatorname{mult}_A(\theta_i)\theta_i^j$ for every $j \geq 0$.
PROOF: Group the trace-power sum of Theorem 3.2 by the unique index in ⟨1⟩3. □
⟨1⟩5. If $s = 0$, the matrix has order zero.
PROOF: The annihilator is $I = 0$; any diagonal index would imply $1 = 0$. Thus the spectrum and all traces are empty sums. □
⟨1⟩6. Q.E.D.
PROOF: Use ⟨1⟩2, ⟨1⟩4, ⟨1⟩5. □

Theorem 9.2 (Moments determine multiplicities on a known support)**.**

ASSUME:
1. A is real symmetric of order n; the nodes $(\theta_i)_{i<s}$ are distinct with $s \geq 0$, and $m_i \in \mathbb{N}$.
2. $\prod_{i<s}(A - \theta_i I) = 0$ and $\operatorname{tr} A^j = \sum_{i<s} m_i \theta_i^j$ for $0 \leq j < s$.

PROVE: $\operatorname{Spec}(A) = [\![\theta_0^{m_0}, \ldots, \theta_{s-1}^{m_{s-1}}]\!]$. In particular $\sum_i m_i = n$, including the empty node list.

PROOF SKETCH: We will compare the assumed moments with the true moments of Lemma 9.1, and conclude by Vandermonde uniqueness as in `LineGraphs/Spectral/Basic.lean`.

⟨1⟩1. The spectrum is supported on the nodes, and its moments have coefficients $d_i = \mathrm{mult}_A(\theta_i)$.
PROOF: Apply Lemma 9.1 to A. □
⟨1⟩2. $\sum_{i<s}(d_i - m_i)\theta_i^j = 0$ for every $j < s$.
PROOF: Subtract the assumed moments from those in ⟨1⟩1. □
⟨1⟩3. The real vector $(d_i - m_i)_{i<s}$ is zero.
PROOF: Apply `Matrix.eq_zero_of_forall_pow_sum_mul_pow_eq_zero` in Facts 16 to the injective nodes and ⟨1⟩2; its finite index size may be zero. □
⟨1⟩4. $d_i = m_i$ as natural numbers for every $i < s$.
PROOF: Evaluate ⟨1⟩3 at i and use injectivity of the natural-number cast into $\mathbb{R}$. □
⟨1⟩5. $\mathrm{Spec}(A) = \llbracket \theta_i^{m_i} : i < s \rrbracket$.
PROOF: Use ⟨1⟩1, ⟨1⟩4 and multiset extensionality in Facts 5; outside the nodes both counts are zero. □
⟨1⟩6. $\sum_i m_i = n$, even when $s = 0$.
PROOF: Take cardinalities in ⟨1⟩5 and apply Theorem 2.2. □
⟨1⟩7. Q.E.D.
PROOF: Use ⟨1⟩5, ⟨1⟩6. □

Corollary 9.3 (Two matrices on a common annihilating support)**.**

ASSUME:

1. A, B are real symmetric on finite index sets, annihilate the product on the same distinct nodes $(\theta_i)_{i<s}$ with $s \geq 0$, and have equal traces for $j < s$.

PROVE: $\mathrm{Spec}(A) = \mathrm{Spec}(B)$ and $\chi_A = \chi_B$.

PROOF SKETCH: We will read the moments of B from Lemma 9.1 and certify A with them by Theorem 9.2.
⟨1⟩1. $\mathrm{tr}\, B^j = \sum_i \mathrm{mult}_B(\theta_i)\theta_i^j$.
PROOF: Apply Lemma 9.1 to B. □
⟨1⟩2. A has these moments for $j < s$.
PROOF: Compose the assumed trace equalities with ⟨1⟩1. □
⟨1⟩3. $\mathrm{Spec}(A) = \left[\!\left[\theta_i^{\mathrm{mult}_B(\theta_i)} : i < s \right]\!\right]$.
PROOF: Apply Theorem 9.2 to A, choosing the natural multiplicities of B and using ⟨1⟩2. □
⟨1⟩4. $\mathrm{Spec}(B)$ is the same multiset.
PROOF: The support assertion of Lemma 9.1 for B identifies all counts, both on and off the nodes. □
⟨1⟩5. $\chi_A = \chi_B$.
PROOF: Apply Proposition 2.6 to ⟨1⟩3 and ⟨1⟩4. □
⟨1⟩6. Q.E.D.
PROOF: Use ⟨1⟩3, ⟨1⟩4, ⟨1⟩5. □

Theorem 9.4 (Similarity and intertwining certificates)**.**

ASSUME:

1. U is finite; $A, B, Q : U \times U \to F$ are matrices over a field F, $\det Q \neq 0$, and $QA = BQ$.
2. The involution variant assumes $Q^2 = I$ and $QAQ = B$ over that field.
3. The integer variant assumes A, B, Q, R are integer matrices of the same finite order, $d \in \mathbb{Z} \setminus \{0\}$, $QA = BQ$ and $QR = RQ = dI$.

PROVE: $\chi_A = \chi_B$. In particular $Q^2 = I$ and $QAQ = B$ suffice. For integer matrices A, B, Q, R and an integer $d \neq 0$, it is enough to check

$$QA = BQ, \qquad QR = RQ = dI.$$

These identities imply equality of the real characteristic polynomials of A and B. For adjacency matrices, they prove graph cospectrality.

PROOF SKETCH: We will conjugate by an invertible matrix and then construct the rational inverse explicitly for the integer certificate.

⟨1⟩1. $\chi_A = \chi_B$ over F.

⟨2⟩1. Q has an inverse and $B = QAQ^{-1}$.
PROOF: The determinant-unit criterion in Facts 4 and right multiplication of $QA = BQ$ by Q^{-1}. □

⟨2⟩2. $\det(tI - B) = \det(tI - A)$.
PROOF: Apply `Matrix.charpoly_units_conj` in Facts 9 to the unit with value Q and inverse Q^{-1} from ⟨2⟩1. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩2. □

⟨1⟩2. $Q^2 = I$ and $QAQ = B$ suffice.

⟨2⟩1. $Q^{-1} = Q$.
PROOF: The equation $Q^2 = I$ gives a two-sided inverse. □

⟨2⟩2. $\chi_A = \chi_B$.
PROOF: Apply `Matrix.charpoly_units_conj` in Facts 9 directly to the unit with value and inverse Q, using $QAQ = B$. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □

⟨1⟩3. The integer certificate implies equality of real characteristic polynomials.

⟨2⟩1. Over $\mathbb{Q}$, $Q((1/d)R) = ((1/d)R)Q = I$.
PROOF: Cast the two identities into $\mathbb{Q}$, distribute $1/d$, and cancel $d \neq 0$. □

⟨2⟩2. The rational characteristic polynomials are equal.
PROOF: Apply `Matrix.charpoly_units_conj` in Facts 9 over $\mathbb{Q}$ to the unit from ⟨2⟩1; $QA = BQ$ gives its conjugation equation. □

⟨2⟩3. The equality persists over $\mathbb{R}$.
PROOF: Apply `Matrix.charpoly_map` in Facts 9 to the inclusion $\mathbb{Q} \hookrightarrow \mathbb{R}$. □

⟨2⟩4. For adjacency matrices this is graph cospectrality.
PROOF: Use Definition 2.5. □

⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩3, ⟨2⟩4. □

⟨1⟩4. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3. □

Theorem 9.5 (Seidel switching)**.**

ASSUME:

1. X is a finite simple graph on a finite vertex set V, and $U \subseteq V$.

2. X^U complements adjacency across $U, V \setminus U$ and retains adjacency within each part; T is diagonal with value -1 on U and 1 outside.

PROVE:

$$S(X^U) = TS(X)T, \qquad T^2 = I.$$

Thus Seidel switching preserves the Seidel spectrum with multiplicity. The matrix used in [CGSS] Corollary 5.10 is $-S = 2A - J + I$, and its spectrum is obtained from that of S by negating every entry.

PROOF SKETCH: We will check the conjugation entry by entry and apply the characteristic-polynomial certificate.

⟨1⟩1. $S(X^U) = TS(X)T$ and $T^2 = I$.

⟨2⟩1.

ASSUME: $u, v \in V$.
PROVE: $S(X^U)(u,v) = (TS(X)T)(u,v) = T(u,u)S(X)(u,v)T(v,v)$.

⟨3⟩1.

CASE: $u = v$.
PROOF: Both entries are zero, since Seidel diagonal entries are zero by Definition 2.5. □

⟨3⟩2.

CASE: $u \neq v$ lie in the same part.
PROOF: The sign product is 1 and adjacency is unchanged by switching. □

⟨3⟩3.

CASE: u, v lie in different parts and uv is an edge of X.
PROOF: Switching removes the edge, so the entry -1 becomes 1; the sign product in the conjugation is -1. □

⟨3⟩4.

CASE: u, v lie in different parts and uv is not an edge of X.
PROOF: Switching inserts the edge, so the entry 1 becomes -1; the sign product is again -1. □

⟨3⟩5. Q.E.D.
PROOF: The four cases are exhaustive. □

⟨2⟩2. $T^2 = I$.
PROOF: The off-diagonal entries vanish and each diagonal sign squares to one. □

⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □

⟨1⟩2. Seidel spectra with multiplicities are preserved, and $-S$ has the negated spectrum.

⟨2⟩1. The two Seidel characteristic polynomials agree.
PROOF: Apply Theorem 9.4 to ⟨1⟩1. □

⟨2⟩2. The two Seidel multisets agree.
PROOF: The matrices are real symmetric; use Proposition 2.6. □

⟨2⟩3. $-S = 2A - J + I$ and its eigenvalues are those of S negated, with repetitions.
PROOF: Negate Definition 2.5 and apply Theorem 3.1 to $-t$. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □

⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2. □

Theorem 9.6 (Rational eigenvalues of integer matrices)**.**

ASSUME:
1. U is finite, $A : U \times U \to \mathbb{Z}$, $b \in \mathbb{Q}$, and a nonzero real vector satisfies $A\mathbf{v} = b\mathbf{v}$ after casting A, b to $\mathbb{R}$.

PROVE: If an integer square matrix has a rational eigenvalue b, viewed as an eigenvalue of its real matrix, then b is an integer.

PROOF SKETCH: We will place the rational root in the monic integer characteristic polynomial and clear its reduced denominator.

⟨1⟩1. The integer characteristic polynomial f is monic of degree $N > 0$ and $f(b) = 0$ over $\mathbb{Q}$.

⟨2⟩1. The real determinant of $bI - A$ is zero.
PROOF: The given nonzero eigenvector lies in its kernel; the determinant criterion in Facts 4 excludes invertibility. □

⟨2⟩2. $f(b) = 0$ in $\mathbb{Q}$.
PROOF: Use `Matrix.charpoly_map` in Facts 9 and injectivity of $\mathbb{Q} \hookrightarrow \mathbb{R}$ on ⟨2⟩1. □

⟨2⟩3. f is monic and its degree N is positive.
PROOF: Characteristic polynomials are monic (Facts 9); a degree-zero monic polynomial is 1 and cannot vanish at b. □

⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3. □

⟨1⟩2. The rational root b is an integer.
PROOF: Apply `isInteger_of_is_root_of_monic` in Facts 19 to the integer characteristic polynomial and the root equation in ⟨1⟩1, with UFD $\mathbb{Z}$ and fraction field $\mathbb{Q}$. □

⟨1⟩3. Q.E.D.
PROOF: Use ⟨1⟩2. □

Lemma 9.7 (Adjacency matrix guards)**.**

ASSUME:
1. U is finite and $A : U \times U \to \mathbb{Z}$ is symmetric, with zero diagonal and entries in $\{0, 1\}$.

PROVE: The relation $A(u, v) = 1$ defines a simple graph whose integer adjacency matrix is A.

PROOF SKETCH: We will recognize the three guards as the fields of Mathlib's `Matrix.IsAdjMatrix`, whose graph has adjacency $A(u, v) = 1$.

⟨1⟩1. A satisfies `Matrix.IsAdjMatrix`.
PROOF: Its three fields, zero-one entries, symmetry and zero diagonal, are the three guards of ⟨0⟩:1 (Facts 10). □

⟨1⟩2. The relation $A(u, v) = 1$ is a simple graph X_A.
PROOF: Use `Matrix.IsAdjMatrix.toGraph` of ⟨1⟩1 in Facts 10: symmetry of A makes the relation symmetric, and the zero diagonal makes it irreflexive. □

⟨1⟩3. $A(X_A) = A$.
PROOF: Apply `Matrix.IsAdjMatrix.adjMatrix_toGraph_eq` in Facts 10 to ⟨1⟩1; on each off-diagonal pair, the zero-one guard makes the test for equality with 1 return the entry itself. □

⟨1⟩4. Q.E.D.
PROOF: Use ⟨1⟩2, ⟨1⟩3. □

Theorem 9.8 (A precise Boolean certificate interface)**.**

ASSUME:

1. Inputs are finite lists of integer rows, integer nodes, natural multiplicities, integer vectors or integer scalars as appropriate; all list lengths and index bounds are checked before arithmetic access.
2. The three Boolean procedures and recursive operations are specified below; every guard is combined by Boolean conjunction.

PROVE: Encode an integer $n \times n$ matrix by a list of n rows of length n. Consider the following three checkers, using exact integers and natural numbers, with no floating-point operations.

(1) A spectral checker first verifies the square shape and symmetry, then a possibly empty list of distinct integer nodes θ_i and an equally long list of natural multiplicities m_i. It verifies the matrix product $\prod_i (A - \theta_i I) = 0$ and the trace identities of Theorem 9.2 for $0 \le j < s$.

(2) A largest-eigenvalue checker verifies square shape, $n > 0$, symmetry, nonnegative entries, a length-n integer vector with strictly positive entries, and $A\mathbf{x} = b\mathbf{x}$ for the proposed integer b.

(3) A similarity checker verifies that A, B, Q, R have the same square shape, that $d \neq 0$, and the three identities in Theorem 9.4.

Returning `true` implies, respectively, the stated real multiset spectrum, the stated largest eigenvalue, or equality of real characteristic polynomials. Graph certificates additionally require zero diagonal, zero-one entries, and symmetry for every proposed adjacency matrix. In the similarity checker these guards apply to both A and B.

PROOF SKETCH: We will prove the list operations agree with matrix arithmetic, maintain the product and power invariants, cast to real matrices, and apply the three soundness theorems.
⟨1⟩1. The recursive list dot product computes the finite coordinate sum.

⟨2⟩1. Define $\mathrm{dot}([],[]) = 0$ and $\mathrm{dot}(h :: v, k :: w) = hk + \mathrm{dot}(v, w)$ for equally long lists.
PROOF: These equations are the specification of the exact integer operation. □
⟨2⟩2. For length zero, $\mathrm{dot}([],[]) = \sum_{i \in \mathrm{Fin}(0)} v(i)w(i)$.
PROOF: Both sides are zero by ⟨2⟩1 and the empty-sum definition. □
⟨2⟩3. If the equality holds on the tails, it holds on the full lists.
PROOF: Both sides split into the same first term hk and the tail sum in ⟨2⟩1. □
⟨2⟩4. Q.E.D.
PROOF: Induction on the common list length using ⟨2⟩2, ⟨2⟩3. □
⟨1⟩2. The guarded operations compute the exact matrices and Boolean predicates.

⟨2⟩1. A true shape guard defines a matrix on $\mathrm{Fin}(n)$ by row and column lookup.
PROOF: Each of the n rows has length n, so every bounded pair of indices has an integer entry. □
⟨2⟩2. Computing each entry as a row-column dot product gives matrix multiplication.
PROOF: Apply ⟨1⟩1 to each row and column; its sum is the defining matrix product. □
⟨2⟩3. Entrywise addition and scalar multiplication, the Kronecker-delta identity, and the diagonal sum give the named matrix operations.
PROOF: These are their coordinate definitions. □
⟨2⟩4. List equality and a universal list test reflect equality and universal quantification.

⟨3⟩1. The empty conjunction is true and the empty equal-length lists agree.
PROOF: These are the base equations for the Boolean tests. □
⟨3⟩2. A nonempty conjunction is true iff its first test and tail conjunction are true.
PROOF: The Boolean truth table for conjunction. □
⟨3⟩3. An equality test is true iff lengths agree and all corresponding integers agree.
PROOF: After the length guard, induct using ⟨3⟩1, ⟨3⟩2 and exact integer equality. □
⟨3⟩4. Q.E.D.
PROOF: List induction proves the universal test; ⟨3⟩3 proves the equality test. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩3. The product accumulator and power accumulator compute exactly the annihilator and all required powers.

⟨2⟩1. The initial product accumulator I is the empty product.
PROOF: This is the definition of a product over an empty prefix. □
⟨2⟩2. Multiplying by the next $A - \theta_i I$ appends that factor to the prefix product.
PROOF: Use the product recursion and exact multiplication from ⟨1⟩2. □
⟨2⟩3. $P_0 = I = A^0$.
PROOF: Definition of the initial accumulator and of zeroth power. □
⟨2⟩4. If $P_j = A^j$, then $P_{j+1} = P_j A = A^{j+1}$.
PROOF: Substitute the induction hypothesis in the specified recurrence. □
⟨2⟩5. The trace tests are precisely the required moment equations, including degree zero.
PROOF: Induct with ⟨2⟩3, ⟨2⟩4 and use the diagonal sum in ⟨1⟩2. □
⟨2⟩6. Q.E.D.
PROOF: Induction on the processed nodes with ⟨2⟩1 and ⟨2⟩2 gives the annihilator; ⟨2⟩5 gives the moments. □
⟨1⟩4. True checks give all real hypotheses of the three certificate theorems.

⟨2⟩1. Integer matrix equations and moments remain valid over $\mathbb{R}$.
PROOF: The inclusion $\mathbb{Z} \hookrightarrow \mathbb{R}$ preserves each finite sum and product defining these operations. □
⟨2⟩2. Nonnegative entries, positive coordinates, distinct nodes and a nonzero scalar remain so after casting.
PROOF: The integer inclusion is order preserving and injective. □
⟨2⟩3. The spectral checker returning true gives the stated multiset.
PROOF: Apply Theorem 9.2 to ⟨1⟩2, ⟨1⟩3, ⟨2⟩1 and ⟨2⟩2. □
⟨2⟩4. The positive-vector checker returning true gives the stated largest eigenvalue.
PROOF: Apply Corollary 4.7 to the reflected shape, symmetry, nonnegativity, positivity and eigenvector checks. □
⟨2⟩5. The similarity checker returning true gives equal real characteristic polynomials.
PROOF: Apply Theorem 9.4 to the reflected three identities and $d \neq 0$. □
⟨2⟩6. Q.E.D.
PROOF: Use ⟨2⟩3, ⟨2⟩4, ⟨2⟩5. □
⟨1⟩5. With the symmetry, zero-diagonal and zero-one guards on every proposed adjacency matrix, the checkers prove the corresponding graph statements; without the symmetry guard they need not.

⟨2⟩1. Zero diagonal and zero-one entries alone do not ensure an undirected graph in the similarity checker.

PROOF: Take $A = B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $Q = R = I$, $d = 1$: all similarity identities and both entry guards hold, but $A(1,2) \neq A(2,1)$. □

⟨2⟩2. For the spectral and positive-vector checkers, the guards on A give a graph with the stated spectrum or largest eigenvalue.

PROOF: Apply Lemma 9.7 to A; the real adjacency matrix of the resulting graph is the cast of A, so ⟨1⟩4 applies to it; use Definition 2.5. □

⟨2⟩3. For the similarity checker, the guards on both A and B give cospectral graphs.

PROOF: Apply Lemma 9.7 separately to A and B; the real adjacency matrices of the two graphs are the casts of A and B, so ⟨1⟩4 gives equal characteristic polynomials, which is cospectrality by Definition 2.5. □

⟨2⟩4. Q.E.D.

PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3. □

⟨1⟩6. Q.E.D.

PROOF: Use ⟨1⟩4, ⟨1⟩5. □

The concrete C13, C15 and C17 data are those of the ordinary version. The sign convention is $S = J - I - 2A$, as in SW2 of `cgss/prerequisites.md`. For C13 the adjacency nodes are $4, 2, 0, -2$, with multiplicities $1, 3, 3, 5$ and moments $12, 0, 48, 48$. Its Seidel nodes are $-5, -1, 3$, with multiplicities $3, 3, 6$ and moments $12, 0, 132$; negating the Seidel matrix negates these nodes. For C15 the nodes are $6, 4, 1, 0, -2$, with multiplicities $1, 5, 4, 5, 15$ and moments $30, 0, 180, 420, 2820$. C17 uses the positive integer vectors in Theorem 8.2. The checks in `checks/check_spectraltools.py` verify these finite identities using exact integers. These data are not themselves a Lean proof.

10. Further spectral applications

Theorem 10.1 (Scaled idempotents and singular principal minors)**.**

ASSUME:
1. U is finite of order n, $C : U \times U \to \mathbb{R}$ is symmetric, $b > 0$ and $C^2 = bC$.
2. For the rank bound Q is finite, $B : U \times Q \to \mathbb{R}$, and $CB = 0$. A principal submatrix is indexed by a subset of U.

PROVE: $C \succeq 0$, its eigenvalues lie in $\{0, b\}$, and $\operatorname{rk} C = \operatorname{tr} C/b$. If B is an $n \times q$ matrix with $CB = 0$, then $\operatorname{rk} C \leq n - \operatorname{rk} B$. Every principal submatrix of order greater than $\operatorname{rk} C$ is singular.

PROOF SKETCH: We will read the two possible eigenvalues, compare image with kernel, and bound the rank of each coordinate restriction.

⟨1⟩1. $C \succeq 0$ and $\operatorname{rk} C = \operatorname{tr} C/b$.

⟨2⟩1. Every eigenvalue μ satisfies $\mu(\mu - b) = 0$.

PROOF: Apply Theorem 3.1 to $C^2 - bC = 0$. □

⟨2⟩2. Every eigenvalue is 0 or $b > 0$, so $C \succeq 0$.

PROOF: Factor cancellation over $\mathbb{R}$ in ⟨2⟩1 and Theorem 3.3. □

⟨2⟩3. $\operatorname{tr} C = b \operatorname{rk} C$.

PROOF: Theorem 3.2 sums the b entries and counts precisely those entries as nonzero. □

⟨2⟩4. $\operatorname{rk} C = \operatorname{tr} C/b$.
PROOF: Divide ⟨2⟩3 by $b > 0$. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩4. □
⟨1⟩2. $\operatorname{rk} C \leq n - \operatorname{rk} B$.

⟨2⟩1. $\operatorname{im} B \subseteq \ker C$.
PROOF: For every $\mathbf{x}$, $C(B\mathbf{x}) = (CB)\mathbf{x} = 0$. □
⟨2⟩2. $\operatorname{rk} B \leq \dim \ker C = n - \operatorname{rk} C$.
PROOF: Dimension monotonicity and rank-nullity in Facts 3. □
⟨2⟩3. Q.E.D.
PROOF: Rearrange the dimension inequality in ⟨2⟩2 over the integers. □
⟨1⟩3. Every principal submatrix of order greater than $\operatorname{rk} C$ is singular.

⟨2⟩1. A principal submatrix factors as coordinate inclusion, C, and coordinate projection.
PROOF: Its entry at retained indices is the corresponding entry of C. □
⟨2⟩2. Its rank is at most $\operatorname{rk} C$.
PROOF: The image in ⟨2⟩1 lies in the projection of $\operatorname{im} C$, whose dimension cannot exceed $\dim \operatorname{im} C$ (Facts 3). □
⟨2⟩3. If its order exceeds that rank, its determinant is zero.
PROOF: Rank-nullity gives a nonzero kernel; use the determinant criterion in Facts 4. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩3. □
⟨1⟩4. Q.E.D.
PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3. □

Theorem 10.2 (Symmetric matrices and rank-one Gram matrices)**.**

ASSUME:
1. $\ell \in \mathbb{N}$ and the matrix space consists of all real symmetric $\ell \times \ell$ matrices.
2. The traceless dimension assertion assumes $\ell > 0$. For the Gram identity $(\mathbf{v}_u)_{u \in U}$ is a finite family of arbitrary real vectors in $\mathbb{R}^\ell$, with $|U| = n$.
3. The last assertion assumes unit vectors and $\langle \mathbf{v}_u, \mathbf{v}_v \rangle \in \{0, 1/2\}$ for all distinct $u, v \in U$.

PROVE: The real symmetric $\ell \times \ell$ matrices form an inner product space for $\langle P, Q \rangle = \operatorname{tr}(PQ)$, of dimension $\ell(\ell+1)/2$. For $\ell > 0$ the traceless subspace has dimension $\ell(\ell+1)/2 - 1$. For arbitrary real vectors $\mathbf{v}_u$, their rank-one matrices $P_u = \mathbf{v}_u \mathbf{v}_u^T$ have Gram matrix $\operatorname{tr}(P_u P_v) = \langle \mathbf{v}_u, \mathbf{v}_v \rangle^2$. Thus, if n unit vectors in $\mathbb{R}^\ell$ have distinct-pair inner products in $\{0, 1/2\}$, then $n \leq \ell(\ell+1)/2$.

PROOF SKETCH: We will use coordinates on unordered pairs for the dimension and a positive definite Gram matrix for the rank-one family.
⟨1⟩1. $\langle P, Q \rangle = \operatorname{tr}(PQ)$ is an inner product on the symmetric matrices.

⟨2⟩1. $\operatorname{tr}(PQ) = \sum_{i,j} P(i,j)Q(i,j)$.
PROOF: Expand trace and multiplication, using $Q(j,i) = Q(i,j)$. □
⟨2⟩2. This form is symmetric and bilinear.
PROOF: Each coordinate product in ⟨2⟩1 is symmetric bilinear; finite sums preserve these equalities. □

⟨2⟩3. $\operatorname{tr}(P^2) = \sum_{i,j} P(i,j)^2 \geq 0$, with equality iff $P = 0$.
PROOF: Facts 8 applied to the coordinate squares in ⟨2⟩1. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3. □
⟨1⟩2. The dimension is $\ell(\ell+1)/2$, and for $\ell > 0$ the traceless dimension is one less.

⟨2⟩1. Use one diagonal unit matrix for each index and one symmetric off-diagonal unit pair for each unordered distinct pair.
PROOF: These are symmetric matrices by their entry definitions. □
⟨2⟩2. This family is independent.
PROOF: A coefficient in a zero combination is recovered by reading its unique diagonal or unordered-pair entry. □
⟨2⟩3. It spans every symmetric matrix.
PROOF: Take the coefficients to be its diagonal and one entry of each off-diagonal pair. □
⟨2⟩4. The number of these matrices is $\ell + \binom{\ell}{2} = \ell(\ell+1)/2$.
PROOF: Apply `Sym2.card` to their index type of unordered pairs with repetitions in Facts 20. □
⟨2⟩5. For $\ell > 0$, the trace map onto $\mathbb{R}$ is surjective.
PROOF: A diagonal unit matrix has trace one; multiplying it by any real number gives that trace. □
⟨2⟩6. The trace kernel has dimension $\ell(\ell+1)/2 - 1$.
PROOF: Rank-nullity in Facts 3, since ⟨2⟩5 gives image dimension one. □
⟨2⟩7. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3, ⟨2⟩4, ⟨2⟩6. □
⟨1⟩3. For $P_u = \mathbf{v}_u \mathbf{v}_u^T$, $\operatorname{tr}(P_u P_v) = \langle \mathbf{v}_u, \mathbf{v}_v \rangle^2$.

⟨2⟩1. $\operatorname{tr}(P_u P_v) = \sum_{i,j} \mathbf{v}_u(i)\mathbf{v}_u(j)\mathbf{v}_v(j)\mathbf{v}_v(i)$.
PROOF: Expand the two rank-one matrices, multiplication and trace. □
⟨2⟩2. This sum equals $(\sum_i \mathbf{v}_u(i)\mathbf{v}_v(i))^2$.
PROOF: Reorder real factors and distribute the square of the finite sum. □
⟨2⟩3. Q.E.D.
PROOF: Use ⟨2⟩1, ⟨2⟩2. □
⟨1⟩4. $n \leq \ell(\ell+1)/2$ under the final hypothesis.

LET: $G \triangleq (\langle \mathbf{v}_u, \mathbf{v}_v \rangle)_{u,v \in U}$ is the Gram matrix of the vectors, and $H \triangleq (\operatorname{tr}(P_u P_v))_{u,v \in U}$ that of the rank-one matrices.
⟨2⟩1. $H = (I + G)/2$.
PROOF: On the diagonal both are 1; off the diagonal, squaring 0 or 1/2 gives half its value, by ⟨1⟩3. □
⟨2⟩2. For $\mathbf{x} \neq 0$, $\mathbf{x}^T (I + G)\mathbf{x}/2 > 0$.
PROOF: $G \succeq 0$ by Theorem 3.5, and $\|\mathbf{x}\|^2 > 0$. □
⟨2⟩3. H has rank n.
PROOF: By ⟨2⟩1 and ⟨2⟩2, H is positive definite; Theorem 3.3 gives n positive eigenvalues, and Theorem 3.2 counts them. □
⟨2⟩4. The span of the P_u has dimension n, at most the symmetric-space dimension.
PROOF: Use Theorem 3.5, ⟨2⟩3, and ⟨1⟩2. □
⟨2⟩5. The argument includes $n = 0$; if $\ell = 0$, a unit-vector family must be empty.

PROOF: The zero space has no unit vector, and its empty family has rank and cardinality zero. □

⟨2⟩6. Q.E.D.

PROOF: Use ⟨2⟩4, ⟨2⟩5. □

⟨1⟩5. Q.E.D.

PROOF: Use ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, ⟨1⟩4. □

Definition 10.3.

ASSUME:

1. H is a real matrix on finite row and column types; the signed matrix uses square H of order n and $x > 0$.

For a real matrix H on a finite row type U and column type V, let $P =$ `Matrix.hadamardPairs` H have rows indexed by the unordered two-element subsets $\{i, j\} \subseteq U$, with $P(\{i, j\}, u) = H(i, u)H(j, u)$. This does not depend on the ordering of i, j. For square H of order n and real $x > 0$, define `Matrix.hadamardSignedAdj` $H\ x = (PP^T - nI)/x$ and write it C_H. The name does not by itself assert that the entries are $0, 1, -1$; that requires an additional hypothesis.

Theorem 10.4 (The Hadamard pair matrix)**.**

ASSUME:

1. $n \geq 2$ and H is a real $n \times n$ matrix with entries ± 1 and $HH^T = nI$.
2. P has one row for each unordered pair of distinct row indices, $P(\{i, j\}, u) = H(i, u)H(j, u)$, and $m = \binom{n}{2}$.
3. For the scaled assertions $x > 0$ and $C_H = (PP^T - nI)/x$. For divisibility $n > 2$, x is a positive integer, and C_H has integer entries.

PROVE:

$$P^T P = \frac{n^2}{2} I - \frac{n}{2} J, \qquad \operatorname{Spec}(PP^T) = \left[\!\left[(n^2/2)^{n-1}, 0^{m-n+1}\right]\!\right], \qquad \operatorname{rk} P = n - 1.$$

For a positive real x, put $C_H = (PP^T - nI)/x$. Then

$$(xC_H + nI)\left(xC_H - \frac{n(n-2)}{2} I\right) = 0,$$

and its spectrum consists of $n(n-2)/(2x)$ with multiplicity $n - 1$ and $-n/x$ with multiplicity $m - n + 1$. If $n > 2$, x is a positive integer, and C_H has integer entries, then $x \mid n$.

PROOF SKETCH: We will derive column orthogonality, expand a sum over row pairs, transfer the nonzero Gram spectrum, and use an occurring rational eigenvalue.

⟨1⟩1. $H^T H = nI$.

⟨2⟩1. $H(H^T/n) = I$.

PROOF: Divide the assumed row-orthogonality equation by $n > 0$. □

⟨2⟩2. H is surjective.

PROOF: For any $\mathbf{y}$, the vector $(H^T/n)\mathbf{y}$ maps to $\mathbf{y}$ by ⟨2⟩1. □

⟨2⟩3. H is invertible with inverse H^T/n.

PROOF: A surjective endomorphism in finite equal dimension is injective (Facts 3); its right inverse is its inverse. □

⟨2⟩4. $H^T H = nI$.
PROOF: Multiply the left-inverse identity from ⟨2⟩3 by n. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩4. □
⟨1⟩2. $P^T P = (n^2/2)I - (n/2)J$.

⟨2⟩1. For fixed u, v, put $b_i = H(i,u)H(i,v)$; then $(P^T P)(u,v) = \sum_{i<j} b_i b_j$.
PROOF: Expand the product and use the unordered distinct-pair indexing of P. □
⟨2⟩2. $2\sum_{i<j} b_i b_j = (\sum_i b_i)^2 - \sum_i b_i^2$.
PROOF: Split the double sum into its diagonal and the two orientations of each distinct pair. □
⟨2⟩3. $\sum_i b_i = n\delta_{uv}$ and $\sum_i b_i^2 = n$.
PROOF: The first is ⟨1⟩1; each squared product of two signs equals one for the second. □
⟨2⟩4. $(P^T P)(u,v) = (n^2\delta_{uv} - n)/2$.
PROOF: Substitute ⟨2⟩3 in ⟨2⟩2 and divide by two; $\delta_{uv}^2 = \delta_{uv}$. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩4. □
⟨1⟩3. $\operatorname{rk} P = n - 1$ and $\operatorname{Spec}(PP^T) = [\![(n^2/2)^{n-1}, 0^{m-n+1}]\!]$.

⟨2⟩1. $P^T P$ has eigenvalue zero once and $n^2/2 > 0$ with multiplicity $n - 1$.
PROOF: Apply Corollary 5.9 to ⟨1⟩2. □
⟨2⟩2. $\operatorname{rk} P = n - 1$.
PROOF: Theorems 3.5 and 3.2 applied to ⟨2⟩1. □
⟨2⟩3. PP^T has the stated positive list and $m - (n-1)$ zeros.
PROOF: Apply Theorem 6.1 to ⟨2⟩1, ⟨2⟩2. □
⟨2⟩4. $m - n + 1 = (n-1)(n-2)/2 \geq 0$, with strict positivity for $n > 2$.
PROOF: Substitute $m = n(n-1)/2$ and factor. □
⟨2⟩5. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □
⟨1⟩4. The polynomial identity and spectrum of C_H hold.

⟨2⟩1. $PP^T(PP^T - (n^2/2)I) = 0$.
PROOF: The polynomial vanishes at both values in ⟨1⟩3; use Theorem 3.1. □
⟨2⟩2. $(xC_H + nI)(xC_H - n(n-2)I/2) = 0$.
PROOF: Substitute $PP^T = xC_H + nI$ in ⟨2⟩1. □
⟨2⟩3. The two values become $n(n-2)/(2x)$ and $-n/x$, with unchanged multiplicities.
PROOF: Apply Theorem 3.1 to $(z - n)/x$, using $x > 0$. □
⟨2⟩4. Q.E.D.
PROOF: Use ⟨2⟩2, ⟨2⟩3. □
⟨1⟩5. Under the integer hypotheses, $x \mid n$.

⟨2⟩1. $-n/x$ actually occurs as a rational eigenvalue of C_H.
PROOF: Its multiplicity is positive for $n > 2$ because $m - n + 1 = (n-1)(n-2)/2 > 0$ in ⟨1⟩3, ⟨1⟩4; n, x are integers and $x \neq 0$. □
⟨2⟩2. $-n/x = z$ for some integer z.
PROOF: Apply Theorem 9.6 to the integer matrix C_H and ⟨2⟩1. □
⟨2⟩3. $n = x(-z)$ in $\mathbb{Z}$.

PROOF: Clear the nonzero denominator in ⟨2⟩2 and use injectivity of the integer cast; this is the divisibility witness. □

⟨2⟩4. Q.E.D.

PROOF: Use ⟨2⟩3. □

⟨1⟩6. Q.E.D.

PROOF: Use ⟨1⟩2, ⟨1⟩3, ⟨1⟩4, ⟨1⟩5. □

Proposition 10.5 (A row-sum assertion in Section 6)**.**

ASSUME:

1. H is the real matrix displayed below, $\mathbf{h}_i$ is its row i, and $n = x = 4$, P is its matrix of unordered distinct row products, $m = 6$, and $C_H = (PP^T - 4I)/4$.

PROVE: The matrix

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

is Hadamard and its unique four-row set agrees in $(n + x)/8 = 1$ columns, but $xC_HJ = -nJ$ is false. The valid identity $P\mathbf{j}_n = 0$ does not imply $P^T\mathbf{j}_m = 0$. In particular the row-sum assertion in Section 6 of CGSS is not a consequence of these hypotheses. More generally, for every real $m \times n$ matrix P, $x > 0$ and $C = (PP^T - nI_m)/x$, we have

$$xCJ = -nJ \quad \Longleftrightarrow \quad PP^T\mathbf{j}_m = 0 \quad \Longleftrightarrow \quad P^T\mathbf{j}_m = 0.$$

If the rows of P are the distinct-row products of a real square matrix H with $HH^T = nI_n$, then $P\mathbf{j}_n = 0$.

PROOF SKETCH: We will list all six pair rows and read the three two-by-two blocks. We will also check the four-row agreement count.

⟨1⟩1. The displayed H has entries ± 1 and $HH^T = 4I$.

⟨2⟩1. Each row has squared norm four.

PROOF: Every row has four sign entries. □

⟨2⟩2. The six distinct-row products are the balanced sign rows $\mathbf{h}_2, \mathbf{h}_3, \mathbf{h}_4, \mathbf{h}_4, \mathbf{h}_3, \mathbf{h}_2$, each of coordinate sum zero.

PROOF: Multiply the four displayed entries for each of the six unordered row pairs. □

⟨2⟩3. $HH^T = 4I$.

PROOF: The diagonal values in ⟨2⟩1 and off-diagonal sums in ⟨2⟩2 give every entry. □

⟨2⟩4. Q.E.D.

PROOF: Use ⟨2⟩3. □

⟨1⟩2. The rows of P consist of each of the three nonconstant rows of H twice.

⟨2⟩1. Writing $\mathbf{h}_i$ for row i, the pairs $(1,2), (1,3), (1,4)$ give $\mathbf{h}_2, \mathbf{h}_3, \mathbf{h}_4$.

PROOF: Row one consists of ones. □

⟨2⟩2. Pair $(2,3)$ gives $(1,-1,-1,1) = \mathbf{h}_4$.

PROOF: Coordinate multiplication of the displayed rows. □

⟨2⟩3. Pair $(2,4)$ gives $(1,1,-1,-1) = \mathbf{h}_3$.

PROOF: Coordinate multiplication of the displayed rows. □

⟨2⟩4. Pair $(3,4)$ gives $(1,-1,1,-1) = \mathbf{h}_2$.

PROOF: Coordinate multiplication of the displayed rows. □

⟨2⟩5. Q.E.D.

PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩3. C_H is, after reordering, three blocks $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; thus $xC_HJ = 4J \neq -4J$.

⟨2⟩1. PP^T is three blocks $4J_2$.

PROOF: The repeated equal rows in ⟨1⟩2 have inner product four, and different nonconstant rows are orthogonal by ⟨1⟩1. □

⟨2⟩2. C_H has the three stated blocks.

PROOF: Subtract $4I$ and divide by four in ⟨2⟩1. □

⟨2⟩3. Each row sum of C_H is 1, so $xC_HJ = 4J$.

PROOF: Each block in ⟨2⟩2 has one entry one in each row. □

⟨2⟩4. The only four-row set agrees in exactly one column, equal to $(n + x)/8$.

PROOF: Column one is all ones; each other column has both signs. Here $(4 + 4)/8 = 1$. □

⟨2⟩5. Q.E.D.

PROOF: Use ⟨2⟩1, ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. □

⟨1⟩4.

ASSUME: P is any real $m \times n$ matrix, $x > 0$, and $C = (PP^T - nI_m)/x$.

PROVE: $xCJ = -nJ$ iff $PP^T\mathbf{j}_m = 0$, iff $P^T\mathbf{j}_m = 0$. If the rows of P are distinct-row products of a real square matrix H with $HH^T = nI_n$, then $P\mathbf{j}_n = 0$.

⟨2⟩1.

ASSUME: P consists of the distinct-row products of a real matrix H with $HH^T = nI_n$.

PROVE: $P\mathbf{j}_n = 0$.

PROOF: Each row of P is a product of two distinct orthogonal rows of H; its coordinate sum is their inner product. □

⟨2⟩2. $xCJ = -nJ$ is equivalent to $PP^T\mathbf{j}_m = 0$, equivalently $P^T\mathbf{j}_m = 0$.

PROOF: Substitute the definition of C from ⟨1⟩ and use the kernel equality in Theorem 3.5 with matrix P^T. □

⟨2⟩3. Q.E.D.

PROOF: Use ⟨2⟩1, ⟨2⟩2. □

⟨1⟩5. Q.E.D.

PROOF: Use ⟨1⟩3, ⟨1⟩4. □

Error 9 of `cgss/errata.md` [Err] records the row-sum error. The shorter divisibility repair is Theorem 10.4: $-n/x$ occurs with positive multiplicity $(n-1)(n-2)/2$ when $n > 2$, so rational-root integrality gives $x \mid n$.

11. Further directions

Not every proposed declaration is in Lean. The Lean formalization is in `LineGraphs/`, and `dependency-diagram.md`, at the top of the repository, maps the numbered results of [CGSS] to its declarations. The spectral certificates, positive-vector bound and similarity lemmas already have proofs in `LineGraphs/Spectral/`; that directory also defines switching equivalence. The Boolean soundness lemmas and their `Matrix` aliases are in `LineGraphs/Computations/`. Each instantiated C13, C15 or C17 certificate requires a kernel-checked proof of its true-checker hypothesis; for C13 and C15 these are in `LineGraphs/Computations/`. No structured proof obligation in this document is deferred to an unproved project declaration.

References

[Err] *Errata to CGSS*, `cgss/errata.md`, Errors 9 and 14.

[CGSS] P. J. Cameron, J. M. Goethals, J. J. Seidel and E. E. Shult, Line graphs, root systems, and elliptic geometry, *J. Algebra* **43** (1976), 305–327.

[L95] L. Lamport, How to write a proof, *Amer. Math. Monthly* **102** (1995), 600–608.